

WORKBOOK OF ANALYTICAL SKILLS-1 PEA-515



Department of Analytical Skills
Centre of Professional Enhancement

PREFACE

Companies that hire students through campus placements have various rounds to shortlist suitable candidates; these rounds include aptitude tests, group discussions and then personal interview. Most, if not all the companies follow this recruitment pattern.

Almost 90% of the applied candidates don't clear the aptitude test. The aptitude test is used to test the candidate on Quantitative Aptitude, Verbal Ability, and Analytical Ability/Logical Reasoning.

Quantitative Aptitude and Reasoning is very important subject to test your problem solving skills. So, in every competitive written exam they asked questions from this subject, not only in written they may ask some brain storming puzzles in interview also. It is the one of the key concept to qualify written exam almost every students who know basic mathematics can solve most of the questions in the exam but the main problem is that the time management, the recruiters does not give enough time to solve the problems so one who has more practice the model questions before exam can easily solve in the exams.

This book is essential for aptitude exams as all the important topics are discussed in this book. This book explains all the concepts clearly and also covers all the types of the questions.

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VEDIC MATHS

Multiplication of Two 2-Digit Numbers

1. Write Your Problem Down. Write down your numbers sitting on top of each other, like you would do when multiplying normally

$$\begin{array}{r} 21 \\ * 23 \\ \hline \end{array}$$

2. Multiply. Multiply the numbers in the ones place and put the product directly under the ones

$$\begin{array}{r} 21 \\ * 23 \\ \hline 3 \end{array}$$

3. Cross Multiply. Cross multiply like you would for fractions by taking the top number's tens digit multiplied by the bottom number's ones place. Then take the top number's ones place multiplied by the bottom number's tens place. Add the two products and place the answer to the left of the ones place's answer.

$$\begin{array}{r} 21 \\ * 23 \\ \hline 8 \quad 3 \end{array} \quad \begin{array}{l} 2*3=6 \\ 1*2=2 \\ (6+2=8) \end{array}$$

4. Multiply Again. Multiply the numbers in the tens place and place the answer to the left of the previous step's answer.

$$\begin{array}{r} 21 \\ * 23 \\ \hline 4 \quad 8 \quad 3 \end{array}$$

The working in our above example can therefore be depicted as

$$\begin{array}{r} 21 \\ \times 23 \\ \hline \end{array}$$

$$2 \times 2 / 1 \times 2 + 2 \times 3 / 1 \times 3 = 483$$

which can be summarized as

$$\begin{array}{r} 21 \\ \times 23 \\ \hline \end{array}$$

$$4 / 2 + 6 / 3 = 483$$

When the units figure is "one" in both the numbers being multiplied, the process of multiplication is simplified further. Consider the following multiplication:

$$\begin{array}{r} 41 \\ \times 81 \\ \hline \end{array}$$

$$4 \times 8 / 8 \times 1 + 1 \times 4 / 1 \times 1$$

You will notice that the middle digit of the answer is $8 \times 1 + 1 \times 4$ i.e. $(8 + 4) \times 1$. So, instead of multiplying "across" for the middle term, you could simply add the tens digit of the two numbers.

Therefore, $41 \times 81 = 32 / (4 + 8) / 1 = 3321$.

Similarly, in 71×91 , you could obtain the middle term as 16 by merely adding 7 and 9.

Multiplication of Two 3-Digit Numbers

Let us consider the multiplicand to be ABC and the multiplier to be DEF, as shown below:

$$\begin{array}{r} A B C \\ \times D E F \\ \hline \end{array}$$

Step 1 $C \times F$

Step 2 $B \times F + C \times E + \text{Carry of Step 1}$

Step 3 $A \times F + C \times D + B \times E + \text{Carry of Step 2}$

Step 4 $A \times E + B \times D + \text{Carry of Step 3}$

Step 5 $A \times D + \text{Carry of Step 4}$

For Example;

$$\begin{array}{r} 234 \\ \times 651 \\ \hline \end{array}$$

$$2 \times 6 / 6 \times 3 + 2 \times 5 / 3 \times 5 + 6 \times 4 + 2 \times 1 / 4 \times 5 + 3 \times 1 / 4 \times 1$$

$$= 12/28/41/23/4 = 152334$$

Squares of Numbers

Numbers	Method
1 – 25	Memorization
Numbers ending in a five 15, 25, 35, 45, ...	Remove the last digit (five), multiply the resulting number (n) by the next number (n + 1), and tag on a 25 at the end of the product. Example: $\begin{array}{r} 65 \\ 6 \times 7 \\ \hline 42 \end{array}$ Tag on a 25 to make 4225. Calculate $35 \times 35 = 1225$
25 – 50	Calculate the difference (d) from 50. First two digits are: 25-d Last two digits are: d^2 (if more than two digits then carry hundred's digit number) Example: $\begin{array}{r} 46 \\ d \times 46 \\ \hline 4 \end{array}$ To calculate = 46 Calculate $46 \times 46 = (25-4)16 = 2116$
50 – 75	Calculate the difference (d) from 50. First two digits are: 25+d Last two digits are: d^2 (if more than two digits then carry hundred's digit number) Example: $\begin{array}{r} 54 \\ d \times 54 \\ \hline 4 \end{array}$ To calculate = 54 Calculate $54 \times 54 = (25+4)16 = 2916$

Step 4: Multiplying the ten's place digit of the square root that we arrived at in step 1(2) and its consecutive number(3) we get, $2 \times 3 = 6$

ten's place digit of original number > Multiplication result
 $7 > 6$

So we need to select the greater number (8) as the unit's place digit of the square root.

Unit's place digit = 8

Ans: $\sqrt{784} = 28$

Cube Root of a Number

Step 1: First of all group the number in pairs of 3 starting from the right.

Step 2: To get the ten's place digit, Find the nearest cube (equivalent or greater than or less than) to the first grouped pair from left and put the cube root of the cube.

Step 3: To get the unit's place digit of the cube root.

Remember the following

Number	Cube	Cubes ends with
1	1	1
2	8	8 (Compliment of 2)
3	27	7 (Compliment of 3)
4	64	4
5	125	5
6	216	6
7	343	3 (Compliment of 7)
8	512	2 (Compliment of 8)
9	729	9
10	1000	0

Thus as seen cubes have distinct ending, there is no overlapping. Thus, if the given number is perfect cube, then the last digit will help to find the cube root.

Example: Find the cube root of 54872?

Step 1: We start by grouping the numbers in pairs of three from right as follows

54 872

Step 2: To get the ten's place digit,

We find that nearest cube to first group (54) is 27 and $(27)^{1/3} = 3$

Therefore ten's place digit = 3

Step 3: To get the unit's place digit,

We notice that the number ends with 2, so the unit's place digit of the cube root is 8 (Refer table).

Ans: cube root of 54872 = 38

Practice Questions

1. Find the product of following numbers:

- | | | |
|---------------------|---------------------|---------------------|
| 1] 53×28 | 2] 62×39 | 3] 27×83 |
| 4] 37×91 | 5] 76×48 | 6] 542×142 |
| 7] 234×471 | 8] 865×713 | 9] 364×901 |

2. Find the squares of numbers:

- | | | |
|-------------------|-------------------|--------------------|
| 1] 15^2 | 2] 75^2 | 3] 45^2 |
| 4] 85^2 | 5] 55^2 | 6] 35^2 |
| 7] 12^2 | 8] 18^2 | 9] 23^2 |
| 11] 29^2 | 11] 37^2 | 12] 44^2 |
| 15] 59^2 | 14] 67^2 | 15] 73^2 |
| 16] 121^2 | 17] 125^2 | 18] 117^2 |
| 19] 78^2 | 20] 83^2 | 21] 88^2 |
| 22] 92^2 | 23] 98^2 | 24] 94^2 |
| 25] 105^2 | 26] 109^2 | 27] 113^2 |
| 28] $65^2 - 37^2$ | 29] $99^2 - 78^2$ | 30] $111^2 - 95^2$ |

3. Find the square root of numbers:

- | | | |
|-----------|-----------|-----------|
| 1] 4624 | 2] 676 | 3] 961 |
| 4] 1521 | 5] 2304 | 6] 6724 |
| 7] 6241 | 8] 9409 | 9] 13456 |
| 10] 11664 | 11] 15129 | 12] 24964 |

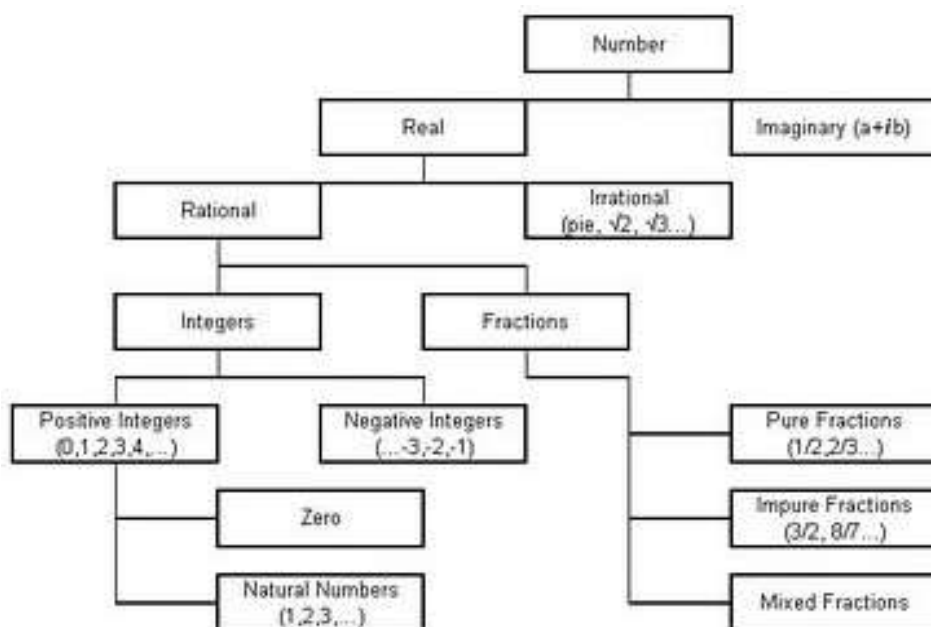
4. Find the cube root of numbers:

- | | | |
|------------|------------|-------------|
| 1] 1331 | 2] 4096 | 3] 6859 |
| 4] 10648 | 5] 21952 | 6] 50653 |
| 7] 79507 | 8] 205379 | 9] 262144 |
| 10] 636056 | 11] 830584 | 12] 1092727 |

NUMBER SYSTEM

Classification of the numbers

The concept of numbers is made clear from the number tree.



1. **Natural Numbers:** The numbers 1, 2, 3, 4, 5....are called natural numbers or positive numbers.
Example: 1, 2, 3, 4, 5 ∞
2. **Whole Numbers:** –The numbers including “0” and all natural numbers are called the whole numbers.
Example: 0, 1, 2, 3, 4, 5... ∞
3. **Integers** – The numbers including 0 and all the positive and negative of the natural numbers are called integers.
Example: $-\infty$-3, -2, -1, 0, 1, 2, 3... ∞
4. **Rational Numbers:** – A number which can be expressed in the form p/q where p and q are integers and $q \neq 0$ is called a rational number.
For example, 4 is a rational number since 4 can be written as $4/1$ where 4 and 1 are integers and the denominator $1 \neq 0$. Similarly, the numbers $3/4$, $-2/5$, etc. are also rational numbers.
Between any two numbers, there can be infinite number of other rational numbers.
Any terminating or recurring decimal is a rational number.
5. **Irrational Numbers:** – Numbers which are not rational but which can be represented by points on the number line are called irrational numbers. Examples for irrational numbers are
Example: $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{8}$, etc.
Numbers like π , e are also irrational numbers.
Between any two numbers, there are infinite numbers of irrational numbers.

Any non-terminating non-recurring decimal is an irrational number.

6. **Real numbers:** – The set of natural numbers, integers, whole numbers, rational numbers, and irrational numbers constitute the set of real numbers.
7. **Even Numbers:** – The numbers that are divisible by 2 are called even numbers.
Example: 2, 4, 6, 8, 16, 32 etc.
8. **Odd Numbers:** – The numbers that are not divisible by 2 are called odd numbers.
Example: 3, 5, 7, 9, 15 etc.
9. **Prime Numbers:** – Those numbers which are divisible by themselves and 1 are called prime numbers or a number which has only two factors 1 and itself is called a prime number.
Example: 2, 3, 5, 7 etc.
10. **Twin Primes:** – A pair of prime numbers when they differ by 2 is called twin prime numbers.
Example: (3, 5), (5, 7), (11, 13), (17, 19) etc.
11. **Co-prime Numbers:** – A pair of two natural numbers (may or may not be prime number) are said to be co-prime if their G.C.D. or H.C.F. is 1.
Example: H.C.F. (3, 4) = 1, H.C.F. (13, 15) = 1 then (3, 4) and (13, 15) are co-prime numbers.
12. **Composite Numbers:** – The natural numbers which are not prime numbers are called composite numbers OR numbers that have factors other than itself and 1, are called composite numbers.
Example: 4, 6, 9, 16, 25 etc.
Note: 1 is neither a composite number nor a prime number.
13. **Perfect Numbers:** – If the addition of all the factors of a number excluding the number itself happens to be equal to the number, it is called a perfect number.
First perfect number is 6.
Factors of 6 are 1, 2, 3, 6.
Now add all the factors excluding 6.
 $1+2+3 = 6$, hence 6 is a perfect number.
Example: 28, 496 and 8128.
14. **Complex Numbers:** – The number which have real and imaginary component is called a complex number.
Example: $3+4i$, $5+6i$, where $i = \sqrt{-1} = \text{a imaginary number}$
15. **Face Value** of a digit in a number is its own value.
Example: 6728, Face Value $\Rightarrow 6 = 6$, $7 = 7$, $2 = 2$ and $8 = 8$
16. **Place Value** of a digit is given by multiplying it with value of place where it is placed.
Example: 6729
Place Value of 9 $\Rightarrow 9 \times 1 = 9$
Place Value of 2 $\Rightarrow 2 \times 10 = 20$
Place Value of 7 $\Rightarrow 7 \times 100 = 700$
Place Value of 6 $\Rightarrow 6 \times 1000 = 6000$
17. **Fractions:** A fraction is a quantity which expresses a part of the whole,
Example: $1/4$ means one fourth of the whole

Types of Fractions

i. Proper Fraction: is one whose numerator is less than its denominator

Example: $2/3$ is proper fraction, as $2 < 3$

ii. Improper Fraction is one whose numerator is equal to or greater than its denominator

Example: $3/2$ is an improper fraction, as $3 > 2$;

$3/3$ is an improper fraction, as $3 = 3$

Some Important Points:

1. Addition or subtraction of any two odd numbers will always result in an even number or zero.
Example: $1 + 3 = 4$.
2. Addition or subtraction of any two even numbers will always result in an even number or zero.
Example: $2 + 4 = 6$.
3. Addition or subtraction of an odd number from an even number will result in an odd number.
Example: $4 + 3 = 7$.
4. Addition or subtraction of an even number from an odd number will result in an odd number.
Example: $3 + 4 = 7$.
5. Multiplication of two odd numbers will result in an odd number.
Example: $3 \times 3 = 9$.
6. Multiplication of two even numbers will result in an even number.
Example: $2 \times 4 = 8$.
7. Multiplication of an odd number by an even number or vice versa will result in an even number.
Example: $3 \times 2 = 6$.
8. An odd number raised to an odd or an even power is always odd.
9. An even number raised to an odd or an even power is always even.
10. The standard form of writing a number is $m \times 10^n$ where m lies between 1 and 10 and n is an integer.
 - If n is odd, $n(n^2 - 1)$ is divisible by 24.
Take $n = 5 \Rightarrow 5(5^2 - 1) = 120$, which is divisible by 24.
 - If n is odd prime number except 3, then $n^2 - 1$ is divisible by 24.
 - If n is odd, $2^n + 1$ is divisible by 3.
 - If n is even, $2^n - 1$ is divisible by 3.
 - If n is odd, $2^{2n} + 1$ is divisible by 5.
 - If n is even, $2^{2n} - 1$ is divisible by 5.
 - If n is odd, $5^{2n} + 1$ is divisible by 13.
 - If n is even, $5^{2n} - 1$ is divisible by 13
11. Some properties of Prime Numbers
 - The lowest prime number is 2.
 - 2 is also the only even prime number.
 - The lowest odd prime number is 3.
 - There are 25 prime numbers between 1 to 100.
 - The remainder of the division of the square of a prime number $p \geq 5$ divided by 24 or 12 is 1.

Divisibility Rules

Divisibility by	Criteria
2	A number is divisible by 2 when its units place is 0 or divisible by 2. Example: 130, 128 etc.
3	A number is divisible by 3 when the sum of its digits is divisible by 3. Example: $6561 \Rightarrow 6+5+6+1 = 18$ is divisible by 3 $17281 \Rightarrow 1+7+2+8+1 = 19$ is not divisible by 3
4	When the last two digits of the number are 0's or divisible by 4. Example: 17400, 132, 12348 etc.
5	If the unit digit is 5 or 0, the number is divisible by 5. Example: 895, 100, 125, 625, 400 etc.
6	A number is divisible by 6, if it is divisible by both 2 and 3.
7	A number is divisible by 7, if and only if the number of tens added to 5 times the number of units, is divisible by 7 Example: 105 is divisible by 7, since $10+5*5= 10+25=35$, which is divisible by 7.
8	If the last three digits of the number are 0's or divisible by 8, the number is divisible by 8. Example: 125128, 135000 etc.
9	If sum of digits is divisible by 9, the number is also divisible by 9. Example: $729 \Rightarrow 7+2+9 = 18$ is divisible by 9. $46377 \Rightarrow 4+6+3+7+7 = 27$ is divisible by 9.
10	A number is divisible by 10 if and only if the unit place digit is 0. Example: 100, 23450, 1100 etc.
11	When difference between sum of digits at odd places and sum of digits at even places is either 0 or 11, the number is divisible by 11. Example: $65967 \Rightarrow (6+9+7) - (5+6) = 22 - 11 = 11$ is divisible by 11.

Important Results:

- $\Sigma n = \frac{n(n+1)}{2}$, Σn is the sum of first n natural numbers.
- $\Sigma n^2 = \frac{n(n+1)(2n+1)}{6}$, Σn^2 is the sum of first n perfect squares.
- $\Sigma n^3 = \frac{n^2(n+1)^2}{4} = (\Sigma n)^2$, Σn^3 is the sum of first n perfect cubes.

Recurring Decimals

A decimal in which a digit or set of digits is repeated continually is called a recurring decimal. Recurring decimals are written in a shortened form, the digits which are repeated being marked by dots placed over the first and the last of them, thus

$$8/3 = 2.666..... = 2.6 \text{ or } 2.\overline{6};$$

$$1/7 = 0.142857142857142857... = 0.\overline{142857}$$

Example: Express $0.\overline{3}$ in the form of a fraction.

Solution: $0.\overline{3} = 0.3333 \dots$ □ (1)

As the period is of one digit, we multiply by 10^1
i.e. 10

$$\therefore 10 \times 0.\overline{3} = 3.333 \dots \quad \square (2)$$

(2) – (1) gives

$$9 \times 0.\overline{3} = 3 \Rightarrow 0.\overline{3} = 3/9 = 1/3$$

Pure Recurring Decimal: It is equivalent to a vulgar fraction which has the number formed by the recurring digits (called the period of the decimal) for its numerator, and for its denominator the number which has for its digits as many nines as their digits in the period.

Thus, $0.\overline{37}$ can be written as equal to $37/99$

$0.\overline{225}$ can be written as equal to $225/999$ which is the same as $25/111$

$$0.\overline{63} = 63/99 = 7/11$$

Mixed Recurring Decimal: In the numerator write the entire given number formed by the (recurring and non-recurring parts) and subtract from it the part of the decimal that is not recurring. In the denominator, write as many nines as the period (i.e., as many nines as the number of digits recurring) and then place next to it as many zeroes as there are digits without recurring in the given decimal.

$$\text{i.e. } 0.\overline{156} = 156 - 1/990 = 155/990 = 31/198$$

$$0.\overline{73} = 73 - 7/90 = 66/90 = 11/15$$

HCF and LCM

Factors and Multiples: If a number a divides another number b exactly, we say that a is a **factor** of b . In this case, b is called a **multiple** of a .

Highest Common Factor (H.C.F.) or Greatest Common Measure (G.C.M.) or Greatest Common Divisor (G.C.D.): The H.C.F. of two or more than two numbers is the greatest number that divides each of them exactly.

Least Common Multiple: The least number which is exactly divisible by each one of the given numbers is called their L.C.M.

H.C.F. and L.C.M. of Fractions:

$$\text{H.C.F.} = \frac{\text{H.C.F. of Numerators}}{\text{L.C.M. of Denominators}}$$

$$\text{L.C.M.} = \frac{\text{L.C.M. of Numerators}}{\text{H.C.F. of Denominators}}$$

The product of the two fractions is always equal to the product of LCM and HCF of the two fractions
i.e. $\text{HCF} \times \text{LCM} = \text{Product of two numbers}$

Example 1: Find the HCF of 24, 30 and 42.

Solution:

2	<u>24</u>	2	<u>30</u>	2	<u>42</u>
2	<u>12</u>	3	<u>15</u>	3	<u>21</u>
2	<u>6</u>	5	<u>5</u>	7	<u>7</u>
3	<u>3</u>	1		1	
	1				

Factors of 24 = $2 \times 2 \times 2 \times 3 = (2^3 \times 3^1)$

Factors of 30 = $2 \times 3 \times 5 = (2^1 \times 3^1 \times 5^1)$

Factors of 42 = $2 \times 3 \times 7 = (2^1 \times 3^1 \times 7^1)$

∴ The product of common prime factors with the least powers = $2^1 \times 3^1 = 6$

Example 2: Find the HCF of 26 and 455.

Solution:

$$\begin{array}{r} 26 \overline{)455} \quad (17 \\ \underline{26} \\ 195 \\ \underline{182} \\ 13 \end{array} \quad \begin{array}{r} 26 \overline{)26} \quad (1 \\ \underline{26} \\ 0 \end{array}$$

∴ Required HCF = 13.

Example 3: Find the HCF of $\frac{36}{51}$ and $3\frac{9}{17}$.

Solution: Here, $\frac{36}{51} = \frac{12}{17}$ and $3\frac{9}{17} = \frac{60}{17}$

Now, we have to find the HCF of $\frac{12}{17}$ and $\frac{60}{17}$.

According to the formula,

$$\text{HCF of fractions} = \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}} = \frac{\text{HCF of 12 and 60}}{\text{LCM of 17 and 17}} = \frac{12}{17}$$

Some important Facts:

1. If a, b and c give remainders p, q and r respectively, when divided by the same number H, then H is HCF of (a-p), (b-q), (c-r).
2. If the HCF of two numbers 'a' and 'b' is H, then, the numbers (a+b) and (a-b) are also divisible by H.
3. If a number N always leaves a remainder R when divided by the numbers a, b and c, then N = LCM (or a multiple of LCM) of a, b and c + R.
4. If a Number when divided by a,b,c leaves a remainders of x,y,z respectively and a-x= b-y= c-z= P, then the smallest number satisfying this condition is L.C.M(a,b,c)- P.

Example: Which is the smallest numbers which leaves a common remainder of 4 when divided by 6, 7, and 9?

Solution: Here you should remember that the smallest number is 4. The next such number will be (LCM of 6,7,9) + 4 i.e. 126+4 or 130.

REMAINDERS

A number M when divided by N leaves remainder R , and quotient is Q can be represented by $M = NQ + R$. where M is dividend, N is divisor, Q is quotient and R is remainder. The above rule is what is commonly called as the Division algorithm.

The concepts required to solve the questions of remainders are enumerated below

- Reducing remainders
- Negative remainders
- Fermat's little theorem
- Chinese remainders
- Wilson's rule.
-

1. Reducing remainders

Some basic rules are given below:

Remainders $(axb)/c = \text{remainder}(a/c) \times \text{remainder}(b/c)$

Remainder $(a+b)/c = \text{remainder}(a/c) + \text{remainder}(b/c)$

Remainder $(a-b)/c = \text{remainder}(a/c) - \text{remainder}(b/c)$

Example 1: $(142+143+145)/7$. What is the remainder.?

Solution: $(2 + 3 + 5)/7 = \text{remainder is } 5$

Example 2: $(142 \times 142 \times 142 \times \dots \times 100 \text{ times})/7$. what is the remainder?

Solution: $(2 \times 2 \times 2 \times \dots \times 100 \text{ times})/7$.

$\Rightarrow (8 \times 8 \times 8 \times \dots \times 33 \text{ times } \times 2)/7$

$\Rightarrow 2$.

Keep on dividing the remainders till the final remainder is less than divisor.

2. Concept of Negative Remainder

Remainder $27/7 = 6$ or its conjugate -1

Remainder $26/7 = 5$ or its conjugate -2

Example: What is the remainder $15^{97}/8$?

Solution: $(15 \times 15 \times 15 \times \dots \times 97 \text{ times})/8$

$(-1 \times -1 \times -1 \times \dots \times 97 \text{ times})/8$

-1 or its conjugate 7

3. Fermat's little theorem

Remainder $(M^{N-1})/N = 1$

Where M and N are co-prime and N is a prime number.

Example 1: Find the remainder of $(2^{100})/101$?

Solution: $(2^{100})/101 = 1$ (using Fermat's Theorem)

Example 2: Find the remainder when (5^{1000}) is divided by 77 ?

Solution: $(5^{1000}) / (7 \times 11)$

Using Fermat's rule $5^6/7 = \text{remainder is } 1$ so $5^{30}/7 (\text{remainder}) = 1$

Using Fermat's rule $5^{10}/11 = \text{remainder } 1$ so $5^{30}/11 (\text{remainder}) = 1$

$5^{30}/77$ remainder =1
 $((5^{30})^{33} \times 5^{10})/77$ = remainder 23;
 Where ($5^{10}/77$) remainder is 23 has to be dealt separately by reducing remainders theory.

4. Chinese remainders

Remainder $N/(axb) = apr_1 + bqr_2$:

Where remainder of $N/a = r_2$ and $N/b = r_1$ and $ap + bq = 1$.

Example: Find the remainder $(5^{1000})/(7 \times 11)$

Solution: remainder $(5^{1000})/7 = ((5^6)^{166} \times 5^4)/7 = 2$ (using Fermat's rule $5^6/7$ =Remainder is 1.)

Remainder $(5^{1000})/11 = (5^{10})^{100}/11 = 1$ (using Fermat's rule $5^{10}/11$ = remainder 1

$7p + 11q = 1$ for $p = -3, q = 2$

So the final remainder is $= 7 \times -3 \times 1 + 11 \times 2 \times 2 = 23$.

5. Wilson's rule:

Remainder $((N-1)! + 1)$ when divided by N has a remainder of 0

Example: $(4! + 1)/5$ = remainder is 0,

$(6! + 1)/7$ = remainder is 0.

Example: Find the remainder for $(96! + 1000)/97$:

$(96! + 1)$ is divisible by 97

So final remainder is remainder $999/97 = 29$.

Important result:

- Theorem: $a^n + b^n$ is divisible by $a + b$ when n is ODD.
- Theorem 2: $a^n - b^n$ is divisible by $a + b$ when n is EVEN.
- Theorem 3: $a^n - b^n$ is ALWAYS divisible by $a - b$.

Cyclicity / Unit Digit

Number	$\wedge 1$	$\wedge 2$	$\wedge 3$	$\wedge 4$	Cyclicity
2	2	4	8	6	4
3	3	9	7	1	4
4	4	6	4	6	2
5	5	5	5	5	1
6	6	6	6	6	1
7	7	9	3	1	4
8	8	4	2	6	4
9	9	1	9	1	2

We can summarize it as:

Cyclicity of 2, 3, 7 and 8 is 4.

Cyclicity of 4 and 9 is 2.

Cyclicity of 0, 1, 5 and 6 is 1.

Steps to find unit digit

1. Consider only Unit Digit of a number. Divide power by the Cyclicity of unit digit of a number or by 4 and find the remainder.

Eg: Find the units place digit of 2^{99}

$$2^{99/4} \Rightarrow 99/4 \Rightarrow \text{remainder is } 3$$

2. Make remainder as a power of a unit digit and consider only last digit.

$$2^3 = 8$$

So, unit digit of $2^{99} = 8$

3. If remainder is 0, then Cyclicity will become the power of unit digit.

Example: Find the units place digit of 252^{84} ?

Solution: Consider only unit digit of a number i.e. 2^{84}

$$84/4 = 0 \text{ (Remainder)}$$

So, power of 2 will become its Cyclicity i.e. 4.

Therefore, Unit digit of $2^4 = 6$.

Example: What is the unit digit in the product $(3^{65} \times 6^{59} \times 7^{71})$?

Solution: firstly, find the unit digit of 3^{65}

$$3^{65/4} = 3^1 = 3$$

Unit digit of $6^{59} = 6$ (Cyclicity of 6 is 1 i.e. unit digit of 6 is always 6)

$$\text{Unit digit of } 7^{71} = 7^{71/4} = 7^3 = 3$$

$$3 \times 6 \times 3 = 4$$

FACTORIAL

Factorial is an important topic in quantitative aptitude preparation. The factorial of a non-negative integer n is denoted as $n!$. The notation was introduced by Christian Kramp in 1808. $n!$ is calculated as the product of all positive integers less than or equal to n .

$$\text{i.e. } 6! = 1 \times 2 \times 3 \times 4 \times 5 \times 6 = 720$$

$$n! = 1 \text{ when } n = 0, \text{ and } n! = (n-1)! \times n \text{ if } n > 0$$

$n!$ is the number of ways we can arrange n distinct objects into a sequence.

$2! = 2$ means numbers 1, 2 can be arranged in 2 sequences (1, 2) and (2, 1).

We can arrange 0 in one way. So $0! = 1$, not zero. Now we know why, and no need to say “its like that” if someone asks ;-)

Find the highest power of a prime number in a given factorial

The highest power of prime number p in $n! = [n/p1] + [n/p2] + [n/p3] + [n/p4] + \dots$ where $[n/p1]$ denotes the quotient when n is divided by p

Example 1: The maximum power of 5 in $60!$

Sol: $60! = 1 \times 2 \times 3 \times \dots \times 60$ so every fifth number is a multiple of 5. So there must be $60/5 = 12$

In addition to this 25 and 50 contribute another two 5's. so total number is $12 + 2 = 14$

Short cut: $[60/5] + [60/25] = 12 + 2 = 14$

Here $[]$ Indicates greatest integer function.

Example 2: How many zero's are there at the end of $100!$

Sol: A zero can be formed by the multiplication of 5 and 2. Since $100!$ contains more 2's than 5's, we can find the maximum power of 5 contained in $100!$

$$\Rightarrow 100/2 + 100/4 + 100/8 + 100/16 + 100/32 + 100/64 = 50 + 25 + 12 + 6 + 3 + 1 = 97$$

$$\Rightarrow 100/5 + 100/25 = 20 + 4 = 24$$

LEVEL - I

- Find the least value of * for which $4832*18$ is divisible by 11.
A] 5 B] 3 C] 7 D] 11
- Is 52563744 divisible by 24?
A] Yes B] No C] can't be determined D] None of these
- What least number must be subtracted from 1672 to obtain a number which is completely divisible by 17?
A] 5 B] 7 C] 3 D] 6
- What least number must be added to 2010 to obtain a number which is completely divisible by 19?
A] 5 B] 4 C] 19 D] None of these
- On dividing 12401 by a certain number, we get 76 as quotient and 13 as remainder. What is the divisor?
A] 163 B] 173 C] 183 D] None of these
- On dividing a certain number by 342, we get 47 as remainder. If the same number is divided by 18, what will be the remainder?
A] 7 B] 9 C] 11 D] 13
- A number when successively divide by 3, 5 and 8 leaves remainders 1, 4 and 7 respectively. Find the respective remainders if the order of divisors be reversed.
A] 5, 4, 2 B] 6, 4, 2 C] 1, 1, 3 D] None of these
- What is the least perfect square divisible by 8, 9 and 10?
A] 4000 B] 6400 C] 3600 D] 14641
- $4a56$ is a four-digit numeral divisible by 33. What is the value of a?
A] 3 B] 4 C] 5 D] 6
- The sum of five distinct whole numbers is 337. If 60 is the smallest of them, what is the maximum value the largest number can have?
A] 91 B] 70 C] 97 D] 274
- The number of 2 digit prime number is
A] 25 B] 17 C] 21 D] None of these
- If $n^2 = 12345678987654321$, what is n?
A] 12344321 B] 1235789 C] 111111111 D] 11111111
- $46917 \times 9999 = ?$
A] 4586970843 B] 469123083 C] 4691100843 D] 584649125
- Which of the following has fractions in ascending order?
A] $\frac{1}{3}, \frac{2}{5}, \frac{4}{7}, \frac{3}{5}, \frac{5}{6}, \frac{6}{7}$ B] $\frac{1}{3}, \frac{2}{5}, \frac{3}{5}, \frac{4}{7}, \frac{5}{6}, \frac{6}{7}$
C] $\frac{1}{3}, \frac{2}{5}, \frac{3}{5}, \frac{4}{6}, \frac{5}{7}, \frac{6}{7}$ D] $\frac{2}{5}, \frac{3}{5}, \frac{1}{3}, \frac{4}{7}, \frac{5}{6}, \frac{6}{7}$

15. Which of the following are in descending order or their value?

A] $\frac{5}{9}, \frac{7}{11}, \frac{8}{15}, \frac{11}{17}$

B] $\frac{5}{9}, \frac{8}{15}, \frac{11}{17}, \frac{7}{11}$

C] $\frac{11}{17}, \frac{7}{11}, \frac{8}{15}, \frac{5}{9}$

D] $\frac{11}{17}, \frac{7}{11}, \frac{5}{9}, \frac{8}{15}$

16. When $0.4\overline{7}$ is converted into a fraction, the result is:

A] $\frac{46}{90}$

B] $\frac{46}{99}$

C] $\frac{47}{90}$

D] $\frac{47}{99}$

17. The value of $0.5\overline{7}$ is:

A] $\frac{57}{10}$

B] $\frac{57}{99}$

C] $\frac{26}{45}$

D] $\frac{52}{9}$

18. The value of $4.1\overline{2}$ is:

A] $4\frac{11}{90}$

B] $4\frac{11}{99}$

C] $\frac{371}{900}$

D] None of these

19. The value of $2.1\overline{36}$ is:

A] $\frac{47}{220}$

B] $\frac{68}{495}$

C] $2\frac{3}{22}$

D] None of these

20. Reduce $\frac{391}{667}$ to lowest terms.

A] $\frac{17}{29}$

B] $\frac{11}{23}$

C] $\frac{13}{25}$

D] $\frac{14}{27}$

LEVEL – II

1. Find the HCF of $2^3 \times 3^2 \times 5 \times 7^5$; $2^2 \times 5^2 \times 7^3$ and $2^3 \times 5^3 \times 7^2$.

A] 940

B] 980

C] 930

D] 925

2. Find the highest common factor of (34, 85)

A] 4

B] 12

C] 17

D] 19

3. The LCM of (198, 252, 308) is:

A] 2072

B] 2772

C] 2727

D] 2770

4. Which of the following is a pair of co-primes?

A] (16, 62)

B] (18, 24)

C] (20, 27)

D] (23, 92)

5. The HCF of $\left(\frac{3}{4}, \frac{5}{6}, \frac{6}{7}\right)$ is

A] $\frac{2}{93}$

B] $\frac{1}{84}$

C] $\frac{1}{83}$

D] $\frac{3}{91}$

6. The product of two numbers is 5476. If the HCF of these numbers is 37. The greater number is:

- A] 107 B] 111 C] 148 D] 185
7. The product of two numbers is 2028 and their HCF is 13. How many pairs of such numbers are possible?
A] 1 B] 2 C] 3 D] 4
8. The greatest possible length which can be used to measure exactly the lengths: 20 m 6 cm, 11m 90 cm and 14 m 45 cm is:
A] 17 B] 18 C] 19 D] 21
9. Find the greatest number that will divide 48, 97 and 188 and leaves the remainder 6 in each case.
A] 4 B] 7 C] 2 D] 6
10. The greatest number, which when divides 1358, 1870, and 2766 leaves the same remainder 14 in each case, is:
A] 124 B] 64 C] 156 D] 260
11. What will be the least number which when doubled becomes exactly divisible by 9, 15, 21, and 30?
A] 196 B] 189 C] 630 D] 315
12. The least five digit number which is exactly divisible by 12, 18, and 21 is:
A] 10010 B] 10015 C] 10080 D] 10020
13. The greatest four digit number which is divisible by 18, 25, 30, and 48 is:
A] 9000 B] 9200 C] 7200 D] 9729
14. Find the least multiple of 23 which when divided by 24, 21, and 18 leaves the remainders 13, 10, and 7 respectively.
A] 3004 B] 3024 C] 3013 D] 3026
15. The least number which when divided by 5, 6, 7, and 8 leaves the remainder 3. But when divided by 9 leaves no remainder, is:
A] 1766 B] 1683 C] 2327 D] 1895
16. Four different devices beep after every half hour, 60 min, 1 and half hours, and 135 min respectively. All the devices beeped together at 12 noon. They will beep together again at:
A] 12 midnight B] 6 AM C] 9 PM D] 7:30 AM
17. X, Y, Z start at the same time in the same direction to run around a circular stadium. X completes a round in 63 seconds, Y in 105 seconds and Z in 210 seconds. If they start at the same time, then at what time will they meet again at the starting point?
A] 9 min 9 seconds B] 10 min 30 seconds
C] 10 min 6 seconds D] 8 min 4 seconds
18. What is the unit digit in the product $(684 \times 759 \times 413 \times 676)$?
A] 6 B] 8 C] 2 D] None of these
19. What is the unit digit in the product $(3547)^{153} \times (251)^{72}$?
A] 1 B] 3 C] 7 D] None of these
20. What is the unit digit in $\{(264)^{102} + (264)^{103}\}$?
A] 0 B] 1 C] 2 D] 4

LEVEL – III

1. Find the total number of prime factors in the product $\{(4)^{11} \times (7)^5 \times (11)^2\}$.
A] 31 B] 10 C] 11 D] 29
2. Find the remainder when 2^{31} is divided by 5.
A] 1 B] 2 C] 3 D] 4
3. Find the Remainder of $\frac{2^{96}}{96}$
A] 16 B] 84 C] 64 D] None
4. Find the Remainder of $\frac{3^{1002}}{33}$
A] 1 B] 3 C] 9 D] 33
5. Find the Remainder of $\frac{100!}{101}$
A] 1 B] 0 C] 101 D] None of these
6. No. of zeros at the end of $(45!)^{450}$
A] 10 B] 450 C] 10^{45} D] 4500
7. No. of zeros at the end of $(31)! \times (42)! \times (100!)$
A] 48 B] 16 C] 6 D] 40
8. Remainder when $(1! + 2! + 3! + \dots + 4000!)$ is divided by 7
A] 7 B] 1 C] 5 D] None of these
9. What will be unit's digit in the expression $78^{5562} \times 56^{256} \times 97^{1250}$?
A] 9 B] 4 C] 6 D] 3
10. Find the first non-zero digit of the number 80^{127} from right side.
A] 8 B] 4 C] 2 D] 6
11. Find the last digit of $1^5 + 2^5 + 3^5 + \dots + 99^5$
A] 1 B] 0 C] 3 D] 5
12. Find the unit's digit of $27^{23} - 23^{27}$
A] 6 B] 4 C] 2 D] 8
13. What is the sum of the factors of 221?
A] 222 B] 251 C] 252 D] 262
14. The number of Zeroes in $18!$ is:
A] 4 B] 6 C] 3 D] 8
15. The number of Zeroes in $29!$ is:

A] 4 B] 3 C] 6 D] 7

16. The digit in the unit place of the number represented by $(7^{95} - 3^{58})$ is :

A] 0 B] 7 C] 6 D] 4

17. If the number 42573* is exactly divisible by 72, then which of the following number should replace *?

A] 4 B] 5 C] 7 D] 6

18. How many numbers between 200 and 600 are divisible by 4, 5 and 6 ?

A] 5 B] 6 C] 7 D] 8

19. X, Y and Z are 3 distinct even integers ranging from 1 to 25. Which of the following is necessarily true?

1. X^Y/Z is an even integer.
2. $17XY^3$ is an even integer
3. $8 Y^X/Z$ is an even integer
4. $3Y(Z^X)$ is an even integer

A] 1 only B] 2 only C] 4 only D] Both 2 & 4

20. Let x, y and z be distinct integers, that are odd and positive. Which one of the following statements cannot be true?

- A] xyz^2 is odd`
- B] $(x - y)^2 z$ is even
- C] $(x + y + z)^2 (x + y)$ is even
- D] $(x - y)(y + z)(x + y - z)$ Cannot be even

AVERAGE

Average is a very simple but effective way of representing an entire group of by a single value.

The term average is also referred to as 'Mean'. Basic formula to calculate the average is as follows

$$\text{Average} = \frac{\text{Sum of Observations}}{\text{Number of Observations}}$$

Average is used quite regular in our day to day life. For example to calculate the average marks of the students, Average height of a particular group etc.

Points to remember

- (i) If the value of each item is increased by the same value P , then the average of the group or items will also increase by p .
- (ii) If the value of each item is decreased by the same value p , then the average of the group or items will also decreased by p .
- (iii) If the value of each item is multiplied by the same value, then the average of the group or items will also be multiplied by p .
- (iv) If the value of each item is divided by the same value P ($P \neq 0$), then the average of the group or items will also be divided by p .
- (v) The average of a group of items will always lie between the smallest value in the group and largest value in the group – i.e., the average will be greater than the smallest value and less than the largest value in the group

Example 1: What is the average of First 10 Prime numbers?

Solution: First 10 Prime numbers are 2,3,5,7,11,13,17,19,23,29.
Hence, Average = $\{2+3+5+7+11+13+17+19+23+29\} / 10$
 $= 129 / 10 = 12.90$

Example 2: Four children have 30, 40, 50 and 60 toffees with them. What is the average number of toffees with them?

Solution: Average = $(30 + 40 + 50 + 60) / 4 = 45$
Using the above point, the average is $(40+50)/2 = 45$

Deviation Method for Calculation of Average

Now take a set of values which are typically cumbersome to deal with.

Example 3: Find average of 213, 227, 233, 223 and 229.

Solution: In the above case take the arbitrary number to be 220.
Then the average can be found as $\{220 + (-7+7+13+3+9)\}/5 = 225$.

Example 4: Find the average of 1093, 1103, 1109, 1089 and 1096?

Solution: This can be done as $(1093 + 1103 + 1109 + 1089 + 1096) / 5$
Or we observe that the numbers are close to 1100. Let us say, the numbers are
 $1093 = 1100 - 7;$ $1103 = 1100 + 3;$ $1109 = 1100 + 9$
 $1089 = 1100 - 11;$ $1096 = 1100 - 4$

$$\text{Sum} = 1100 \times 5 - 10$$

$$\text{So, average} = (1100 \times 5 - 10) / 5 = (1100/5) - 2 = 1098$$

This can be written as

$$P = P + \frac{\sum(P_i - P)}{n}$$

Here it doesn't really make a difference whether we assume the mean as 1100 or 1095 etc.

This method would come in handy while dealing with weighted averages or in DI.

Weighted Average

When two groups of items are combined together, then we can talk of the average of the entire group. However, if we know only the average of the two groups individually, we cannot find out the average of the combined group of items.

For example, there are two sections A and B of a class where the average height of section A is 150 cm and that of section B is 160 cm. On the basis of this information alone, we cannot find the average of the entire class (of the two sections).

Important Formulae Related to Average of numbers

1. Average of first n natural number = $(n+1)/2$
2. Average of first n even number = $(n+1)$
3. Average of first n odd number = n
4. Average of consecutive number = $(\text{First number} + \text{Last number})/2$
5. Average of 1 to n odd numbers = $(\text{Last odd number} + 1)/2$
6. Average of 1 to n even numbers = $(\text{Last even number} + 2)/2$
7. Average of squares of first n natural numbers = $[(n+1)(2n+1)]/6$
8. Average of the cubes of first n natural number = $[n(n+1)^2]/4$
9. Average of n multiples of any number = $[\text{Number} \times (n+1)]/2$

Example 5: Find out the average of 2, 4, 6, 8, 10, 12 and 14.

Solution: As we know that average of 1st n' even numbers = $(n + 1)$

$$\therefore \text{Required average} = (7 + 1) = 8$$

Example 6: Calculate the average of 1, 3, 5, 7, 9, 11, 13, 15 and 17.

Solution: As we know, average of 1st n' odd numbers = n

$$\therefore \text{Required average} = 9$$

Example 7: What will be the average of 1, 2, 3, 4... 51, 52, 53?

Solution: As we know, average of consecutive numbers = $\frac{\text{First Number} + \text{Last Number}}{2}$

Where first number = 1 and last number = 53

$$\therefore \text{Required average} = \frac{1+53}{2} = \frac{54}{2} = 27$$

Example 8: Calculate the average of the squares of natural numbers from 1 to 25.

Solution: According to the formula, average of squares of first ' n ' natural numbers = $\frac{(n+1)(2n+1)}{6}$

Where $n = 25$.

$$\therefore \text{Required average} = \frac{(25+1)(2 \times 25+1)}{6} = \frac{26 \times 51}{6} = \frac{1326}{6} = 221$$

Example 9: The average weight of a group of 15 friends increases by 1 kg, when a person joins the group. Find the weight of the person who joins the group, if the initial average weight of the group is 48 kg.

Solution: Let the weight of the person joining the group be x kg.

$$\text{Given, } (15 \times 48 + x) / 16 = 49$$

$$\Rightarrow x = 16 \times 49 - 15 \times 48 = 784 - 720 = 64$$

Hence the weight of the new person is 64 kg.

Alternate Method:

When the new person joins group as the average weight of the group increases by 1 kg, we can understand that this person is bringing 1 kg additionally not only for himself but even for others. Hence his weight should be $48 + 1$ (16) i.e., 64 kg

Example 10: In a class there are two sections $\rightarrow$ A and B. Section A contains 50 students with an average of 30 marks in Maths. Section B contains 75 students with an average of 40 marks in Maths. What is the average mark of the whole class?

Solution: For finding out the average mark of the whole class we will need the sum of marks of the whole class.

The sum of marks of all students in section A is $50 \times 30 = 1500$

The sum of marks of all students in section B is $75 \times 40 = 3000$

Thus total sum of marks for the class is $(1500+3000) = 4500$

So the average mark of the class is $4500 / (50+75) = 36$

Example 11: The average age of a group of friends is 34 years. If five new friends with an average age of 30 years join the group, the average of the entire group becomes 32 years. How many people were there in the group initially?

Solution: Let there be ' n ' people initially in the group. Then the total age of the group after the five new friends joined the group is $34n + (5 \times 30)$

But this is also equal to $32(n + 5)$

$$\therefore 32(n+5) = 34n + 150 \Rightarrow n = 5$$

So, there were 5 friends in the group initially.

Example 12: A batsman scored an average of 55 runs in the first 6 tests. If the first test is not counted and the seventh is counted then his average score goes, up to 57. If the score in the first test is 50 runs. Find his score in the seventh test.

Solution: Total score in the first six tests = $6 \times 55 = 330$

Total score in the last six tests = $6 \times 57 = 342$

Score in all seven tests = Total score in the last

Six tests + score in the 1st test = $342 + 50 = 392$

$$\therefore \text{Score in the 7th test} = \text{Total score in all 7 test} - \text{total score in the first 6 tests} \\ = 392 - 330 = 62$$

Example 13: Twelve years ago, the average of the ages of the members of a joint family having ten members was 25 years. Four years later a member aged 50 years died and a child was born in the

family that year. Four years after that, another member aged 50 years died and another child was born. Find the present average age of the members of the family (in years).

Solution: Had there been no alterations, the current average would have been $25 + 12 = 37$.

The first person who goes out takes with him 50 years, Similarly, the second person who goes out takes with him another 50 years.

$$\begin{aligned}\text{Present average} &= 37 - (50 + 50)/10 = 37 - 10 \\ &= 27 \text{ years.}\end{aligned}$$

Example 14: The average of five positive numbers is 213. The average of the first two numbers is 233.5 and the average of last two numbers is 271. What is the third number?

Solution: The sum of the five numbers $= 5 \times 213 = 1065$

$$\text{The sum of the first two numbers} = 2 \times 233.5 = 467$$

$$\text{The sum of the last two numbers} = 542$$

$$\text{Then the sum of the four numbers} = 467 + 542 = 1009$$

$$\text{So, the third number will be} = 1065 - 1009 = 56.$$

Example 15: The average marks of 65 students in a class were calculated as 150. It was later realized that the marks of one of the students was calculated as 142, whereas his actual average marks were 152. What is the actual average mark of the group of 65 students? (Rounded off to two digits after decimal)

Solution: Increase in total marks $= 152 - 142 = 10$

$$\text{Therefore the New average} = 150 + 10/65 = 150.15$$

Example 16: In a class there are 32 boys and 28 girls. The average age of the boys in the class is 14 years and the average age of the girls in the class is 13 years. What is the average age of the whole class? (Rounded off to two digits after decimal)

Solution: The sum of the ages of 32 boys $= 32 \times 14 = 448$

$$\text{The sum of the ages of 28 girls} = 28 \times 13 = 364$$

$$\text{Therefore, the sum of the ages of the whole class of 60 students} = 812$$

$$\text{The average age of the whole class of 60 students} = 812/60 = 13.53$$

Example 17: Srikanth earned an average of Rs. 16,00 per month from January to June. Then, he earned Rs. 1500, Rs, 1800, Rs, 1900 and Rs. 2150 respectively during the months July, through October. During November he earned 5-% of what he earned in December. If his average earnings for the entire year is Rs. 1,600. Find his earnings in the month of November.

Solution: Let earnings for the month of November be 'x'.

Then total annual earnings will be

$$(1600 \times 6) + 1500 + 1800 + 1900 + 2150 + 3X = 16950 + 3x$$

Total earnings for the entire year

$$= 12 \times 1600 \text{ (given)}$$

$$= 19200 \text{ which is equal to } 16950 + 3x$$

$$16950 + 3x = 19200 \Rightarrow 3x = 2250$$

$$\therefore x = \text{Rs } 750$$

Hence, the earnings in the month of November is Rs 750.

Level - I

1. The average of first five prime numbers is:
A] 4.5 B] 5 C] 5.6 D] 7.5
2. The average of first five multiples of 3 is:
A] 3 B] 9 C] 12 D] 15
3. The average height of 30 boys out of a class of 50 is 160 cm. If the average height of the remaining boys is 165 cm, the average height of the whole class (in cm) is:
A] 161 B] 162 C] 163 D] 164
4. The average of three numbers is 20. If the two numbers are 16 and 22, the third number is:
A] 22 B] 20 C] 19 D] 18
5. The average of five results is 46 and that of the first four is 45. The fifth result is:
A] 1 B] 10 C] 12.5 D] 50
6. The average of Radhika's marks in 7 subjects is 75. His average in six subjects excluding science is 72. How many marks did he get in Science?
A] 72 B] 90 C] 93 D] None of these
7. The average of eight numbers is 14. The average of six of these numbers is 16. The average of the remaining two numbers is:
A] 4 B] 8 C] 16 D] Data inadequate
8. Average of four consecutive odd numbers is 106. What is the third number in ascending order?
A] 107 B] 111 C] 113 D] cannot be determined
9. Of the three numbers, the average of the first and the second is greater than the average of the second and the third by 15. What is the difference between the first and the third of the three numbers?
A] 15 B] 45 C] 60 D] None of these
10. The ratio of roses and lilies in a garden is 3:2 respectively. The average number of roses and lilies is 180. What is the number of lilies in the garden?
A] 144 B] 182 C] 216 D] 360
11. The average price of three items of furniture is Rs. 15000. If their prices are in the ratio 3:5:7, the price of the cheapest item is:
A] R9000 B] R15000 C] R18000 D] R21000
12. Of three numbers, second is twice the first and is also thrice the third. If the average of the three numbers is 44, the largest number is:
A] 24 B] 36 C] 72 D] 108
13. The average of ten numbers is 7. If each number is multiplied by 12, then the average of new set of numbers is:
A] 7 B] 19 C] 82 D] 84

14. The average age of 30 students of a class is 12 years. The average age of a group of 5 of the students is 10 years and that of another group of 5 of them is 14 years. What is the average age of the remaining students?
A] 8 years B] 10 years C] 12 years D] 14 years
15. The average of 50 numbers is 38. If two numbers 45 and 55 are discarded, the average of the remaining numbers is:
A] 36.5 B] 37 C] 37.5 D] 37.52
16. The mean of 100 observations was calculated as 40. It was found later on that one of the observations was misread as 83 instead of 53. The correct mean is:
A] 39 B] 39.7 C] 40.3 D] 42.7
17. The average of six numbers is 30. If the average of first four is 25 and that of last three is 35, the fourth number is:
A] 25 B] 30 C] 35 D] 40
18. The average of 11 observations is 60. If the average of first five observations is 58 and that of the last five is 56, the sixth observation is:
A] 90 B] 110 C] 85 D] 100
19. Harish has twice as much money as Rohan and Rohan has 50% more money than what Anita has. If the average money with them is Rs.110, then Harish has:
A] Rs.55 B] Rs. 60 C] Rs.90 D] Rs. 180
20. A motorist has travels to a place 150 km away at an average speed of 50 km per hour and returns at 30 km per hour. His average speed for whole journey in km per hour is
A] 35 B] 37 C] 37.5 D] 40

LEVEL – II

1. The average of 5 numbers is 7. When 3 new numbers are added, the average of the eight numbers is 8.5. The average of three new numbers is:
A] 11 B] 7.75 C] 8.5 D] 7
2. The average age of 30 students is 9 years. If the age of their teacher is included, it becomes 10 years. The age of the teacher (in years) is:
A] 27 B] 31 C] 35 D] 40
3. The average age of 24 boys and the teacher is 15 years. When the teacher's age is excluded, the average decreases by 1. What is the age of the teacher?
A] 38 years B] 39 years C] 40 years D] Data inadequate
4. The average salary per month of 30 employees in a company is Rs. 4000. If the manager's salary is added, the average salary increases to Rs. 4300, what is the salary of the manager?
A] Rs. 10000 B] Rs. 13000 C] Rs. 12000 D] Rs. 13300

5. The average age of 40 students of a class is 15 years. When 10 new students are admitted, the average is increased by 0.2 years. The average age of new students is:
A] 15.2 years B] 16 years C] 16.2 years D] 16.4 years
6. The average weight of 8 men is increased by 1.5 kg when one of the men who weigh 65 kg is replaced by a new man. The weight of the new man is:
A] 76 kg B] 76.5 kg C] 76.7 kg D] 77 kg
7. The average weight of 6 men decreases by 3 kg when one of them weighing 80 kg is replaced by a new man. The weight of the new man is:
A] 56 kg B] 58 kg C] 62 kg D] 76 kg
8. The average age of a committee of eight members is 40 years. A member aged 55 years retired and his place was taken by another member aged 39 years. The average age of the present committee is:
A] 39 years B] 38 years C] 36 years D] 35 years
9. A cricketer has a certain average for 9 innings. In the tenth innings, the score is 100 runs, thereby increasing his average by 8 runs. His new average is:
A] 20 runs B] 24 runs C] 28 runs D] 32 runs
10. A man whose bowling average is 12.4 takes 5 wickets for 26 runs and thereby decreases his average by 0.4. The number of wickets, taken by him before his last match is:
A] 85 B] 78 C] 72 D] 64
11. The mean temperature of Monday to Wednesday was 37°C and of Tuesday to Thursday was 34°C . If the temperature on Thursday was $\frac{4}{5}$ that of Monday, the temperature on Thursday was:
A] 36.5°C B] 36°C C] 35.5°C D] 34°C
12. Three years ago, the average age of X, Y and Z was 27 years and that of Y and Z, 5 years ago was 20 years. X's present age is:
A] 30 years B] 35 years C] 40 years D] 48 years
13. Three years ago, the average age of a family of 5 members was 17 years. A baby having been born, the average age of the family is the same today. The present age of the baby is:
A] 2 years B] 2.4 years C] 3 years D] 1.5 years
14. The average monthly income of a family of four earning members was Rs.15130. One of the daughters in the family got married and left home, so the average monthly income of the family came down to Rs.14660. What is the monthly income of the married daughter?
A] Rs.15350 B] Rs.12000 C] Rs.16540 D] Cannot be determined
15. The average of four positive integers is 73.5. The highest integer is 108 and the lowest integer is 29. The difference between the remaining two integers is 15. Which of the following is the smaller of the remaining two integers?
A] 80 B] 86 C] 73 D] None of these

16. The average age of a woman and her daughter is 46 years. The ratio of their ages is 15:8 respectively. What will be the respective ratio of their ages after 8 years?
A] 8 : 5 B] 10 : 17 C] 17 : 10 D] 5 : 8
17. There are 50 boys in a class. One boy weighing 40 kg goes away and at the same time another boy joins the class. If the average weight of the class is thus decreased by 100 g, find the weight of the new boy.
A] 35 kg B] 43 kg C] 36 kg D] 30 kg
18. Kamlesh bought 65 books for Rs.1050 from one shop and 50 books for Rs.1020 from another. What is the average price he paid per book?
A] Rs. 36.40 B] Rs.18.20 C] Rs. 24 D] Rs.18
19. A car covers the first 39 kms of its journey in 45 minutes and covers the remaining 25 kms in 35 minutes. What is the average speed of the car?
A] 40 kms/hr B] 64 kms/hr C] 49 kms/hr D] 48 kms/hr
20. The average marks in Science subject of a class of 20 students is 68. If the marks of two students were misread as 48 and 65 of the actual marks 72 and 61 respectively, then what would be the correct average?
A] 68.5 B] 69 C] 69.5 D] 70

LEVEL – III

1. The average speed of a car is 75 kms/hr. What will be the average speed of the car if the driver decreases the average speed of the car by 40 percent?
A] 50 kms/hr B] 45 kms/hr C] 40 kms/hr D] 55 kms/hr
2. The average marks of a student in seven subjects are 41. After re-evaluation in one subject the marks were changed to 42 from 14 and in remaining subjects the marks remained unchanged. What are the new average marks?
A] 45 B] 44 C] 46 D] 47
3. The average marks of nine students in a group are 63. Three of them scored 78, 69 and 48 marks. What are the average marks of remaining six students?
A] 63.5 B] 64 C] 63 D] 62
4. Out of the three given numbers, the first number is twice the second and thrice the third. If average of the three numbers are 154. What is the difference between the first and the third number?
A] 126 B] 42 C] 52 D] 168
5. The average of four positive integers is 124.5. The highest integer is 251 and the lowest integer is 65. The difference between the remaining two integers is 26. Which of the following integers is higher of the remaining two integers?
A] 78 B] 102 C] 100 D] 104
6. In an examination, a student's average marks were 63. If he had obtained 20 more marks for his Geography and 2 more marks for his history, his average would have been 65. How many subjects were there in the examination?
A] 12 B] 11 C] 13 D] 14

7. The average age of a husband and his wife was 23 years at the time of their marriage. After five years they have a one year old child. What is the average age of the family?
A] 21 years B] 20 years C] 18 years D] 19 years
8. A painter is paid x rupees for painting every 10 metres of a wall and y rupees for painting every extra metre. During one week, he painted 10 metres on Monday, 13 metres on Tuesday, 12 metres on Wednesday, 11 metres on Thursday and 12 metres on Friday. What is his average daily earning in rupees for the five day week?
A] $x + (8/5)y$ B] $(5x + 9y)/5$ C] $10x + (8/5)y$ D] $5x + 8y$
9. In a class with a certain number of students, if one new student weighing 50 kg is added, then the average weight of the class increased by 1 kg. If one more student weighing 50 kg is added, then the average weight of the class increases by 1.5 kg over the original average. What is the original weight (in kg) of the class?
A] 46 B] 42 C] 27 D] 47
10. There was one mess for 30 boarders in a certain hostel. If the number of boarders was increased by 10, the expenses of the mess increased by Rs. 40 per month, while the average expenditure per head diminished by Rs. 2. Find the original monthly expenses.
A] Rs. 390 B] Rs. 360 C] Rs. 410 D] Rs. 480
11. There are three baskets of fruits. First basket has twice the number of fruits in the 2nd basket. Third basket has three-fourths of the fruits in the first. The average of the fruits in all brackets is 30. What is the number of fruits in the first basket?
A] 20 B] 30 C] 35 D] 40
12. The average weight of 3 men, X, Y and Z is 84 kg. Another man T joins the group and the average now becomes 80 kg. If another man S, whose weight is 3 kg more than that of T, replaces X then the average weight of Y, Z, T and S becomes 79 kg. The weight of X is:
A] 70 kg B] 72 kg C] 75 kg D] 80 kg
13. If the three numbers, the average of the first and the second is greater than the average of the second and the third by 15. What is the difference between the first and the third if the three numbers?
A] 15 B] 45 C] 60 D] None of these
14. Mr. Anant Roy, the renowned author, recently got his new novel released. To his utter dismay he found that for the 1,007 pages on an average there were 2 mistakes every page. While, in the first 612 pages there were only 434 mistakes, they seemed to increase for the latter pages. Find the average number of mistakes per page for the remaining pages.
A] 6 B] 4 C] 2 D] None of these
15. The average age of a couple is 25 years. The average age of the family just after the birth of the first child was 18 years. The average age of family just after the second child was born was 15 years. The average age of the family after the third and the fourth children (who are twins) were born was 12 years. If the present average age of the family of six persons is 16 years, how old is the eldest child?
A] 6 years B] 7 years C] 8 years D] 9 years

16. Eleven years earlier the average age of a family of 4 members was 28 years. Now the age of the same family with six members is yet the same, even when 2 children were born in this period. If they belong to the same parents and the age of the first child at the time of the birth of the younger child was same as there were total family members just after the birth of the youngest members of this family, then the present age of the youngest member of the family.
A] 3 years B] 5 years C] 6 years D] None of these
17. The average price of 3 diamonds of same weights is Rs. 5 crore, where the average price of the two costliest diamonds is double the price of the cheapest diamond. The price of the cheapest diamond is
A] 3 crore B] 5 crore C] 1.66 crore D] can't be determined
18. A set of consecutive positive integers beginning with 1 is written on the blackboard. A student came along and erased one number. The average of the remaining numbers is $35\frac{7}{17}$.
What was the number erased?
A] 7 B] 8 C] 9 D] None of these
19. In a class of 100 students the average marks obtained by a student in Maths in 44. If we remove the highest and the lowest the average of the class becomes 43.92. A total of three students get either highest or lowest. What is the average of these three students?
A] 73.33 B] 46.66 C] 59.99 D] cannot be determined
20. Neeta's attendance for first two semesters out of four was 60% and 70% respectively. What is the minimum attendance required in third semester so that her average attendance will be 80% throughout for semesters?
(Assume equal number of days among the four semesters)
A] 70 % B] 80% C] 90% D] None of these

PERCENTAGE

In mathematics, a Percentage is a number or ratio expressed as a fraction of 100. Basically percent means per hundred. It is often denoted using the percent sign, "%".

Eg: If a person scores 96 out of 100 marks, his marks percentage is 96 percent.

Note: If you score 60 marks Information technology does that mean you score 60%. The answer is NO; as you don't know the base. Percentage is basically a game of base. If you don't know the base than you can't calculate the percentage.

Conversion of a Fraction into Percentage- To convert a fraction into percentage, multiply the fraction by 100 and put % sign.

Ex. If fraction is $\frac{1}{4}$ then $\frac{1}{4} \times 100 = 25\%$

If fraction is $\frac{1}{8}$ then $\frac{1}{8} \times 100 = 12.5\%$

Conversion of a Percentage into Fraction- To convert a percentage into fraction, replace the % sign with $\frac{1}{100}$ and reduce the fraction into simplest form.

Ex. $10\% = \frac{10}{100} = \frac{1}{10}$

$75\% = \frac{75}{100} = \frac{3}{4}$

$300\% = \frac{300}{100} = 3$

Relation between Fraction and Percentage

Fraction	Percentage
$\frac{1}{2}$	50%
$\frac{1}{3}$	33.33%
$\frac{1}{4}$	25%
$\frac{1}{5}$	20%
$\frac{1}{6}$	$16.66\% = 16\frac{2}{3}\%$
$\frac{1}{7}$	$14.28\% = 14\frac{2}{7}\%$
$\frac{1}{8}$	$12.5\% = 12\frac{1}{2}\%$
$\frac{1}{9}$	$11.11\% = 11\frac{1}{9}\%$
$\frac{1}{10}$	10%
$\frac{1}{11}$	$9.09\% = 9\frac{1}{11}\%$

1/12	8.33 % = $8\frac{1}{3}\%$
1/13	7.69% = $7\frac{9}{13}\%$
1/14	7.14 % = $7\frac{1}{7}\%$
1/15	6.67 % = $6\frac{2}{3}\%$
1/16	6.25 % = $6\frac{1}{4}\%$
1/17	5.88 % = $5\frac{15}{17}\%$
1/18	5.55% = $5\frac{5}{9}\%$
1/19	5.26 % = $5\frac{5}{19}\%$
1/20	5%

Commodity Price Increase/Decrease

- If the price of a commodity increases by R%, then the reduction in consumption so as not to increase the expenditure is = $[(R / (100+R)) \times 100] \%$
- If the price of the commodity decreases by R%, then to maintain the same expenditure by increasing the consumption is = $[(R / (100-R)) \times 100] \%$
- If A is R% more than B, then B is less than A by = $[(R / (100+R)) \times 100] \%$
- If A is R% less than B, then B is more than A by = $[(R / (100-R)) \times 100] \%$

Example 1: If A is 20% more than B, by what percent is B less than A?

Solution: Let B=100 and A = 120

Using formula, $[20/120] \times 100 = 100/6 = 16.66\%$

Example 2: If the price of a commodity be raised by 20% then by how much % a house holder reduce his consumption so that the expenditure does not change?

Solution: Expenditure = rate x consumption

Here the expenditure remains constant in both the cases

Initially $1 \times 1 = 1$

After change $1.2 \times a = 1$

This means $1.2a = 1$ and $a = .833$ and hence decrease will be 16.66%

Example 3: Two numbers are 25% and 40% less than the third number. What % is the second of the first?

Solution: let three numbers be A, B and C. If C=100 it means B=60 and A=75

Hence B is $60/75 \times 100$ of A = 80%

Percentage Change

Percentage increase/decrease in a quantity = $(\text{change in quantity} / \text{original quantity}) \times 100 \%$

Example 4: Salary of Raja in 2001 was Rs 1000 per day and his salary in 2002 was Rs 1250 per day. Again in 2003 his salary was Rs 1000 per day

- What is the % increase in salary in 2002?
- What is the % decrease in salary in 2003 over 2002?

Solution: In a) part the increase is $125 - 100 = 25$ hence % increase will be $25 / 100 \times 100 = 25\%$
 In b) part the decrease is $125 - 100 = 25$ hence % decrease will be $25 / 125 \times 100 = 20\%$

Results on Population Increase/Decrease

- If the original population is P and increase in population is at the rate of r % every year then, the population after n years will be $= P (1+r/100)^n$
- Similarly, If the original population is P and decrease in population is at the rate of r % every year then the population after n years will be $= P (1-r/100)^n$
- Let the population of the town be P now and suppose it increases at the rate of R% per annum, then = Population n years ago $= P / [1+(R / 100)]^n$

Results on Depreciation

Let the present value of a machine be P. Suppose it depreciates at the rate R% per annum. Then,

- Value of the machine after n years $= P [1-(R/100)]^n$
- Value of the machine n years ago $= P / [1-(R/100)]^n$

Example 5: If the present population of a town is 10000 and annual increase is 20%. Then what will be the population after 3 years?

Solution: Population after three years $= 10000(1+20/100)^3 = 10000(1.2)^3 = 17280$

Successive Percent changes

- If A is increase by x% in the first year and increase y % in the second year, then the net percentage change $= +x+y+xy/100$.
Where; + stands for increase and – stands for decrease
- Generalize, if a quantity increases by p% and then decreases by p% then there is net reduction of $(p^2)/100$ %.

Example 6: In an examination it is required to get 65% of the aggregate marks to pass. A student gets 522 marks and is declared failed by 7% marks. What are the maximum aggregate marks a student can get?

Solution: Pass marks of the examination = 65%
 Student failed by 7%, so marks secured by student $= 65\% - 7\% = 58\%$
 $58\% = 522$; $100\% = 900$

Example 7: Ajay spends 25 per cent of his salary on house rent, 5 per cent on food, 15 per cent on travel, 10 per cent on clothes and the remaining amount of ` 27,000 is saved. What is Ajay's income?

Solution: Ajay's total income be 100%
 His total expenditure $= 25\% + 5\% + 15\% + 10\% = 55\%$
 Savings $= 100\% - 55\% = 45\%$
 $45\% = 27,000$
 Therefore, $100\% = 27000 * 100/45 = 60,000$

Example 8: When the price of eggs is reduced by 20%, it enables a man to buy 20 more eggs for 40. What is the reduced price per egg?

Solution: Saving due to reduction $= 20/100 \times 40 = 8$ So the sum of Rs 8 enables the man to purchase 20 more eggs at the reduced price (R.P); Reduced price per egg $= 8/20 = 0.4 = 40$ paisa

Example 9: Two successive discounts of 10% and 20% are equal to a single discount of?

Solution: Discount is same as decrease of price. So, decrease $= 0.9 \times 0.8 = 0.72 \Rightarrow 28\%$ decrease (Since only 72% is remaining)

Example 10: A fruit seller had some apples. He sells 40% apples and still has 420 apples. Originally, he had?

Solution: Suppose originally he had x apples.
Then, $(100 - 40)\%$ of $x = 420$
 $60/100 \times x = 420$
 $x = 420 \times 100/60 = 700$.

Example 11: In a group of students, 70% can speak English and 65% can speak Hindi. If 27% of the students can speak none of the two languages, then what per cent of the group can speak both the languages?

Solution: 27% students speak neither of the languages.
Number of students speaking either of the languages $= 100\% - 27\% = 73\%$
 $n(E \cup H) = 73\%$ $n(E) = 70\%$ $n(H) = 65\%$ $n(E \cap H) = ?$
 $n(E \cup H) = n(E) + n(H) - n(E \cap H)$
 $73 = 70 + 65 - n(E \cap H)$
 $n(E \cap H) = 135 - 73 = 62$

LEVEL – I

1. What is 30 % of 80?
A] 30 B] 24 C] 27 D] 40
2. In a class of 50 students, 40 % are girls. Find the number of girls and number of boys in the class?
A] 30, 20 B] 20, 30 C] 25, 25 D] 40, 10
3. Ron scored 344 marks out of 400 marks and his elder brother Ben scored 582 marks out of 600 marks. Who scored percentage is better?
A] Ben B] Ram C] both same D] None of these
4. Victor gets 92 % marks in examinations. If these are 460 marks, find the maximum marks.
A] 480 B] 600 C] 500 D] 700
5. The price of rice is increased from \$10 to \$12.50 per kg. Find the percentage increase in price?
A] 30% B] 20% C] 25% D] 40%
6. The population in a small town increases from 20000 to 21250 in one year. Find the percentage increase in population?
A] 10% B] 15% C] 12% D] 6.25%
7. By what number must the given number be multiplied to increase the number by 50 %?
A] $\frac{3}{2}$ B] $\frac{1}{2}$ C] $\frac{1}{4}$ D] $\frac{2}{3}$
8. A television manufacturing company declares that a television is now available for \$5600 as against \$8400 one year before. Find the percentage reduction in the price of television offered by the company?
A] 33.33% B] 20% C] 25% D] 66.67%
9. Find the number which when decreased by 12 % becomes 198?
A] 300 B] 210 C] 250 D] 225
10. Aaron had \$ 2100 left after spending 30 % of the money he took for shopping. How much money did he take along with him?
A] 3000 B] 2100 C] 2500 D] 2200
11. A shopkeeper bought 600 oranges and 400 bananas. He found 15% of oranges and 8% of bananas were rotten. Find the percentage of fruits in good condition.
A] 90% B] 80% C] 87.8% D] 66.67%
12. Mike needs 30% to pass. If he scored 212 marks and falls short by 13 marks, what was the maximum marks he could have got?
A] 700 B] 750 C] 850 D] 725

13. A number is increased by 40 % and then decreased by 40 %. Find the net increase or decrease percent?
A] 10% B] 15% C] 16% D] 0%
14. Max scored 6 marks more than what he did in the previous examination in which he scored 30. Maria scored 30 marks more than she did in the previous examination in which she scored 60. Who showed less improvement?
A] Max B] Maria C] Both D] None
15. An alloy contains 26 % of copper. What quantity of alloy is required to get 260 g of copper?
A] 800 B] 750 C] 900 D] 1000
16. In a basket of apples, 12% of them are rotten and 66 are in good condition. Find the total number of apples in the basket.
A] 80 B] 75 C] 90 D] 100
17. An increase of 30% in the price of oranges enables a man to buy 6 kg less for Rs. 300. Find the increased price per kg.
A] 20 B] 15 C] 30 D] 10
18. Conclusion shows that an angle is $37\frac{1}{2}$ %. The size used by drawing and measurement is 36%. The error percentage scenario is?
A] 4 B] 1 C] 8 D] 9
19. A digit has increased $37\frac{1}{2}$ % gives 33. The actual digit is?
A] 35 B] 24 C] 15 D] 40
20. A particular digit is subtracting 15 from it, reduces to its 80%. What is 40% of that digit?
A] 30 B] 03 C] 35 D] 20

LEVEL - II

1. In an election, candidate A got 75% of the total valid votes. If 15% of the total votes were declared invalid and the total numbers of votes is 560000, find the number of valid vote polled in favour of candidate?
A] 375000 B] 357000 C] 325000 D] 350000
2. In an examination, 300 students appeared. Out of these students; 28 % got first division, 54 % got second division and the remaining just passed. Assuming that no student failed; find the number of students who just passed?
A] 80 B] 74 C] 60 D] 54
3. In an election, candidate A got 70 % of the total valid votes. 20 % of the total votes were declared invalid. If the total number of votes is 600000, find the number of valid votes polled in favour of the candidate.
A] 350000 B] 357000 C] 336000 D] 340000

4. In a college, among all students 48% comes by bus, 62% takes canteen lunch and 40% comes by bus and takes canteen lunch. Find the percentage of students neither having canteen lunch nor coming by bus.
A] 30% B] 42% C] 51% D] 32%
5. The difference between 78% of a number and 59% of the same number is 323. What is 62% of that number?
A] 1071 B] 1173 C] 1037 D] 1054
6. 40% of 60% of $\frac{3}{5}$ th of a number is 504. What is 25% of $\frac{2}{5}$ th of that number?
A] 180 B] 175 C] 360 D] 350
7. Tanya obtained 95% marks in Physics, Chemistry and Maths together and she obtained 225 marks in Social Science, English and Hindi together. If the maximum marks one can obtain in each of the subjects is 100, what is Tanya's overall percentage in all the six subjects together?
A] 80 B] 85 C] 90 D] 95
8. In a test consisting of 80 questions carrying one mark each, Arpita answers 65% of the first 40 questions correctly. What percent of the other 40 questions does she need to answer correctly to score 75% on the entire test?
A] 60 B] 80 C] 75 D] None of these
9. A sum of Rs.2236 is divided among A, B and C such that A receives 25% more than C and C receives 25% less than B, what is A's share in the amount?
A] Rs. 460 B] Rs. 890 C] Rs. 780 D] Rs. 900
10. Vaishali spent Rs.31897 on air-conditioner for her home, Rs.38789 on buying plasma tv and the remaining 23% of the total amount she had as cash with her. What was the total amount?
A] Rs.74625 B] Rs.86750 C] Rs.91800 D] None of these
11. In an examination it is required to get 40% of the aggregate marks to pass. A student gets 265 marks and is declared fail by 55 marks. What is the maximum aggregate marks a student can get?
A] 800 B] 750 C] 650 D] None of these
12. Ramu gave 35% of the amount he had to Shakeel. Shakeel in turn gave two-thirds of what he received from Ramu to Joseph. After spending Rs.150 on lunch out of the amount he got from Shakeel, Joseph now has Rs.550 left with him. How much amount did Ramu have?
A] Rs.2800 B] Rs.2000 C] Rs.3000 D] None of these
13. If the numerator of a fraction is increased by 200% and the denominator is increased by 300%, the resultant fraction is $\frac{15}{26}$. What was the original fraction?
A] $\frac{8}{11}$ B] $\frac{10}{11}$ C] $\frac{9}{13}$ D] $\frac{10}{13}$
14. A cricketer has batted in 105 innings. He has scored 17 centuries, crossed the 50- run mark 28 times and crossed the 25-run mark 39 times. The no. of times he has scored below 25 is what percentage of the number of times he scored between 25 and 49 (both 25 and 49 included)?
A] 450% B] 600% C] 500% D] None of these

15. Sunil scored 54 percent marks in a test. Ravi scored 450 marks in the same test which is 300 less than Sonu. Sunil's score is 60 more than Sonu. If Ram scored 900 marks in the test, what is Ram's percentage?
A] 80 B] 65 C] 75 D] 60
16. In a school there are 800 students out of whom 45 percent are girls. Monthly fee of each boy is Rs.600 and monthly fee of each girl is 30 percent less than each boy. What is the total monthly fee of girls and boys together?
A] Rs.425400 B] Rs.414600 C] Rs.419600 D] None of these
17. In a class of 50 students and 5 teachers, each student got sweets that are 12% of the total number of students and each teacher got sweets that are 20% of the total number of students. How many sweets were there?
A] 345 B] 365 C] 350 D] 330
18. In a school there are 250 students out of whom 12 per cent are girls. Each girl's monthly fee is Rs.450/- and each boy's monthly fee is 24 percent more than a girl. What is the total monthly fee of girls and boys together?
A] Rs.136620/- B] Rs.136260/- C] Rs.132660/- D] Rs.132460/-
19. If the height of a triangle is decreased by 40% and its base is increased by 40%, what will be the effect on its area?
A] No change B] 16% decrease C] 8% decrease D] 16% increase
20. An HR Company employs 4800 people, out of which 45 per cent are males and 60 per cent of the males are either 25 years or older. How many males are employed in HR Company who is younger than 25 years?
A] 864 B] 2160 C] 1296 D] 2640

LEVEL – III

1. Six-eleventh of a number is equal to twenty two per cent of second number. Second number is equal to the one-fourth of third number. The value of the third number is 2400. What is the 45% of first number?
A] 108.9 B] 111.7 C] 117.6 D] 123.4
2. Raman scored 456 marks in an exam and Seeta got 54 percent marks in the same exam which is 24 marks less than Raman. If the minimum passing marks in the exam is 34 per cent, then how much more marks did Raman score than the minimum passing marks?
A] 184 B] 196 C] 190 D] 180
3. The length, breadth and height of a room in the shape of a cuboid are increased by 10%, 20% and 50% respectively. Find the percentage change in the volume of the cuboid.
A] 77% B] 75% C] 88% D] 98%

4. The price of the sugar is reduced by 25% but inspite of the decrease, Aayush ends up increasing his expenditure on sugar by 20%. What is the percentage change in his monthly consumption of sugar?
A] +60% B] -10% C] +33.33% D] -50%
5. The population of the village of Gavas is 10,000 at this moment. It increases by 10% in the first year. However, in the second year, due to immigration, the population drops by 5%. Find the population at the end of the third year if in third year the population increase by 20%
A] 12,340 B] 12,540 C] 1, 27, 540 D] 12, 340
6. Ram spends 20% of his monthly income on his household expenditure, 15% of the rest on books, 30% of the rest on clothes and saves the rest. On counting, he comes to know that he has finally saved Rs. 9520. Find his monthly income.
A] 10000 B] 15000 C] 20000 D] 12000
7. The population of a village is 5500. If the number of males increases by 11% and the number of females increases by 20%, then the population becomes 6330. Find the population of females in the town.
A] 2500 B] 3000 C] 2000 D] 3500
8. Last, year the Indian Cricket team played 40 one-day cricket matches out of which they managed to win only 40%. This year, so far it has played some matches, which made it mandatory for it to win 80% of the remaining matches to maintain its exiting winning percentage. Find the number of matches played by India so far this year.
A] 30 B] 25 C] 28 D] Data Insufficient
9. In the recent, climate conference in New York, out of 700 men, 500 women, 800 children present inside the building premises, 20% of the men, 40% of the women and 10% of the children were Indians. Find the percentage of people who were not Indian.
A] 73% B] 77% C] 79% D] 83%
10. In an examination, 48% students failed in Hindi and 32% students in History, 20% students failed in both the subjects. If the number of students who passed the examination was 880, how many students appeared in the examination if the examination consisted only of these two subjects?
A] 2000 B] 2200 C] 2500 D] 1800
11. A machine depreciates in value year at the rate of 10% of its previous value. However, every second year there is some maintenance work so that in that particular year, depreciation is only 5% of its previous value. If at the end of the fourth year, the value of the machine stands at Rs. 1,46,205; then find the value of machine at the start of the first year.
A] Rs. 1, 90, 000 B] Rs. 2, 00, 000 C] Rs. 1, 95, 000 D] 2,10,000
12. After three successive equal percentage rise in the salary the sum of 100 rupees turned into 140 rupees and 49 paisa. Find the percentage rise in the salary.
A] 12% B] 22% C] 66% D] 82%

13. King Dashratha, at his eleventh hour, called his three queens and distributed his gold in the following way: HE gave 50% of his wealth to his first wife, 50% of the rest to his second wife and again 50% of the rest to his third wife. If their combined share is worth 1,30,9000 kilograms of gold, find the quantity of gold King Dashratha was having initially?
A] 1, 50, 000 kg B] 1,49,600 kg C] 1,51,600 kg D] 1,52,600 kg
14. The population of New Foundland increases with a uniform rate of 80% per annum, but due to immigration, there is a further increase of population by 1% (however, this 1% increase in population is to be calculated on the population after the 8% increase and not on the previous year's population).
A] 18.984 B] 18.081 D] 18.24 D] 17.91
15. A's salary is first increased by 25% and then decreased by 20%. The result is the same as B's salary increased by 20% and then reduced by 25%. Find the ratio of B's salary to that of A's.
A] 4 : 3 B] 11 : 10 C] 10 : 9 D] 12 : 13
16. In order to maximize his gain, a theatre owner decides to reduce the price of tickets by 20% and as a result of this, the sales of tickets increase by 40%. If, as a result of these changes, he is able to increase his weekly collection by Rs. 1,68,000. Find by what value did the gross collection increase per day.
A] 14,000 B] 18,000 C] 24,000 D] 20,000
17. In a town consisting of three localities A, B and C, the population of the three localities A, B and C are in the ratio 9 : 8 : 3. In locality A, 80% of the people are literate, in locality B, 30% of the people are illiterate. If 90% people in locality C are literate, find the percentage literacy in that town.
A] 61.5% B] 78 % C] 75 % D] None of these
18. The population of a town increase by 5% annually frails present population is 64000. What will it be in 3 years' time?
A] 72044 B] 74088 C] 75042 D] None of these
19. Let A and B be two solid spheres such that the surface area of B is 300% higher than the surface area of A. the volume of A is found to be k% lower than the volume of B. the value of k must be?
A] 85.5 B] 92.5 C] 90.5 D] 87.5
20. In a factory there are three types of Machines M1, M2 and M3 which produces 25%, 35% and 40% of the total products respectively. M1, M2 and M3 produce 2%, 4% and 5% defective products, respectively. What is the percentage of non-defective products?
A] 89% B] 97.1% C] 96.1% D] 86.1%

PROFIT AND LOSS

Cost Price: The price (amount) paid to purchase a product or the cost incurred in manufacturing a product is known as the cost price (CP) of that product.

Selling Price: The price at which a product is sold is called the selling price (SP) of the product.

List Price: List price or the tag price is the price that is printed on the tag of the article. For all practical purposes, we assume it to be same as the marked-price.

Margin: The profit percentage on selling price is known as MARGIN.

Profit

If the Selling Price exceeds the Cost Price, then there is Profit.

$$\text{Profit or gain} = \text{SP} - \text{CP}$$

$$\text{Profit \%} = \text{Profit} / (\text{CP}) \times 100$$

$$\text{S P} = (100 + \text{gain \%}) / 100 \times \text{C P}$$

$$\text{C P} = 100 / (100 + \text{gain \%}) \times \text{S P}$$

Loss

If the overall Cost Price exceeds the selling price of the buyer then he is said to have incurred loss.

$$\text{Loss} = \text{C P} - \text{S P}$$

$$\text{Loss \%} = \text{Loss} / (\text{CP}) \times 100$$

$$\text{S P} = (100 - \text{loss \%}) / 100 \times \text{CP}$$

$$\text{C P} = 100 / (100 - \text{loss \%}) \times \text{SP}$$

Mark-up Price

Generally the SP is less than the marked price (MP) the difference MP – SP is known as discount, D.

$$\text{Discount} = \text{M P} - \text{S P}$$

$$\text{Discount \%}, \text{D\%} = (\text{Discount}) / (\text{MP}) \times 100$$

$$\text{Mark up percentage} = \frac{\text{MP} - \text{CP}}{\text{CP}} \times 100$$

Successive Discount

Sometimes more than one discount is offered by the shopkeeper on a single item or article. When two or more discounts are applicable successively to the list price of an article, they form the discount series.

When there are two successive Profit of x % and y % then the resultant profit percent is given by

$$[x + y + \frac{xy}{100}]$$

If there is a Profit of x% and loss of y % in a transaction, then the resultant profit or loss% is given by

$$[x - y - \frac{xy}{100}]$$

Note: For profit use sign + in previous formula and for loss use – sign.

If resultant came + then there will be overall profit, if it came – then there will be overall loss.

False Weight Problems

Shown or indicate weight is always equivalent to selling price, and actual/true weight is equivalent to cost price.

If a trader professes to sell his goods at cost price, but uses false weights, then

$$\text{Gain}\% = \left(\frac{\text{Error}}{\text{True value} - \text{Error}} \right) \times 100\%$$

If a cost price of m articles is equal to the selling Price of n articles, then Profit percentage

$$\frac{m - n}{n} \times 100\%$$

Solved Examples

Example 1: Marked price of a dining table is Rs 1350. It is sold at Rs. 1188 after allowing certain discount. Find the rate of discount.

Solution: MP of the dining table = Rs. 1350
 SP of the dining table = Rs. 1188
 Discount allowed = Rs. (1350 - 1188) = Rs. 162
 Discount percent = $162/1350 \times 100 = 12$
 This the rate of discount is 12%

Example 2: If two articles are sold at same selling price one at 30% profit another at 30% loss then what is his overall percentage profit or loss?

Solution: Overall loss = $-x^2/100\%$
 $= -900/100 = -9\%$ loss

Example 3: A shopkeeper takes 20%, extra quantity while purchasing the milk, and gives 25% less than the indicated weight while selling the milk. Find the profit percentage of he sells at the cost price only.

Solution: Suppose the price of milk = 1 Rs per ml shopkeeper takes 120 ml, and pays only Rs. 100
 While selling he gives only 75 ml and shows 100 ml.
 Total selling price of 120 ml
 $100/75 \times 120 = 160$, hence percentage profit = 60%

Example 4: A sells an item at a profit of 20% to B and B sells it to C at a profit of 10%. Find the resultant profit percent?

Solution: When there are two successive profit of x% and y% , net profit percentage
 $= \{x+y+(xy/100)\}$
 $= 20+ 10 + \{(20 \times 10)/100\} = 32\%$

Example 5: The cost price of 25 articles is equal to selling price of 20 articles. The gain percent is?

Solution: Profit percentage = $(x-y/y) * 100\%$
 $\% \text{Gain} = \{(25-20)/20\} \times 100 = 5/20 \times 100 = 25\%$

Example 6: A man sold an article at a loss of 20%. If he has sold that article for Rs. 12 more he would have gained 10%. Find the cost price of that article

Solution: Let the CP be x
 SP at 20% loss = Rs 0.8x

$$\begin{aligned}
0.8x + 12 &= 1.1x \\
\Rightarrow 0.3x &= 12 \\
\Rightarrow x &= \text{Rs } 40
\end{aligned}$$

Example 7: In a transaction, the profit percentage is 80% of the cost. If the cost further increases by 20% but the selling price remain the same, how much is the decrease in profit percentage?

Solution: Let us assume CP = Rs. 100.
Then Profit = Rs. 80 and selling price = Rs. 180.
The cost increases by 20% $\rightarrow$ New CP = Rs. 120, SP = Rs. 180.
Profit % = $60/120 \times 100 = 50\%$.
Therefore, Profit decreases by 30%.

Example 8: A man bought some toys at the rate of 10 for Rs. 40 and sold them at 8 for Rs. 35. Find his gain or loss percent.

Solution: Cost price of 10 toys = Rs. 40 $\rightarrow$ CP of 1 toy = Rs. 4.
Selling price of 8 toys = Rs. 35 $\rightarrow$ SP of 1 toy = Rs. $35/8$
Therefore, Gain = $35/8 - 4 = 3/8$.
Gain percent = $(3/8)/4 \times 100 = 9.375\%$

Example 9: A shopkeeper allows a discount of 10% on the marked price and still gains 17% on the whole. Find at what percent above the cost price he marked his goods.

Solution: Let the cost price be 100. Then SP = 117.
Let the marked price be x.
So, 90% of x = 117 $\rightarrow x = 130$.
Therefore, he marked his goods 30% above the cost price.

Example 10: A shopkeeper offers a discount of 20% on the selling price. On a special sale day, he offers an extra 25% off coupon after the first discount. If the article was sold for Rs. 3600, find

- I. The marked price of the article and
- II. The cost price if the shopkeeper still makes a profit of 80% on the whole after all discounts are applied.

Solution: Let the marked price of the article be x.
First a 20% discount was offered, on which another 25% discount was offered.
So, 75% of 80% of x = 3600
 $75/100 \times 80/100 \times x = 3600 \rightarrow x = 6000$.
So the article was marked at Rs. 6000.
Cost price of the article = $[100 / (100+80)] \times 3600 = \text{Rs. } 2000$.

LEVEL – I

1. Alfred buys an old scooter for Rs. 4700 and spends Rs. 800 on its repairs. If he sells the scooter for Rs. 5800, his gain percent is?
A. $4\frac{4}{7}\%$ B. $5\frac{5}{11}\%$ C. 10% D. 12%
2. The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then the value of x is?
A. 15 B. 16 C. 18 D. 25
3. If selling price is doubled, the profit triples. Find the profit percent?
A. $200/3$ B. $105/3$ C. 100 D. 120
4. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?
A. 30% B. 70% C. 100% D. 250%
5. A vendor bought toffees at 6 for a rupee. How many for a rupee must he sell to gain 20%?
A. 3 B. 4 C. 5 D. 6
6. The percentage profit earned by selling an article for Rs. 1920 is equal to the percentage loss incurred by selling the same article for Rs. 1280. At what price should the article be sold to make 25% profit?
A. Rs. 2000 B. Rs. 2200 C. Rs. 2400 D. Data inadequate
7. A shopkeeper expects a gain of 22.5% on his cost price. If in a week, his sale was of Rs. 392, what was his profit?
A. Rs. 18.20 B. Rs. 70 C. Rs. 72 D. Rs. 88.25
8. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the cycle?
A. Rs. 1090 B. Rs. 1160 C. Rs. 1190 D. Rs. 1202
9. Sam purchased 20 dozens of toys at the rate of Rs. 375 per dozen. He sold each one of them at the rate of Rs. 33. What was his percentage profit?
A. 3.5 B. 4.5 C. 5.6 D. 6.5
10. Some articles were bought at 6 articles for Rs. 5 and sold at 5 articles for Rs. 6. Gain percent is:
A. 30% B. $33\frac{1}{3}\%$ C. 35% D. 44%
11. A shopkeeper sells one transistor for Rs. 840 at a gain of 20% and another for Rs. 960 at a loss of 4%. His total gain or loss percent is?
A. $5\frac{15}{17}\%$ gain B. $20/3$ C. $5\frac{15}{23}$ loss D. None of these
12. A trader mixes 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sells the mixture at Rs. 30 per kg. His profit percent is:
A. No profit, no loss B. 15% C. 8% D. 5%

13. When a plot is sold for Rs. 18,700, the owner loses 15%. At what price must that plot be sold in order to gain 15%?
 A. Rs. 21,000 B. Rs. 22,500 C. Rs. 25,300 D. Rs. 25,800
14. 100 oranges are bought at the rate of Rs. 350 and sold at the rate of Rs. 48 per dozen. The percentage of profit or loss is:
 A. $14\frac{2}{7}\%$ gain B. 15% gain C. $14\frac{2}{7}\%$ loss D. 15 % loss
15. On selling 17 balls at Rs. 720, there is a loss equal to the cost price of 5 balls. The cost price of a ball is?
 A. Rs. 45 B. Rs. 50 C. Rs. 55 D. Rs. 60
16. By selling 45 lemons for Rs 40, a man loses 20%. How many should he sell for Rs 24 to gain 20% in the transaction?
 A. 16 B. 18 C. 20 D. 22
17. A shopkeeper cheats to the extent of 10% while buying and selling, by using false weights. His total gain is.
 A. 20% B. 21% C. 22% D. 23%
18. If the cost price of 12 pens is equal to the selling price of 8 pens, the gain percent is?
 A. 12% B. 30% C. 50% D. 60%
19. The cost price of 24 articles is the same as the selling price of x articles. If the profit is 20%, then the value of x is?
 A. 15 B. 20 C. 18 D. 25
20. If books bought at prices ranging from Rs. 200 to Rs. 350 are sold at prices ranging from Rs. 300 to Rs. 425, what is the greatest possible profit that might be made in selling eight books?
 A. 600 B. 1200 C. 1800 D. none of these

LEVEL – II

1. How much percent more than the cost price should a shopkeeper mark his goods so that after allowing a discount of 25% on the marked price, he gains 20%?
 A. 60% B. 55% C. 70% D. 50%
2. A dishonest dealer professes to sell his goods at the cost price but uses a false weight of 850 g instead of 1 kg. His gain percent is?
 A. $71\frac{11}{17}\%$ B. $11\frac{11}{17}\%$ C. $17\frac{12}{17}\%$ D. $17\frac{11}{17}\%$
3. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?
 A. 30% B. 70% C. 100% D. 250%

4. The percentage profit earned by selling an item for Rs. 832 is equal to the percentage loss incurred by selling the same item for Rs. 448. At what price should the item be sold to make 50% profit?
A. Rs.920 B. Rs. 960 C. Rs. 1060 D. Rs. 1200
5. 100 oranges are bought at the rate of Rs. 350 and sold at the rate of Rs. 48 per dozen. The percentage of profit or loss is:
A. 14 $\frac{2}{7}$ % gain B. 15% gain C. 14 $\frac{2}{7}$ % loss D. 15 % loss
6. A trader mixes 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sells the mixture at Rs. 30 per kg. What is his profit percentage?
A. 6% B. 5% C. 4% D. 7%
7. A trader gives 12% additional discount on the discounted price, after giving an initial discount of 20% on the labelled price of an item. The final sale price of the item is Rs.704. Find out the labelled price?
A. 1000 B. 2000 C. 1200 D. 920
8. A milkman buys some milk contained in 10 vessels of equal size. If he sells his milk at Rs 5 a litre, he loses Rs 200; while selling it at Rs6 a litre, he would gain Rs150 on the whole. Find the number of litres contained in each vessel.
A. 30 B. 35 C. 40 D. 45
9. A watch passes through three hands and each gain 25%. If the third sells it for Rs250, what did the first pay for it?
A. 128 B. 130 C. 145 D. 150
10. If by selling an article for Rs 60, a person loses $\frac{1}{7}$ of outlay (cost), what would he have gained or lost per cent by selling it for Rs77?
A. 5% B. 10% C. 11% D. 15%
11. I sold a book at a profit of 7%. Had I sold it for Rs 7.50 more, 22% would have been gained. Find the cost price?
A. 25 B. 30 C. 50 D. 55
12. A reduction of 40 per cent in the price of bananas would enable a man to obtain 64 more for Rs40. What is the reduced price per dozen?
A. 3 B. 6 C. 5 D. 4
13. A man purchased an article at $\frac{3}{4}$ th of the list price and sold at half more than the list price. What was his gain per cent?
A. 25% B. 50% C. 75% D. 100%
14. I lose 9 per cent selling pencils at the rate of 15 a rupee. How many for a rupee must I sell them to gain 5 per cent?
A. 10 B. 13 C. 15 D. 18
15. Goods are sold so that when 4 per cent is taken off the list price, a profit of 20% is made. How much per cent is the list price more than the cost price?
A. 25% B. 50% C. 75% D. 100%

16. A man sells an article at 5% profit. If he had bought it at 5% less and sold it for Re 1 less, he would have gained 10%. Find the cost price.
A. 100 B. 150 C. 200 D. 250
17. A profit of 20% is made on goods when a discount of 10% is given on the marked price. What profit per cent will be made when a discount of 20% is given on the marked price?
A. $6\frac{2}{3}\%$ B. $7\frac{2}{3}\%$ C. $3\frac{1}{4}\%$ D. $7\frac{4}{5}\%$
18. A dealer sells a table for 400, making a profit of 25%. He sells another table at a loss of 10%, and on the whole he makes neither profit nor loss. What did the second table cost him?
A. 700 B. 800 C. 900 D. 950
19. Each of the two horses is sold for Rs720. The first one is sold at 25% profit and the other one at 25% loss. What is the % loss or gain in this deal?
A. 7.25% B. 6.25% C. 8.5% D. 9.25%
20. Each of the two cars is sold at the same price. A profit of 10% is made on the first and a loss of 7% is made on the second. What is the combined loss or gain?
A. $150/203\%$ B. $160/203\%$ C. $180/203\%$ D. $170/203\%$

LEVEL – III

1. A Camera shop allows a discount of 10% on the advertised price of a camera. What price must be marked on the camera that costs him Rs. 600, so that he makes a profit of 20%?
A. Rs. 800 B. Rs. 720 C. Rs. 750 D. Rs. 850
2. In the land of the famous milkman Merghese Durian, a milkman sells his buffalo for Rs. 720 at some profit. Had he sold his buffalo at Rs. 510, the quantum of the loss incurred would have been double that of the profit earned. What is the cost price?
A. Rs. 600 B. 625 C. 675 D. None of these
3. A trader purchases apples at Rs. 60 per hundred. He spends 15% on the transportation. What should be the selling price per 100 to earn a profit of 20%?
A. Rs. 72 B. Rs. 81.8 C. Rs. 82.8 D. Rs. 83.8
4. A tradesman fixed his selling price of goods at 30% above the cost price. He sells half the stock at this price, one-quarter of his stock at a discount of 15% on the original selling price and rest at a discount of 30% on the original selling price. Find the gain percent altogether.
A. 14.875% B. 15.375% C. 15.575% D. 16.375%
5. The percentage profit earned by selling an article for Rs. 900 is double the loss incurred by selling the same article for Rs. 450. At what price should the article be sold to make 25% profit?
A. Rs.600 B. Rs75 C. Rs.800 D. Data Inadequate

6. Tarun got 30% concession on the labelled price of an article and sold it for Rs. 8750 with 25% profit on the price he bought. What was the labelled price?
A. 10000 B. 12000 C. 13000 D. 14000
7. The spring balance of a trader shows 1 kg for an actual weight of 750 gm. Find the profit or loss percentage if the trader marks the price at 10% above the cost price.
A. 46.67% profit B. 46.67% loss C. 43.33% profit D. 43.33% loss
8. A milkman makes a profit of 10% on the sale of milk. If we were to add 10% water to milk, what is the percentage increase in profit %, assuming water is free of cost?
A. 21% B. 10% C. 11% D. 110%
9. A shopkeeper marks articles at 30% above than the cost price. He offers 20% discount for sale on cash and 10% discount for sale on credit. If he sells 70% of the articles on cash and remaining on credit then what is his profit percent over the total business?
A. 4.2% B. 8.4% C. 7.9% D. 7.6%
10. Some lollipops are bought at 11 for a rupee and the same number at 9 for a rupee. If the whole is sold at 10 for a rupee, find the gain or loss percent.
A. 1% gain B. No profit no loss C. 2% loss D. 1% loss
11. A table is offered for Rs300 with 20% and 10% off. If in addition, a discount of 5% is offered on cash payment, then the cash price of the table is:
A. Rs240 B. Rs216 C. Rs210 D. Rs205.20
12. The difference between a discount of 40% on Rs500 and two successive discounts of 36% and 4% on the same amount is:
A. Nil B. Rs2 C. Rs7.20 D. Rs1.93
13. A dealer marks his goods 20% above C.P. he then allows some discount on it and makes a profit of 8%. The rate of discount is:
A. 4% B. 6% C. 10% D. 12%
14. Nandlal purchased 20 dozen notebooks at Rs. 48 per dozen. He sold 8 dozen at 10% profit and the remaining 12 dozen with 20% profit. What is his profit percentage in this transaction?
A. 15% B. 16% C. 7.68% D. 19.2%
15. An article when sold for Rs. 840 earns a profit which is double the amount of loss when the same article is sold for Rs. 600, what is the C.P. of the article?
A. Rs. 500 B. Rs. 680 C. Rs. 720 D. Data inadequate
16. A milkman has 20 liters of milk. If he mixes 5 liters of water, which is freely available in 20 liters of pure milk. If the cost of pure milk is Rs. 18 per litre, then the profit of the milkman, when he sells all the mixture at its cost price, is:
A. 20% B. 25% C. 33.33% D. 18%

17. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?

- A. 70% B. 80% C. 90% D. none

18. A man bought an article and sold it at a gain of 5 %. If he had bought it at 5% less and sold it for Re 1 less, he would have made a profit of 10%. The C.P. of the article was

- A. Rs 100 B. Rs 150 C. Rs 200 D. Rs 250

19. Ashok bought an article and spent Rs. 110 on its repairs. He then sold it to Bhushan at a profit of 20%. Bhushan sold it to Charan at a loss of 10%. Charan finally sold it for Rs. 1188 at a profit of 10%. How much did Ashok pay for the article.

- A. Rs. 890 B. Rs. 1000 C. Rs. 780 D. 840

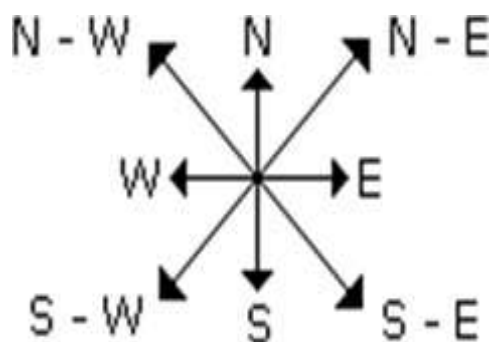
20. A merchant bought two transistors, which together cost him Rs. 480. He sold one of them at a loss of 15% and other at a gain of 19%. If the selling price of both the transistors are equal, find the cost of the lower priced transistor.

- A. Rs. 300 B. Rs. 180 C. Rs. 200 D. Rs. 280

DIRECTION SENSE TEST

The Concept behind the Directions is same that we use in our day to day life. This direction sense test is nothing but a precise of sensing the direction. To solve the direction sense test first you need to make a sketch of the data provided. Directions questions asked in the exam are based on two principles- Distance and Direction.

The first step for solving the questions using the concept of 'directions' is to understand the direction chart, which has 8 directions. Take a look at the direction chart given below:

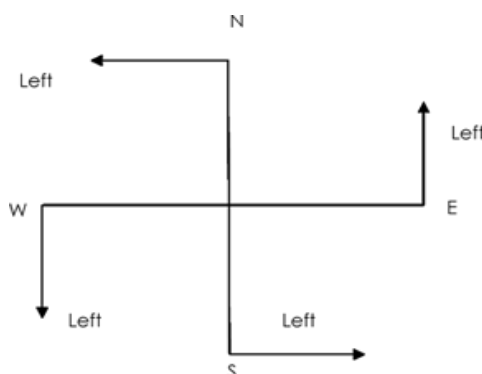


The first point you need to remember is that each main direction change undergoes a 90° change in direction e.g. from North to West/East it will be 90° change. But the change between North and North-east is only 45° .

Left Right Movement

A person facing north, on taking left will face towards west and on taking the right turn towards east. A person facing west, on taking left will face towards south and on taking right turn towards north.

When a question says moved towards left or right side, we assume that the movement is at an angle of 90 degrees.



Hence, we can sum up the above points as:

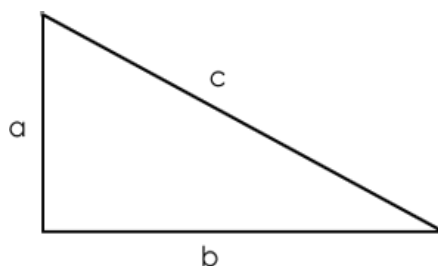
- Whenever a person moves to his left side, he will move towards anti- clockwise direction.
- Whenever a person moves to his right side, he will move towards clockwise direction.

Concept of Pythagoras Theorem

We'll use the concept of Pythagoras theorem to solve the questions on directions.

In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

$$c^2 = a^2 + b^2$$

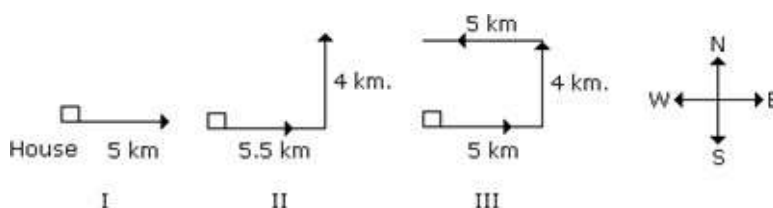


Points to Remember

- At the time of sunrise if a man stands facing the east, his shadow will be towards the west.
- At the time of sunset the shadow of an object is always in the east.
- If a man stands facing the North, at the time of sunrise his shadow will be towards his left and at the time of sunset it will be towards his right.
- At 12:00 noon, the rays of the sun are vertically downward hence there will be no shadow.

Example 1: A man starting from his home walks 5 km towards East, and then he turns left and goes 4 km. At last, he turn to his left and walks 5 km. Now find the distance between the man and his home and also find at which direction he is facing?

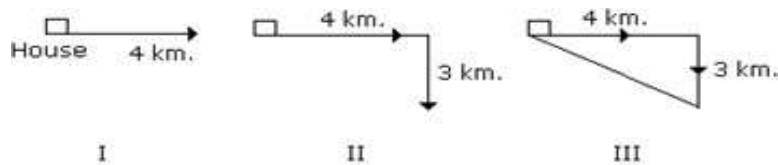
Solution:



From the above diagram we can find he is 4 km from his house and facing the West Direction.

Example 2: A man starting from his home moves 4 km towards East, then he turns right and moves 3 km. Now what will be the minimum distance covered by him to come back to his home?

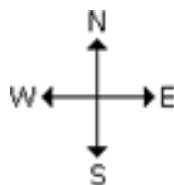
Solution:



$$\begin{aligned}
 \text{Minimum distance} &= \sqrt{(4)^2 + (3)^2} \\
 &= \sqrt{16 + 9} \\
 &= \sqrt{25} \\
 &= 5 \text{ km.}
 \end{aligned}$$

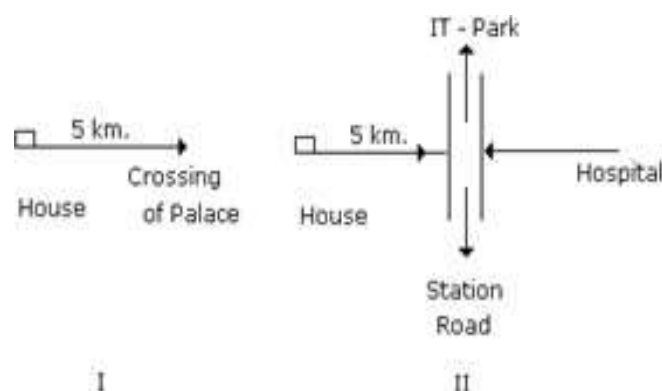
Example 3: After Sunrise, Prakash while going to college suddenly met with Lokesh at a crossing point. Lokesh's Shadow was exactly right of Prakash. If they were facing each other on which direction was Prakash facing?

Solution: Always Sun rises in the East Direction. So Shadow falls towards West



Example 4: Prem started from his home and moved 5 km to reach the crossing point of the palace. In which direction was Prem going, if the road opposite to his direction goes to the hospital. The road to the right of Prem goes to the station. If the road which goes to station is just opposite to the road of the IT-Park, then in which direction is Prem which leads to the IT- Park?

Solution: From the below diagram its shows that the road which goes towards the IT-Park is left of Prem.



Example 5: A child is looking for his father. He went 90 metres in the East before turning to his right. He went 20 meters before turning to his right again to look for his father at his uncle's place 30 metres from this point. His father was not there. From here he went 100 metres to the North before meeting his father in a street. What is the smallest distance between the starting point and his father's position?

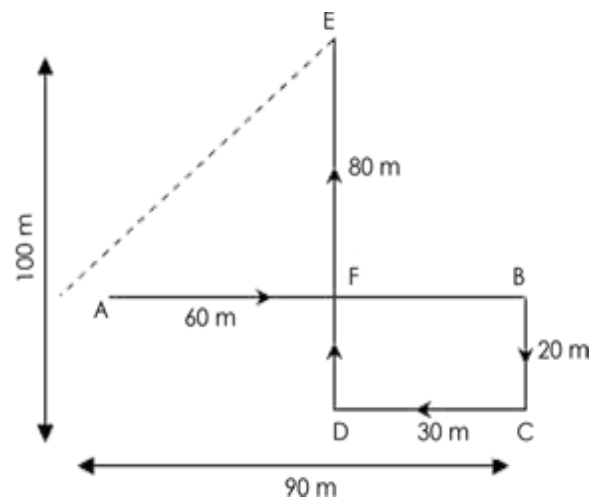
Solution: The movement of the child from A to E is as shown in fig.

Clearly, the child meets his father at E.

Now, $AF = (AB - FB) = (AB - DC) = (90 - 30) \text{ m} = 60 \text{ m}$.

$EF = (DE - DF) = (DE - BC) = (100 - 20) \text{ m} = 80 \text{ m}$.

Now the distance is square root of $(60^2 + 80^2)$, which will be 100 metres.



LEVEL – I

1. Siva Reddy walked 2 km west of his house and then turned south covering 4 km. Finally, He moved 3 km towards east and then again 1 km west. How far is he from his initial position?
A. 10 km B. 9 km C. 2 km D. 4 km
2. A man went 10 kms towards South. Then turned East and covered 10 kms and turned to the right. Again after 10 kms, he turned to left and covered 10 kms to reach the destination. How far is he from his starting point?
A. 18.8 km B. 28.28 km C. 16 km D. 20 km
3. Rajesh's school bus is facing North when reaches his school. After starting from Rajesh's house, it turning twice right and then left before reaching the school. What direction the bus facing when it left the bus stop in front of Rajesh's house?
A. East B. North C. South D. West
4. Anil wants to go the university. He starts from his house which is in the East and comes to a crossing. The road to his left ends in a theatre, straight ahead is the hospital. In which direction is the University?
A. East B. North C. South D. West
5. If South-East becomes North, North-East becomes West and so on, what will West become?
A. North B. East C. South-East D. North-West
6. A man walks 1 km towards East and then he turns to South and walks 5 km. Again he turns to East and walks 2 km, after this he turns to North and walks 9 km. Now, how far is he from his starting point?
A. 10 km B. 9 km C. 5 km D. 4 km
7. I am facing South. I turn right and walk 20 m. Then I turn right again and walk 10 m. Then I turn left and walk 10 m and then turning right walk 20 m. Then, I turn right again and walks 60 m. In which direction am I from the starting point?
A. North-East B. North-West C. North D. West
8. Dharma walks 10 km toward North. From there, he walks 6 km towards South. Then, he walks 3 km towards East. How far and in which direction is he with reference to his starting point?
A. 2 km South-East B. 5 km South-East C. 5 km North-East D. 5 km West
9. Kumar walks 10 meters in front and 10 meters to the right. Then every time turning to his left he walks 5, 15 and 15 meters respectively. How far is he now from his starting point?
A. 15m B. 10m C. 12m D. 5m
10. The time on the watch is quarter to three. If the minute-hand points to North-East, If the minute-hand points to North-East, in which direction does the hour hand point?
A. South-West B. South-East C. North-West D. North-East

11. Starting from a point X, Ravi walked 20 m towards South. He turned left and walked 30m. He then turned left and walked 20 m. He again turned left and walked 40 m and reached at a point Y. How far and in which direction is the point Y from the point X?

- A. 15m East of X B. 10m West of X C. 15m West of X D. 10m East of X

12. Lakshman went 15 kms to the West of his house, then turned left and walked 20 kms. He then turned East and walked 25 kms and finally turning left covered 20 kms. How far was he from his house?

- A. 5 kms B. 10 kms C. 40 kms D. 80 kms

13. Ravi started walking towards North. After walking 30 m, he turned left and walked 40 m. He then turned left and walked 30 m. He again turned left and walked 50 m. How far was he from his original position?

- A. 50m B. 30m C. 10m D. 60m

14. I am standing at the center of a circular field .I go down south to the edge of the field and then turning left I walk along the boundary of the field equal to three-eighth of its length .Then I turn west and go right across to the opposite point on the boundary .In which direction am I from the starting point?

- A. South-West B. West C. North-West D. North

15. A villager went to meet his uncle in another village situated 5 km away in the North-East direction of his own village. From there he came to meet his father-in-law living in a village situated 4 km in the south of his uncle village. How far away and in what direction is he now?

- A. 4 km in the East B. 3 km in the East C. 4 km in the west D. 3 km in the North

16. Dhanumjay walks 10 m towards the South .Turning to the left , he walks 20 m and then moves to his right. After moving a distance of 20 m , he turns to the right and walks 20 m. Finally, he turns to the right and moves a distance of 10 m .How far and in which direction is he from the starting point?

- A. 20 m North B. 20 m South C. 10 m North D. 10 m South

17. Arjun walked 30 m towards East, took a right turn and walked 40 m .Then he took a left turn and walked 30 m .In which direction is he now from the starting point?

- A. South-East B. South C. North-East D. East

18. A man leaves for his office from his house. He walks towards East. After moving a distance of 20 m, he turns South and walks 10 m. Then he walks 35 m towards the West and further 5 m towards the North .He then turns towards East and walks 15 m .What is the straight distance between his initial and final positions?

- A. 0 B. 5 C. 10 D. 15

19. Murari walked 40 m towards North, took a left turn and walked 20 m .He again took a left turn and walked for 40 km. How far and in which direction is he from the starting point?

- A. 20 m West B. 20 m South C. 20 m East D. 20 m North

20. Nishitha walks 14 m towards west, then turns to her right and walks 14 m and then turns to her left and walks 10 m. Again turning to her left she walks 14 m .What is the shortest distance between her starting point and the present position?

- A. 14 B. 24 C. 34 D. 44

LEVEL – II

1. One morning Udai and Vishal were talking to each other face to face at a crossing. If Vishal's shadow was exactly to the left of Udai, which direction was Udai facing?

- A. East B. West C. North D. South

2. Y is in the East of X which is in the North of Z. If P is in the South of Z, then in which direction of Y, is P?

- A. North B. South C. South-East D. None of these

3. If South-East becomes North, North-East becomes West and so on. What will West become?

- A. North-East B. North-West C. South-East D. South-West

4. A man walks 5 km toward south and then turns to the right. After walking 3 km he turns to the left and walks 5 km. Now in which direction is he from the starting place?

- A. West B. South C. North-East D. South-West

5. Rahul put his timepiece on the table in such a way that at 6 P.M. hour hand points to North. In which direction the minute hand will point at 9.15 P.M.?

- A. South-East B. South C. North D. West

6. Rasik walked 20 m towards north. Then he turned right and walks 30 m. Then he turns right and walks 35 m. Then he turns left and walks 15 m. Finally he turns left and walks 15 m. In which direction and how many metres is he from the starting position?

- A. 15 m West B. 30 m East C. 30 m West D. 45 m East

7. Two cars start from the opposite places of a main road, 150 km apart. First car runs for 25 km and takes a right turn and then runs 15 km. It then turns left and then runs for another 25 km and then takes the direction back to reach the main road. In the meantime, due to minor break down the other car has run only 35 km along the main road. What would be the distance between two cars at this point?

- A. 65 km B. 75 km C. 80 km D. 85 km

8. Starting from the point X, Jayant walked 15 m towards west. He turned left and walked 20 m. He then turned left and walked 15 m. After this he turned to his right and walked 12 m. How far and in which directions is now Jayant from X?

- A. 32 m, South B. 47 m, East C. 42 m, North D. 27 m, South

9. One evening before sunset Rekha and Hema were talking to each other face to face. If Hema's shadow was exactly to the right of Hema, which direction was Rekha facing?

- A. North B. South C. East D. Data is inadequate

10. A boy rode his bicycle Northward, then turned left and rode 1 km and again turned left and rode 2 km. He found himself 1 km west of his starting point. How far did he ride northward initially?

- A. 1 km B. 2 km C. 3 km D. 5 km

Directions to Solve (11-12): Dev, Kumar, Niles, Ankur and Pintu are standing facing to the North in a playground such as given below:

Kumar is at 40 m to the right of Ankur.

Dev is at 60 m in the south of Kumar.

Niles is at a distance of 25 m in the west of Ankur.

Pintu is at a distance of 90 m in the North of Dev.

11. Which one is in the North-East of the person who is to the left of Kumar?

- A. Dev B. Niles C. Ankur D. Pintu

12. If a boy starting from Niles, met to Ankur and then to Kumar and after this he to Dev and then to Pintu and whole the time he walked in a straight line, then how much total distance did he cover?

- A. 215 m B. 155 m C. 245 m D. 185 m

Directions (13-15): These questions are based on the following information:

Seven villages A, B, C, D, E, F and G are situated as follows:

E is 2 km to the west of B.

F is 2 km to the north of A.

C is 1 km to the west of A.

D is 2 km to the south of G.

G is 2 km to the east of C.

D is exactly in the middle of B and E.

13. A is in the middle of

- A. E and C B. E and G C. F and G D. G and C

14. Which two villages are west of G?

- A. D and C B. F and E C. C and A D. G and E

15. How far is E from D (in km)?

- A. 1 B. $\sqrt{20}$ C. 5 D. $\sqrt{26}$

16. A man is facing west. He turns 45 degree in the clockwise direction and then another 180 degree in the same direction and then 270 degree in the anticlockwise direction. Find which direction he is facing now?

- A. South-West B. West C. South D. East-South

17. A man is facing north. He turns 45 degree in the clockwise direction and then another 180 degree in the same direction and then 45 degree in the anticlockwise direction. Find which direction he is facing now?

- A. North B. East C. West D. South

18. One day, Raviraj left home and cycled 20 Km southwards, turned right and cycled 10 km and turned right and cycled 20 Km and turned left and cycled 20 Km. How many kilometres will he have to cycle to reach his home straight?

- A. 50 Km B. 30 Km C. 40 Km D. 60 Km

19. Kunal walks 10 km towards North. From there he walks 6 Km towards South. Then, he walks 3 Km towards east. How far and in which direction is he with reference to his starting point?

- A. 5 Km North B. 5 Km South C. 5 Km East D. 5 Km North-East

20. Gaurav walks 20 metres towards North. He then turns left and walks 40 metres. He again turns left and walks 20 metres. Further, he moves 20 metres after turning to the right. How far is he from his original position?

- A. 40 metres B. 50 metres C. 60 metres D. 70 metres

LEVEL - III

Directions to Solve (1-3): Each of the following questions is based on the following information:

A # B means B is at 1 metre to the right of A.

A \$ B means B is at 1 metre to the North of A.

A * B means B is at 1 metre to the left of A.

A @ B means B is at 1 metre to the south of A.

In each question first person from the left is facing North.

1. According to X @ B * P, P is in which direction with respect to X?

- A. North B. South C. North-East D. South-West

2. According to M # N \$ T, T is in which direction with respect to M?

- A. North-West B. North-East C. South-West D. South-East

3. According to P # R \$ A * U, in which direction is U with respect to P?

- A. East B. West C. North D. South

Directions to Solve (4-7): Each of the following questions is based on the following information:

Six flats on a floor in two rows facing North and South are allotted to P, Q, R, S, T and U.

Q gets a North facing flat and is not next to S.

S and U get diagonally opposite flats.

R next to U, gets a south facing flat and T gets North facing flat.

4. If the flats of P and T are interchanged then who's flat will be next to that of U?

- A. P B. Q C. R D. T

5. Which of the following combination get south facing flats?

- A. QTS B. UPT C. URP D. Data is inadequate

6. The flats of which of the other pair than SU, is diagonally opposite to each other?

- A. QP B. QR C. PT D. TS

7. Whose flat is between Q and S?

- A. T B. U C. R D. P

8. A child is looking for his father. He went 90 meters in the east before turning to his right. He went 20 meters before turning to his right again to look for his father at his uncle's place 30 meters from this point. His father was not there. From there, he went 100 meters to his north before meeting his father in a street. How far did the son meet his father from starting point?

- A. 80 metre B. 90 metre C. 100 metre D. 110 metre

9. Rohit walked 25 metres towards South. Then he turned to his left and walked 20 metres. He then turned to his left and walked 25 metres. He again turned to his right and walked 15 metres. At what distance is he from the starting point and in which direction?

- A. 35 metre, North B. 30 metre, South C. 35 metre, East D. 30 metre, North

10. Starting from a point P, Sachin walked 20 metres towards South. He turned left and walked 30 metres. He then turned left and walked 20 metres. He again turned left and walked 40 metres and reached a point Q. How far and in which direction is the point Q from the point P?

- A. 30 metres, West B. 10 metres, West C. 30 metres, North D. 10 metres, North

11. Amit starts from a point A and walks 5 m towards North-East direction and reaches point B. From here he travels 8 m in East direction and reaches point C. From C he travels towards South-West direction and reaches point D after traveling a distance equal to AB. At last, he turns towards West direction and reaches point A. How much distance has been covered by Amit and which geometrical figure has been formed by path travelled by him?

- A. 26m, square B. 26m, parallelogram
C. 26m, trapezium D. 16m, parallelogram

12. Vinod starts from his house and travels 4 km in East direction, after that he turns towards left and moves 4 km. Finally, he turns towards left and moves 4 km. At what distance and in which direction he finally stands from his original point?

- A. North, 4 km B. North-East, 4 km
C. South, 12 km D. West, 4 km

13. A and B starts from a point in opposite directions. A travels 3 km and B also travels 3 km. Then, A turns towards right and travels 4 km and B turns towards right and travels 4 km. What is the distance between A and B?

- A. 8 km B. 10 km C. 12 km D. 14 km

14. An object is projected from South-East direction to North-West direction with a certain force. Air exerts an equal force from South-West direction to North-East direction. What will be the new direction of the object?

- A. Towards South-East B. Towards North-East
C. Towards East D. None of these.

Directions (15-18): Each of the following questions is based on the following information:

All the roads of the city is either parallel or perpendicular to each other. A, B, C, D and E are parallel to each other. Roads G, H, I, J, K and M are parallel to one another.

I. Road A is 1 km east of Road B.

- II. Road B is $\frac{1}{2}$ km west of Road C.
- III. Road D is 1 km west of Road E.
- IV. Road G is $\frac{1}{2}$ km south of Road H.
- V. Road I is 1 km north of Road L.
- VI. Road K is $\frac{1}{2}$ km north of Road L.
- VII. Road K is 1 km south of Road M.

15. Which is essentially true?

- A. E and B intersect
- B. D is 2 km west of B
- C. D is at least 2 km west of A
- D. M is 1.5 km north of L

16. If E is between B and C, then which of the option is false?

- A. D is 2 km west of A
- B. C is less than 1.5 km from D
- C. E is less than 1 km from A
- D. D is less than 1 km from B.

17. If E is between B and C, then distance between A and D is

- A. $\frac{1}{2}$ km
- B. 1 km
- C. 1.5 km
- D. $1.5 - 2$ km

18. Which of the possibilities would make two roads coincide?

- A. L is $\frac{1}{2}$ km north of I
- B. C is 1 km west of D
- C. D is $\frac{1}{2}$ km east of A
- D. E and B are $\frac{1}{2}$ km apart

Directions (19-20): Each of the following questions is based on the following information:

- I. There are 6 check-posts A, B, C, D, E and F.
- II. Check-post F is 25 kms to the north of D which is 35 kms to the north-east of B.
- III. Check-post A is 15 kms west of E and 35 kms to the south-west of C.
- IV. B, A and E are in straight line.
- V. The check-post B and E are 70 kms apart from each other.

19. Which check-post is the farthest to the south-west of D?

- A. A
- B. B
- C. C
- D. D

20. Which check-post is the nearest to the north-east of E?

- A. A
- B. B
- C. C
- D. D

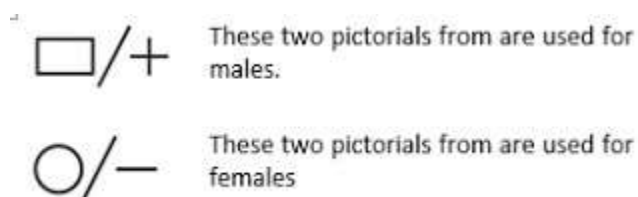
BLOOD RELATION

Blood relation is an important topic keeping in view the competitive exams. Blood relation shows the different relations among the members of a family. Based on the information given, we have to find relation between particular members of the family.

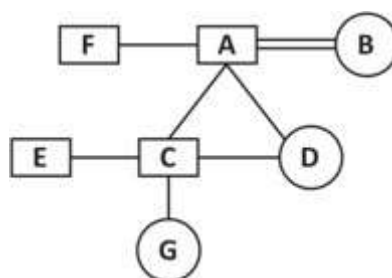
Now, take a look at below given “Generation Table” which will help you to understand the different relationship.

Generation	Male	Female
Three generations above	Great Grandfather, Maternal great Grandfather, Great Grandfather-in-law	Great Grandmother, Maternal great Grandmother, Great Grandmother-in-law
Two generations above	Grandfather Maternal Grandfather Grandfather-in-law	Grandmother Maternal Grandmother Grandmother-in-law
One generation above	Father, Uncle Maternal Uncle Father-in-law	Mother, Aunt, Maternal Aunt Mother-in-law
current generation (self)→	Husband, Brother Cousin, Brother-in-law	Wife, Sister, Cousin, Sister-in-law
One generation below ↓	Son Nephew Son-in-law	Daughter Niece Daughter-in-law
Two generations below ↓↓	Grandson Grandson-in-law	Granddaughter Granddaughter-in-law
Three generations below ↓↓↓	Great Grandson Great Grandson-in-law	Great Granddaughter Great Granddaughter-in-law

The questions of Blood Relations can be solved easily with the help of “Generation Tree”. Different pictorial form, which are used to define the relationship among them.



Representation of different relations:



From given generation tree we can deduce some important relationship between family members:

1. A is Father of C, E and D
2. B is Mother of C, E and D
3. F is Brother of A
4. F is Brother in law of B
5. A is Husband of B
6. B is Wife of A
7. F is Uncle of E, C and D
8. C and E are Son of A and B
9. D is Daughter of A and B
10. D is Sister of E and C
11. E is Brother of C and D
12. C is Brother of E and D
13. A is Grandfather of G
14. B is Grandmother of G
15. G is Granddaughter of A and B.

Types of questions asked from Blood Relations:

1. Based on Dialogue or Conversation
2. Based on Puzzles
3. Based on Symbolically coded

Based on Conversation or Dialogue- In this type of questions, the one person talking to or doing chit-chat with other person giving information by pointing to some picture or person.

Example 1: Pointing to a lady on the stage, Monika said, “She is the sister of the son of the wife of my husband.” How is the lady related to Monika?

Solution: Find who you can easily relate to and be that person-then go about creating one relation after another.

In this question, be Monika-then start from the end of the sentence.

“My husband” = Monika’s husband

‘Wife of my husband’ = is me = Monika

‘Son of the wife of my husband’ = My Son

‘Sister of the Son of the wife of my Husband’ = My Son’s Sister = My daughter

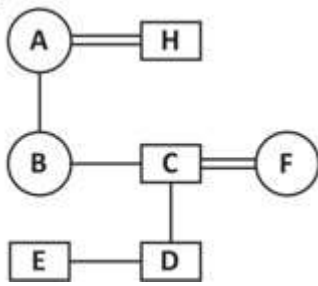
‘She’ is the sister of the son of the wife of my husband’ = the lady on the stage = the lady being pointed out = my daughter.

So, lady on the stage is Monika’s daughter.

Based on Puzzles: In this type of question, you have to conclude the relations between two given person based on more than one information given in the question.

Example 2: A is the mother of B. B is the sister of C. D is the son of C. E is the brother of D. F is the mother of E. H has only two children B and C. How is F related to E?

Solution: In this question, first we will draw the generations tree:



So, F is mother of E.

Based on Symbols: In this type of question, information are coded in the form of symbols like □, #, \$, % etc.

Example 3: Direction: Read the following information carefully and then answer the question given below:

- (a) A □ B means A is mother of B.
- (b) A \$ B means A is sister of B.
- (c) A * B means A is father of B.
- (d) A # B means A is brother of B.

Question: Which of the following means R is uncle of T?

- (a) R * P # S □ Q \$ T
- (b) S * P # R * U # T
- (c) P * R \$ Q \$ S * T
- (d) P * R \$ Q \$ S * T
- (e) None of these

Solution: From option, C, we will get R is uncle of T.

LEVEL – I

1. Q is the brother of R; P is the sister of Q; T is the brother of S; S is the daughter of R. How is Q related to T?
A. Uncle B. Father C. Brother-in-law D. Nephew
2. Z; the son-in-law of K, is the brother-in-law of G who is the brother of E. E is the daughter of K. How is G related to K?
A. Brother B. Son C. Father D. Data inadequate
3. Pointing to a lady in the photograph, Monika said, “Her son’s father is the son-in-law of my mother.” How is Monika related to the lady?
A. Aunt B. Sister C. Mother D. Cousin
4. Anil introduces Akash as the son of the only brother of his father’s wife. How is Akash related to Anil?
A. Cousin B. Son C. Uncle D. Son-in-law
5. Shikha told Aarushi, “The girl I met, yesterday at the beach was youngest daughter of the brother-in-law of my friend’s mother.” How is the girl related to Shikha’s friend?
A. Cousin B. Daughter C. Niece D. Friend
6. Pointing to a man on the stage, Natasha said, “He is the brother of the daughter of the wife of my husband.” How is the man on the stage related to Natasha?
A. Son B. Husband C. Cousin D. Nephew
7. Pointing to a man in a photograph, Aarohi said, “His mother’s only daughter is my mother.” How is Aarohi related to that man?
A. Nephew B. Sister C. Wife D. Niece
8. Pointing to a girl in the photograph, Umesh said, “Her mother’s brother is the only son of my mother’s father.” How is the girl’s mother related to Umesh?
A. Mother B. Sister C. Niece D. Grandmother
9. X introduces Y saying “He is the husband of the granddaughter of the father of my father”. How is Y related to X?
A. Brother B. Son C. Brother-in-law D. Nephew
10. Pointing to a woman, Manish said, “She is the daughter of the only child of my grandmother.” How is the woman related to Manish?
A. Sister B. Niece C. Cousin D. Data inadequate
11. Pointing to a person, a man said to a woman, “His mother is the only daughter of your father.” How was the woman related to the person?
A. Aunt B. Mother C. Wife D. Daughter
12. K and L are brothers. M and N are sisters. K’s son is N’s brother. How is L related to M?
A. Father B. Brother C. Grandfather D. Uncle

13. Pointing to a person, Deepak said, "His only brother is the father of my daughter's father". How is the person related to Deepak?
A. Father B. Grandfather C. Uncle D. Brother-in-law
14. Pointing to a photograph Anjali said, "He is the son of the only son of my grandfather." How is the man in the photograph related to Anjali?
A. Brother B. Uncle C. Son D. Data is inadequate
15. Amit said - "This girl is the wife of the grandson of my mother". How is Amit related to the girl?
A. Brother B. Grandfather C. Husband D. Father-in-law
16. Pointing towards a man, a woman said, "His mother is the only daughter of my mother." How is the woman related to the man?
A. Mother B. Grandmother C. Sister D. Daughter
17. Introducing Sonia, Aamir says, "She is the wife of only nephew of only brother of my mother." How Sonia is related to Aamir?
A. Wife B. Sister C. Sister-in-law D. Data is inadequate
- Directions(18-20): A is the son of B. C, B's sister has a son D and a daughter E. F is the maternal Uncle of D.
18. How is A related to D?
A. Cousin B. Nephew C. Uncle D. Brother
19. How is E related to F?
A. Sister B. Daughter C. Niece D. Wife
20. How many Nephews does F have?
A. None B. One C. Two D. Three

LEVEL – II

Directions (Questions 1 to 5): Read the following information carefully and answer the questions given below it:

In a family, there are six members A, B, C, D, E and F. A and B are a married couple, A being the male member. D is the only son of C, who is the brother of A. E is the sister of D. B is the daughter-in-law of F, whose husband has died.

1. How is F related to A?
A. Mother B. Sister-in-law C. Sister D. Mother-in-law
2. How is E related to C?
A. Sister B. Daughter C. Cousin D. Aunt
3. Who is C to B?
A. Brother B. Brother-in-law C. Nephew D. Son-in-law

4. How Many male members are there in the family?
 A. One B. Two C. Three D. Four

5. How is F related to C?
 A. Mother-in-law B. Sister-in-law C. Mother D. Aunt

Directions (Q.No. 6 to 10): Read the following information and answer the questions given below it:

‘A+B’ means ‘A is the daughter of B’

‘A*B’ means ‘A is the son of B’

‘A-B’ means ‘A is the wife of B’

6. If P*Q-S, which of the following is true?
 A. S is wife of Q B. S is father of P
 C. P is daughter of Q D. Q is father of P

7. If T-S*B-M, which of the following is not true?
 A. B is mother of S B. M is husband of B
 C. T is wife of S D. S is daughter of B

8. If Z*T-S*U+P, what is U to Z?
 A. Mother B. Grandmother
 C. Father D. Cannot be determined

9. If ‘P \$ Q’ means ‘P is father of Q’; ‘P # Q’ means ‘P is mother of Q’; ‘P * Q’ means ‘P is sister of Q’, then how is D related to N in N#A\$B*D ?
 A. Nephew B. Grandson
 C. Grand-daughter D. Data inadequate

10. If ‘P*Q’ means ‘P is the daughter of Q’; ‘P+Q’ means ‘P is the father of Q’; ‘P@Q’ means ‘P is the mother of Q’ and ‘P-Q’ means ‘P is the brother of Q’, then in the expression A@B+C-E*F, how is A related to F ?
 A. Mother B. Aunt C. Daughter-in-law D. None of these

Directions(11-15):

A \$ B means A is mother of B

A # B means A is father of B

A @ B means A is husband of B

A % b means A is daughter of B

11. P @ Q \$ M # T indicates what relation of P with T?
 A. Paternal grandmother B. Maternal grandmother
 C. Paternal grandfather D. Maternal grandfather

12. Which of the following expressions indicates R is the sister of H
 A. H \$ D @ F # R B. R % D @ F \$ H
 C. R \$ D @ F # H D. H % D @ F \$ R

13. If F @ D % K # H, then how F related to H
 A. Brother- in-law B. Sister
 C. Sister-in-law D. Cannot be determined

14. Which of the following expressions indicates H is the brother of N
- A. H # R \$ D \$ N B. N % F @ D \$ H # R
- C. N % F @ D \$ H D. N % F @ D % H

15. If G \$ M @ K, how is K related to G
- A. Daughter-in-law B. Mother-in-law
- C. Daughter D. None of these

Directions (16-18): Six persons A, B, C, D, E and F are travelling together. B is the son of C but C is not the mother of B. A and C are married couple. E is the brother of C. D is the daughter of A and F is the brother of B.

16. How many male members are there in the Family?
- A. 1 B. 2 C. 3 D. 4
17. Who is the mother of B?
- A] D B. F C. E D. A
18. How many children does A have?
- A. One B. Two C. Three D. Four
19. Rita told Mani, "The girl I met yesterday at the beach was the youngest daughter of the brother-in-law of my friend's mother." How is the girl related to Rita's friend?
- A. Cousin B. Daughter C. Niece D. Friend
20. Pointing to a photograph a woman says "This man's son's sister is my mother-in-law" How is the woman's husband related to the man in this photograph?
- A. Grandson B. Son C. Son-in-law D. Nephew

LEVEL – III

Directions (1-5):

- P = Q means Q is the father of P
- P * Q means P is the sister of Q
- P ? Q means Q is the mother P
- P \$ Q means P is the brother of Q
- P @ Q means Q is son of P
- P # Q means P is the daughter of Q

1. Which of the following is not correct?
- A. R # S ? T means R is granddaughter of T
- B. P = Q ? R means R is grandmother of P
- C. L \$ M * O means O is sister of L
- D. All are correct
2. Which of the following is correct?
- A. V # T * P means P is maternal uncle of V
- B. D ? V # T means D is granddaughter of T
- C. L @ M \$ R means R is paternal uncle of L

$$M \times L + N - O$$

- A. M is daughter of O
C. N is husband of L

- B. L is father of O
D. L is son of N

13. What will come in place of question mark (?) if it is given that V is mother-in-law of W in the following expression?

$$W \div P + C - T ? V - Q$$

- A. $\div$ B. + C. $\times$ D. -

Directions (14-15): These questions are based on the following information.

'P@Q' means 'P is the mother of Q'

'P\$Q' means 'P is the husband of Q'

'P#Q' means 'P is the sister of Q'

'P*Q' means 'P is the son of Q'.

14. If F#J*T\$R@L, then which of the following is definitely true?

- A. L is the brother of F B. F is the sister of L
C. F is the brother of J D. L is the brother of J

15. Which of the following indicates the relationship 'R' is the daughter of T'?

- A. R#F*B@T B. R#F*B\$T
C. T@B#R*F D. T@B#F*R

Directions (16-20): A, B, C, D, E, F, G and H are sitting around a circle facing the centre but not necessarily in the same order. Each of them has a relationship with A.

G is sitting second to the left of father of A. F is immediate neighbour of A. D, mother of A is sitting opposite to the sister of A. B is sitting to the immediate right of wife of A. E who is a male is sitting second to the right of mother of C. Brother of A is sitting third to right of B. Daughter of A is sitting to third to right of sister of A. A is sitting second to the right of daughter of E. E is sitting to the immediate left of sister of A.

16. Who is the mother of H?

- A. F B. G C. C D. D

17. Who is the grand-daughter of E?

- A. B B. C C. D D. G

18. Who is sitting second to the right of F' sister?

- A. wife of A B. brother of C C. daughter of A D. father of B

19. How many persons are sitting between A's wife and D's husband when counted from right of A's wife?

- A. None B. One C. Two D. Three

20. What is the position of G's daughter with respect to D's daughter?

- A. third to left B. second to left C. third to right D. second to right

NUMBER, RANKING AND TIME SEQUENCE

In this chapter, we deal with questions which are followed with a sequence consisting numbers, ranking and time. We have to find answers on the basis of given condition. The importance of such types of questions cannot be over-emphasised as their presence in a test of reasoning is almost certain.

Ordering and Ranking

In ordering and ranking arrangement questions, position/rank of a person from left-right/top-bottom of a row/class is to be determined or rank/position is given & total no. of persons is to be calculated.

Here, different types of ordering & ranking arrangement questions are explained below.

$$\text{Left} + \text{Right} = \text{Total} + 1$$

Type 1

1) Total number of persons = {(sum of positions of same person from both sides i.e. left and right side) – 1}

2) Position of a person from opposite side = {(Total no. of persons – Position of same person from given side) + 1}

Example 1: In a row of persons, position of A from left side of the row is 27th and position of A from right side of the row is 34th. Find total no. of persons in the row?

Solution: Total no. of students = (Position of A from left + Position of A from right) - 1

$$\Rightarrow \text{Total no. of students} = (27 + 34) - 1 = 61 - 1 = 60$$

Type 2

1) Total no. of persons = No. of persons after or before the given person in a row + Position of same person from the other side

Example 2: In a row of persons, position of A from left side of the row is 27th and there are 5 persons after A in the row. Find total no. of persons in the row?

Solution: No. of persons in the row = Position of A from left + No. of persons after A

$$\Rightarrow \text{Total no. of persons} = 27 + 5 = 32$$

Type 3

When the positions of two persons are given from opposite ends and we know the total number of persons, then two cases arise when trying to determine the number of persons between these two persons –

When there is no overlapping: i.e. the sum of positions of the two persons from opposite ends < total number of persons

Case I: No. of students between two different persons = Total no. of students – (Sum of positions of two different persons from opposite sides)

When there is overlapping: i.e. the sum of positions of the two persons from opposite ends > total number of persons

Case II: No. of students between two different persons = (Sum of positions of two different persons from opposite sides) – Total no. of students – 2

Example 3: In a row of 54 persons, A is 35th from the left side of the row and B is 22nd from the right side of the row. Find the no. of persons sitting between A and B?

Solution: Here Sum of positions of A & B from opposite ends = $35 + 22 = 57 >$ Total no. of persons
 $\therefore$ No. of persons between A & B = (Position of A from left + Position of B from right) – Total no. of students – 2

$\Rightarrow$ No. of persons between A & B = $(35+22) - 54 - 2 = 57 - 54 - 2 = 1$

Type 4

Positions of two persons is given and their positions are interchanged and after interchanging position of 1st person is given from same side as before interchanging

1) Position of 2nd person from the same side as before interchanging = Position of 2nd person from same side before interchanging + (Position of 1st person after interchanging – position of 1st person before interchanging from same side)

2) To find total no. of students, find the person whose position from both sides can be depicted from the statement. Add both his positions from opposite ends and subtract 1.

3) To find no. of persons between them, Difference in the position of common person whose position from same side before and after interchanging is given then subtract 1

Example 4: A and B are standing in a row of persons. A is 18th from left side of the row and B is 24th from right side of the row. If they interchange their positions A becomes 31st from left. Find No. of persons between A & B

Solution: No. of persons between A & B = (Position of A from left after interchanging – Position of A from left before interchanging) – 1

$\Rightarrow$ No. of persons between A & B = $(31 - 18) - 1 = 13 - 1 = 12$

Number Test

In these types of questions, a number, a set of numbers, series of digit is given and the student is asked to trace out digit following certain given conditions.

Example 5: How many 5s are there in the following number sequence which are immediately preceded by 7 and immediately followed by 6?

Terms : 7 5 5 9 4 5 7 6 4 5 9 8 7 5 6 7 6 4 3 2 5 6 7 8

Solution: Preceded by 7 and followed by 6 So, there is only one such 5.

Example 6: How many even numbers are there in the following series of numbers, each of which is immediately preceded by an odd number, but not immediately followed by an even number?

Terms : 5 3 4 8 9 7 1 6 5 3 2 9 8 7 3 5

Solution: There are three such even numbers 6, 2, 8 each of which is preceded by an odd number and not followed by an even number. 5 3 4 8 9 7 1 6 3 2 9 8 7 3 5

Example 7: The positions of the first and the sixth digit in the number 5 1 0 9 2 3 8 6 7 4 are interchanged. Similarly, the positions of the second and the seventh digit are interchanged and so on. Which of the following will be the third digit from the right end after the rearrangement?

Solution: Given number = 5109238674

According to the question, after interchanging digits, new number = 3867451092

Hence, third digit from the right end = 0

LEVEL – I

1. In a row of trees, one tree is fifth from either end of the row. How many trees are there in the row?
A. 8 B. 9 C. 10 D. 11
2. Raman ranks sixteenth from the top and forty ninth from the bottom in a class. How many students are there in the class?
A. 64 B. 65 C. 66 D. Cannot be determined
3. Sanjeev ranks seventh from the top and twenty eight from the bottom in a class. How many students are there in the class?
A. 37 B. 36 C. 35 D. 34
4. Some boys are sitting in a row. P is sitting fourteenth from the left and Q is seventh from the right. If there are four boys between P and Q, how many boys are there in the row?
A. 25 B. 23 C. 21 D. 19
5. Aruna ranks twelfth in a class of forty-six. What will be her rank from the last?
A. 33 B. 34 C. 35 D. 37
6. Manoj and Sachin are ranked seventh and eleventh respectively from the top in a class of 31 students. What will be their respective ranks from the bottom in the class?
A. 20th and 24th B. 24th and 20th
C. 25th and 21st D. 26th and 22nd
7. Ravi is 7 ranks ahead of Sumit in a class of 39. If Sumit's rank is seventeenth from the last, what is Ravi's rank from the start?
A. 14th B. 15th C. 16th D. 17th
8. How many 3's are there in the following sequence which are neither preceded by 6 nor immediately followed by 9?
9 3 6 6 3 9 5 9 3 7 8 9 1 6 3 9 6 3 9
A. One B. Two C. Three D. Four
9. How many 6's are there in the following series of numbers which are preceded by 7 but not immediately followed by 9?
6 7 9 5 6 9 7 6 8 7 6 7 8 6 9 4 6 7 7 6 9 5 7 6 3
A. One B. Two C. Three D. Four
10. How many 7's are there in the following series which are not immediately followed by 3 but immediately preceded by 8 ?
8 9 8 7 6 2 2 6 3 2 6 9 7 3 2 8 7 2 7 7 8 7 3 7 7 9 4
A. 10 B. 3 C. 2 D. 0
11. In the following list of numerals, how many 2's are followed by 1's but not preceded by 4?
4 2 1 2 1 4 2 1 1 2 4 4 4 1 2 2 1 2 1 4 4 2 1 4 2 1 2 1 2 4 1 4 2 1 2 4 1 4 6
A. Two B. Three C. Four D. Five

12. How many 7's are there in the following sequence which are preceded by 9 and followed by 6 ?
7 8 9 7 6 5 3 4 2 8 9 7 2 4 5 9 2 9 7 6 4 7
- A. 2 B. 3 C. 4 D. 5
13. What will be the 13th letter from right side if 1st half of series is reversed?
- A. M B. O C. N D. A
14. What will be 16th letter from left side if 2nd half is written in reversed order?
- A. W B. X C. V D. U
15. Which alphabet will be mid-way b/w 7th & 11th letter counting from left end of the normal alphabet?
- A. G B. I C. H D. M
16. Which letter is 5th to the left of 7th to the right of 12th letter from the right?
- A. Z B. Q C. D D. R
17. How many pairs of letters are there in the given words which have as many letter b/w them in the word as in the alphabet-> CREATIVE
- A. 1 B. 2 C. 3 D. 4
18. How many such digits are there in the number 7346285 which are as far away from the beginning of the number, as they will be when arranged in ascending order within the number?
- A. None B. one C. Two D. Three
19. Nitin was counting down from 32. Sumit was counting upwards the numbers starting from 1 and he was calling out only the odd numbers. What common number will they call out at the same time if they were calling out at the same speed?
- A. 19 B. 21
C. 22 D. They will not call out the same number
20. If possible to make a meaningful word from 2nd, 5th, 10th & 12th letters of the word METROPOLITAN, which is 3rd letter of the word? If no such word can be made, X is answer. If more than one word formed, give M as answer.
- A. Q B. T C. X D. M

LEVEL - II

1. In a queue, Amrita is 10th from the front while Mukul is 25th from behind and Mamta is just in the middle of the two. If there be 50 persons in the queue. What position does Mamta occupy from the front ?
- A. 20th B. 19th C. 18th D. 17th
2. If Atul finds that he is twelfth from the right in a line of boys and fourth from the left, how many boys should be added to the line such that there are 28 boys in the line?
- A. 12 B. 13 C. 14 D. 20

3. Manisha ranked sixteenth from the top and twenty ninth from the bottom among those who passed an examination. Six boys did not participate in the competition and five failed in it. How many boys were there in the class?
A. 40 B. 44 C. 50 D. 55
4. Kailash remembers that his brother Deepak's birthday falls after 20th May but before 28th May, while Geeta remembers that Deepak's birthday falls before 22nd May but after 12th May. On what date Deepak's birthday falls?
A. 20th May B. 21st May C. 22nd May D. Cannot be determined
5. Sangeeta remembers that her father's birthday was certainly after eighth but before thirteenth of December. Her sister Natasha remembers that their father's birthday was definitely after ninth but before fourteenth of December. On which date of December was their father's birthday?
A. 10th B. 11th C. 12th D. Data inadequate
6. Standing on a platform, Amit told Sunita that Aligarh was more than ten kilometers but less than fifteen kilometers from there. Sunita knew that it was more than twelve but less than fourteen kilometers from there. If both of them were correct, which of the following could be the distance of Aligarh from the platform?
A. 11 km B. 12 km C. 13 km D. 14 km
7. Ashish leaves his house at 20 minutes to seven in the morning, reaches Kunal's house in 25 minutes, they finish their breakfast in another 15 minute and leave for their office which takes another 35 minutes, At what time do they leave Kunal's houses to reach their office ?
A. 7.40 a.m. B. 7.20 a.m. C. 7.45 a.m. D. 8.15 a.m.
8. Ajay left home for the bus stop 15 minutes earlier than usual. It takes 10 minutes to reach the stop. He reached the stop at 8.40 a.m. What time does he usually leave home for the bus stop?
A. 8.30 a.m. B. 8.45 p.m. C. 8.55 a.m. D. None of these
9. Which is the third number to the left of the number which is exactly in the middle of the following sequence of numbers?
1 2 3 4 5 6 7 8 9 2 4 6 8 9 7 5 3 1 9 8 7 6 5 4 3 2 1
A. 3 B. 4 C. 5 D. 6
10. Count each 7 which is not immediately preceded by 5 but is immediately followed by either 2 or 3. How many such 7's are there ?
5 7 2 6 5 7 3 8 3 7 3 2 5 7 2 7 3 4 8 2 6 7 8
A. 2 B. 3 C. 4 D. 5
11. In the following series, how many such odd nummbers are there which are divisible by 3 or 5, then followed by odd numbers and then also followed by even numbers ?
12, 19, 21, 3, 25, 18, 35, 20, 22, 21, 45, 46, 47, 48, 9, 50, 52, 54, 55, 56
A. Nil B. One C. Two D. Three

12. In the following number sequence, how many such even numbers are there which are exactly divisible by its immediate preceding number but not exactly divisible by its immediate following number ?

3 8 4 1 5 7 2 8 3 4 8 9 3 9 4 2 1 5 8 2

- A. One B. Two C. Three D. Four

13. Series : 5 1 4 7 3 9 8 5 7 2 6 3 1 5 8 6 3 8 5 2 2 4 3 4 9 6

How many even numbers are there in the sequence which are immediately preceded by an odd number but immediately followed by an even number?

- A. 1 B. 2 C. 3 D. 4

14. Make a meaningful word from 2nd, 5th, 8th letter of the word CARETAKER. Which will be the 1st letter of that word? If no such word can be made, give X as answer. If more than one such word formed, give M as an answer

- A. E B. T C. X D. M

15. How many digits are there which are either immediately followed by a digit divisible by 3 or immediately preceded by a digit divisible by 5?

7 3 2 7 6 5 4 8 7 6 3 2 3 5 4 3 7 6 3 2

- A. Nine B. Eight C. Four D. Two

Directions for questions 16 to 18: Consider the letters of the alphabet written in the order from left to right i.e. from A to Z

16. The letter which is fourth to the left of the letter, which is fifth to the right of F is

- A. C B. W C. E D. G

17. The letter which is 6th to the left of the letter which is 8th to the right of P is

- A. E B. D C. S D. R

18. The letter which is third to the right of the letter, which is seventh to the left of O is

- A. P B. C C. K D. L

19. In the following set of numbers, if 1 is added to the last digit and then the order of digits is reversed, which number will be fourth if arranged in ascending order?

567 284 696 865 738

- A. 567 B. 284 C. 865 D. 738

20. How many numbers amongst the numbers 9 to 54 are there which are exactly divisible by 9 but not by 3 ?

- A. 5 B. 6 C. 0 D. 9

LEVEL - III

1. If the numbers from 1 to 45 which are exactly divisible by 3 are arranged in ascending order, minimum number being on the top, which would come at the ninth place from the top?

- A. 18 B. 21 C. 24 D. 27

2. If the numbers from 5 to 85 which are exactly divisible by 5 are arranged in descending order, which would come at the eleventh place from the bottom?
A. 35 B. 45 C. 50 D. None of these
3. In a class of 60, where girls are twice that of boys, Kamal ranked seventeenth from the top. if there are 9 girls ahead of Kamal, how many boys are after him in rank ?
A. 3 B. 7 C. 12 D. 23
4. In a row of ten boys, when Rohit was shifted by two places towards the left, he became seventh from the left end. What was his earlier position from the right end of the row?
A. First B. Second C. Fourth D. Sixth
5. In a queue, Vijay is fourteenth from the front and Jack is seventeenth from the end, while Mary is in middle of Vijay and Jack. If Vijay be ahead of Jack and there be 48 persons in the queue, how many persons are there between Vijay and Marry?
A. 8 B. 7 C. 6 D. 5
6. In a row of girls, Rita and Monika occupy the ninth places from the right end and tenth place from the left end, respectively. If they interchange their places, Rita and Monika occupy seventeenth places from the right and eighteenth place from the left, respectively. How many girls are there in the row?
A. 25 B. 26 C. 27 D. Data inadequate
7. In a row of girls, Shilpa is eighth from the left and Reena is seventeenth from the right. If they interchange their positions, Shilpa becomes fourteenth from the left. How many girls are there in the row?
A. 25 B. 27 C. 29 D. None of these
8. In a row of boys, Kapil is eighth from the right and Nikunj is twelfth from the left. When Kapil and Nikunj interchange positions, Nikunj becomes twenty first from the left. Which of the following will be Kapil's position from the right?
A. 8th B. 17th C. 21st D. Canont be determined
9. If Thursday was the day after the day before yesterday five days ago, what is the least number of days ago when Sunday was three days before the day after tomorrow ?
A. Two B. Three C. Four D. Five
10. If the positions of the first and the third digit within each number are interchanged, which of the following will be the third digit of the second lowest number?
987, 514, 658, 487, 404, 269
A. 8 B. 9 C. 2 D. 4
11. In the number 76534218 each digit is replaced next digit, ie '1' is replaced by '2' , '2' is replaced by '3' and so on and then the digits are arranged in ascending order from left to right, which digit will be fifth from the left end?
A. 6 B. 5 C. 7 D. 4

12. If it is possible to form a number with the second, the fifth and the eighth digits of the number 31549786, which is the perfect square of a two - digit even number, which of the following will be the second digit of that even number?

- A. 1 B. 4 C. 6 D. None of these

13. Which of the following is fifth to the right of the twelfth digit from the right end of the below arrangement?

1 8 5 9 4 7 1 2 5 8 3 6 5 9 2 7 6 4 5 2 9 2 6 4 1 2 3 5 1 4 2 8 3

- A. 3 B. 1 C. 2 D. 7

14. In a row, Kumar is at 7th place from the left and Pawan is at 9th place from the right. When they interchange the positions Kumar becomes 11th from left. How many were seated in the row?

- A. 19 B. 20 C. 21 D. 27

15. If the following series is written in the reverse order, which number will be fourth to the right of the seventh number from the left?

7, 3, 9, 7, 0, 3, 8, 4, 6, 2, 1, 0, 5, 11, 13

- A. 5 B. 11 C. 9 D. 0

16. In a row of girls, Nithya and Suganya occupy the ninth place from the right end and tenth place from the left end, respectively. If they interchange their places, then Nithya and Suganya occupy seventeenth place from the right and eighteenth place from the left respectively. How many girls are there in the row?

- A. 22 B. 24 C. 26 D. 28

17. In a row of girls facing North, Reena is 10th to the left of Pallavi, who is 21st from the right end. If Malini, who is 17th from the left end, is fourth to the right of Reena, how many girls are there in the row?

- A. 37 B. 41 C. 43 D. 49

18. If it is possible to make a meaningful word with the fourth, the eighth and the tenth letters of the word 'COUNTERACT', which of the following will be the last letter of that word ? If no such word can be made, give X as the answer. If more than one such word can be made, give M as the answer.

- A. A B. N C. X D. M

19. In a row of boys, Haran is eleventh from the left and manoj is seventeenth from the right. When they exchange their places then Haran will be thirteenth from the left. Which of the following will be the new position of Manoj from the right?

- A. 11 B. 19 C. 13 D. 17

20. Read the arrangement carefully and give the answer of following questions?

K\$23DBE8HM4@5JF4%K1+WR#AA*415

How many such symbols are there which is not immediately preceded by a letter but immediately followed by a number?

- A. 1 B. 2 C. 3 D. 0

RATIO AND PROPORTION

The ratio of two quantities a and b in the same units, is the fraction $\frac{a}{b}$ and we write it as $a : b$.
 Ratio of any number is expressed after removing all the common factors in the terms. For example, if there are two quantities having values of 8 and 6, then their ratios will be “4 : 3” because a common factor of 2 was removed from both the terms.

- In the ratio $a : b$, we call a as the first term or antecedent and b , the second term or consequent.
 Eg: In the ratio 5 : 9, antecedent = 5 and consequent = 9.
- The multiplication or division of each term of a ratio by the same non-zero number does not affect the ratio.
 Eg. $4 : 5 = 8 : 10 = 12 : 15$. Also, $4 : 6 = 2 : 3$.

Types of Ratio

1. **Duplicate Ratio:** If $a : b$ is a ratio, then its duplicate ratio is $a^2 : b^2$
 Example: If 2 : 3 is a ratio, then its duplicate ratio is $2^2 : 3^2$ i.e. 4 : 9
2. **Sub-duplicate Ratio:** If $a : b$ is a ratio, then its sub-duplicate ratio is $\sqrt{a} : \sqrt{b}$
 Example: If 16 : 25 is a ratio, then its sub-duplicate ratio is $\sqrt{16} : \sqrt{25} = 4 : 5$
3. **Triplicate Ratio:** If $a : b$ is a ratio, then its triplicate ratio is $a^3 : b^3$
 Example: If 2 : 3 is a ratio, then its triplicate ratio is $2^3 : 3^3 = 8 : 27$
4. **Sub-triplicate Ratio:** If $a : b$ is a ratio, then its sub-triplicate ratio is $a^{1/3} : b^{1/3}$
 Example: If 8 : 27 is a ratio, then its sub-triplicate ratio is $8^{1/3} : 27^{1/3} = 2 : 3$
5. **Inverse or Reciprocal Ratio:** The inverse ratio of $a : b$ is $1/a : 1/b$
 Example: If 2 : 3 is a ratio, then its inverse ratio is $(1/2) : (1/3)$
6. **Compounded Ratio:** Compound ratio is the ratio of the products, of the corresponding terms of two or more simple ratios.
 Example: The compounded ratio of the ratios: $(A : B)$, $(C : D)$, $(E : F)$ is $(ACE : BDF)$.

Proportion

The equality of two ratios is called proportion.

If $a : b = c : d$, we write $a : b :: c : d$ and we say that a, b, c, d are in proportion.

Here a and d are called **extreme terms**, while b and c are called **mean terms**.

Product of mean terms = Product of extreme terms.

Thus, $a : b :: c : d \Rightarrow (b \times c) = (a \times d)$.

If $a/b = c/d$, then:

- i. Invertendo - $b : a = d : c$
- ii. Alternendo - $a : c = b : d$
- iii. Componendo - $(a+b) : b = (c+d) : d$
- iv. Dividendo - $(a-b) : b = (c-d) : d$
- v. Componendo & Dividendo - $(a+b)(a-b) = (c+d)(c-d)$

Types of Proportions

1. Fourth Proportional: If $a : b = c : d$, then d is called the fourth proportional to a, b, c .
2. Third Proportional: If $a : b = c : d$, then c is called the third proportion to a and b .
 Similarly, If the given proportion is $a : b :: b : c$ then c is said to be the third proportion of a and b .
3. Mean Proportional: If the given ratio is $a : b :: b : c$, then b is said to be the mean proportion.
 $b = \sqrt{ac}$

Example 1: If $a : b = 2 : 3$ and $b : c = 4 : 3$, then find $a : b : c$?

Solution: $a : b = 2 : 3$

$$b : c = 4 : 3 = (4 \times (3/4)) : (3 \times (3/4)) = 3 : (9/4)$$

$$a : b : c = 2 : 3 : (9/4) = 8 : 12 : 9$$

Example 2: The sum of two numbers is 72. If the two numbers are in the ratio of 5:3. Find the two numbers.

Solution: As discussed in the theory of this topic, if the two numbers are in the ratio 5:3, let the actual number is $5k$ and $3k$. the sum of two numbers is 72. We have,

$$5k + 3k = 72$$

$$K = (72/8) = 9$$

$$\text{Hence } 5k = 45 \text{ and } 3k = 27$$

Example 3: A number is divided into parts such that 4 times the first part, 3 times the second part, 6 times the third part and the 8 times the four parts are all equal. In what ratio is the number divided?

Solution: Let the four parts into which the number is divided is a, b, c and d .

$$4a = 3b = 6c = 8d = e \quad (\text{let})$$

$$A = (e/4), b = (e/3), c = (e/6), d = (e/8)$$

$$\begin{aligned} \text{Hence, } a : b : c : d &= (e/4) : (e/3) : (e/6) : (e/8) \\ &= (1/4) : (1/3) : (1/6) : (1/8) \\ &= (24/4) : (24/3) : (24/6) : (24/8) \\ &= 6 : 8 : 4 : 3 \end{aligned}$$

Example 4: $a : b = 3 : 4$; $b : c = 6 : 7$. Find $a : b : c$?

$$\begin{array}{ccc} \text{Solution:} & a & b & c \\ & 3 & 4 & \\ & & 6 & 7 \end{array}$$

$$a : b : c = 3 \times 6 : 6 \times 4 : 4 \times 7 = 9 : 12 : 14$$

Example 5: 36% of first number is 28% of the second number. What is the respective ratio of the first number to the second number?

Solution: Let the numbers be x and y .

$$36\% \text{ of } x = 28\% \text{ of } y$$

$$\frac{x}{y} = \frac{28}{36} = \frac{7}{9}$$

$$\therefore x : y = 7 : 9$$

Example 6: The average of their ages is 30 years. What will be the ratio of their ages after 4 years?

Solution: Average age = 30 years

$$\text{Total age} = 2 \times 30 = 60 \text{ years.}$$

Let their present ages be $7x$ and $3x$ years

$$\therefore 7x + 3x = 60 \Rightarrow x = \frac{60}{10} = 6$$

$$\therefore \text{Their present ages are } 7 \times 6 \text{ and } 3 \times 6 = 42 \text{ and } 18.$$

$$\begin{aligned} \therefore \text{Their ages after 4 years} \\ &= 42 + 4 \text{ and } 18 + 4 = 46 \text{ and } 22 \text{ years} \end{aligned}$$

$$\therefore \text{ratio} = 46 : 22 = 23 : 11$$

Example 7: In a bowl there is 30 litre mixtures of milk and water. The ratio of milk and water is 7:3. How much water must be added to it so that the ratio of milk to the water be 3:7?

Solution: Milk quantity in the mixture

$$= \frac{7}{10} \times 30 = 21 \text{ litres}$$

$$\therefore \text{Water} = 30 - 21 = 9 \text{ litres}$$

$$\text{New ratio} = 3 : 7$$

$\therefore$ 3 parts of milk is 21 litres (There is no difference in the milk quantity of new mixture)

$\therefore$ Water quantity in the mixture

$$= \frac{7}{3} \times 21 = 49 \text{ litres}$$

$$\therefore 49 - 9 = 40 \text{ litres water is to be added in the new mixture}$$

Example 8: A bag contains of one rupee, 50 paise and 25 paise coins. if these coins are in the ratio of 5 : 6 : 8, and the total amount of coins is Rs. 210, find the number of 50 paise coins in the bag.

Solution: Let the number of one rupee, 50 paise, 25 paise coins be 5, 6 and 8 respectively

The value of one rupee coins

$$= \text{Rs. } 1 \times 5 = \text{Rs. } 5$$

The value of fifty paise coins

$$= \text{Rs. } 0.50 \times 6 = \text{Rs. } 3$$

The value of twenty five paise coins

$$= \text{Rs. } 0.25 \times 8 = \text{Rs. } 2$$

$$\text{Total value} = 5 + 3 + 2 = \text{Rs. } 10$$

If the total value is Rs. 10

there are 6 coins of fifty paise

$$\text{if the total value is Rs. 210, then the number of 50 coins} = \frac{210}{10} \times 6 = 126$$

VARIATION

Direct Variation

One quantity A is said to vary directly as another quantity B if the two quantities depend upon each other in such a manner that if B is increased in a certain ratio, A also increases in the same ratio.

This is denoted as $A \propto B$ (A varies directly as B).

If $A \propto B$ then $A = kB$, where k is a constant. It is called the constant of proportionality.

Inverse Variation

A quantity A is said to vary inversely as another quantity B if the two quantities depend upon each other in such a manner that if B is increased in a certain ratio, A gets decreased in the same ratio and if B is decreased in a certain ratio, then A gets increased in the same ratio.

It is the same as saying that A varies directly with $1/B$. It is denoted as $A \propto 1/B$ i.e., $A = k/B$ where k is k the constant of proportionality.

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Joint Variation

If there are three quantities A, B and C such that A varies with B when C is constant and varies with C when B is constant, then A is said to vary jointly with B and C when both B and C are varying i.e. $A \propto B$ when C is constant and $A \propto C$ when B is constant $\Rightarrow A \propto BC$ where k is the constant of proportionality.

Examples 9: The volume of a cylinder varies jointly as its height and the area of its base. When the area of the base is 64 sq. ft. and the height is 10 ft., the volume is 640 cu. ft.. What is the height of the cylinder, whose volume is 360 cu. ft. and area of the base is 72 sq.ft.

Solution: Let V be the volume, a be the area of the base and h be the height.

$V = m a h$ (m is a proportionality constant)

We know $a = 64$, $h = 10$ and $V = 640$

$$640 = m (64) (10)$$

$$m = 1; V = ah$$

$$\text{Therefore, } 360 = 72 \times h$$

$$\Rightarrow h = 360/72 = 5 \text{ ft.}$$

Hence the height of the cylinder is 5 ft.

Examples 10: If x varies directly as $y^4 + 9$ and $x = 3$ when $y = 3$, find x when $y = 9$.

Solution: $x \propto (y^4 + 9)$.

$$c = \frac{x}{y^4 + 9}.$$

when $x = 3$, $y = 3$ (given)

$$\text{Hence } c = \frac{3}{3^4 + 9} = \frac{3}{90} = \frac{1}{30};$$

$$\text{and } x = \frac{1}{30}(y^4 + 9)$$

When $y = 9$

$$x = \frac{1}{30}(y^4 + 9) = \frac{1}{30}(6561 + 9) = 219.$$

PARTNERSHIP

When two or more than two persons run a business jointly, they are called partners and the deal is known as partnership. The money put in by each of the partners is called his “investment or capital”.

Ratio of Division of Gains:

1. If the partners invest *different* amounts each for the *same* period of time, then the profits at the end of the year are shared in the ratio of their investments

Suppose A and B invest Rs x and Rs y respectively for a year in a business, then at the end of the year: (A's share of profit) : (B's share of profit) = $x : y$

2. If the partners invest the *same* amounts for *different* periods of time, then the profits at the end of the year are shared in the ratio of the time periods for which respectively investments have been in business.

Suppose A and B invest Rs x and Rs x respectively for a time period t_1 and t_2 in a business, then at the end of the year: (A's share of profit) : (B's share of profit) = $t_1 : t_2$

3. If the partners invest *different* amounts and the time period for which their investments are in the business are also *different*, then the profits at the end of the year are shared in the ratio of the product (investment $\times$ time period) for each partner.

Suppose A invests Rs x for p months and B invests Rs y for q months, then

$$(A's \text{ share of profit}) : (B's \text{ share of profit}) = xp : yq$$

4. Working and sleeping partners: A partner who manages the business is known as working partner and the one who simply invests the money is a sleeping partner.

Example 11: P and Q started a business investing Rs 85,000 and Rs 15,000 respectively. In what ratio the profit earned after 2 years be divided between P and Q respectively?

Solution: As time period is same for both, ratio of profit is directly proportional to investment
 $85,000 : 15,000 = 17 : 3$

Example 12: A,B and C started a business by investing Rs 1,20,000, Rs 1,35,000 and Rs 1,50,000. Find the share of each, out of an annual profit of Rs 56,700?

Solution: Ratio of shares of A, B and C = Ratio of their investments

$120,000 : 135,000 : 150,000 = 8 : 9 : 10$

A's share = $\text{Rs } 56,700 \times \frac{8}{27} = \text{Rs } 16,800$

B's share = $\text{Rs } 56,700 \times \frac{9}{27} = \text{Rs } 18,900$

C's share = $\text{Rs } 56,700 \times \frac{10}{27} = \text{Rs } 21,000$

Example 13: Ram and Krishna entered into a partnership with Rs 50,000 and Rs 60,000, after 4 months Ram invested Rs 25,000 more while Krishna withdrew Rs 20,000. Find the share of Ram in the annual profit of Rs 289,000.

Solution: Ram : Krishna = $50,000 \times 4 + 75,000 \times 8 : 60,000 \times 4 + 40,000 \times 8 = 10 : 7$

Ram's annual profit = $289,000 \times \frac{10}{17} = \text{Rs } 170,000$

Example 14: Prerna starts a business with Rs.45,000. Three months later Sanjna joins her with Rs, 30,000. At the end of the year in what ratio should they shared profits?

Solution: Sharing of profits will be in the ratio of investments multiplied by the time period.

Hence the ratio is

$(45,000 \times 12) : (30,000 \times 9) = 2 : 1$

Example 15: The working partner of a business gets as his commission 10% of the profits left after his commission is paid. If the working partner's commission is Rs. 30,000 then, find the total profit.

Solution: Let total profit be P.

The profit left after the working partner's commission of Rs. 30,000 is $(P - 30,000)$.

10% of this is the working partner's commission. So we have $(0.1)(P - 30,000) = 30,000$

$$\Rightarrow (0.1)P = 33,000$$

$$\therefore P = \text{Rs. } 3,30,000$$

LEVEL – I

1. If 15% of x is the same as 20% of y , then $x : y$ is?
A] 3 : 4 B] 17 : 16 C] 4 : 3 D] 16 : 17
2. In a college, the ratio of the number of boys to girls is 8 : 5. If there are 160 girls, the total number of students in the college is:
A] 100 B] 260 C] 250 D] 416
3. The ratio of income of A to that of B is 5 : 4 and the expenditure of A to that of B is 3: 2. If at the end of the year, each saves Rs, 800, the income of A is:
A] Rs. 1600 B] Rs. 2000 C] Rs. 1800 D] Rs. 2200
4. If $p : q = 3 : 4$ and $q : r = 8 : 9$, then $p : r$ is:
A] 1: 3 B] 2 : 3 C] 3 : 2 D] 1: 2
5. If $a+b : b+c : c+a = 6 : 7 : 8$ and $a + b + c = 14$, then the value of c is : .
A] 6 B] 8 C] 7 D] 2
6. Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is?
A] 5:4 B] 3:2 C] 4:5 D] 2:3
7. If three numbers in the ratio 3 : 2: 5 be such that the sum of their squares is 1862, the middle number will be?
A] 7 B] 21 C] 14 D] 35
8. A certain amount was divided between Salim and Rahim in the ratio of 4 : 3. If Rahim's share was Rs. 2400, the total amount was.
A] Rs. 5600 B] Rs. 9600 C] Rs. 3200 D] Rs. 16800
9. A profit of Rs. 30000 is to be distributed among A, B, C in the proportion 3 : 5 : 7. What will be the difference between B's and C's shares?
A] Rs. 2000 B] Rs. 10000 C] Rs. 4000 D] Rs. 14000
10. An amount of money is to be distributed among P, Q and R in the ratio 3 : 5 : 7. If Q's share is Rs. 1500, what is the difference between P's and R's shares?
A] Rs. 1200 B] Rs. 1600 C] Rs. 1500 D] Rs. 1900
11. Rs. 120 are divided among A, B, C such that A's share is Rs. 20 more than B's and Rs. 20 less than C's. What is B's share?
A] Rs. 10 B] Rs. 20 C] Rs. 15 D] Rs. 25
12. The compounded ratio of (2 : 3), (6: 11) and (11 :2) is.
A] 1: 2 B] 11: 24 C] 2: 1 D] 36: 121
13. What number should be added to each of the numbers 8, 21, 13 and 31 so that the resulting numbers, in this order form a proportion?
A] 2 B] 5 C] 3 D] 7
14. An alloy is to contain copper and zinc in the ratio 9 :4. The zinc required (in kg) to be melted with 24 kg of copper, is
A] 10.66 B] 9.66 C] 10.33 D] 9

15. What number should be subtracted from both the terms of the ratio 15 : 19 so as to make it as 3 : 4 ?
A] 3 B] 6 C] 5 D] 9
16. Rs.432 is divided amongst three workers A, B and C such that 8 times A's share is equal to 12 times B's share which is equal to 6 times C's share. How much did A get?
A] Rs.192 B] Rs.133 C] Rs.144 D] Rs.128
17. The price of Computer and CD player are in the ratio 6:5. If the computer costs Rs 5000 more than CD player. What is the price of the computer?
A] 25,000 B] 15,000 C] 50,000 D] 30,000
18. If Rs. 582 be divided into three parts, proportional to $\frac{1}{2}:\frac{2}{3}:\frac{3}{4}$, then the first part is?
A] 161 B] 151.8 C] 142 D] 153
19. The price of mixer, grinder and washing machine are in the ratio 2:3:6. After one year the price of the items are increased 20%, 15%, 25% respectively. Then what will be ratio after one year?
A] 16:23:50 B] 17:22:23 C] 16:10:22 D] 18:22:24
20. A person has 25p, 10p and 5p in the ration 2:3:4 in his pocket. If the person has Rs. 45 in all, how many 5 paisa coins are there?
A] 110 B] 100 C] 130 D] 180
21. If the sum two numbers is 31 and their product is 240, then find the absolute difference between the numbers.
A. 1 B. 3 C. 4 D. 5
22. The product of fraction A and B is $\frac{3}{49}$. Find the value of fraction A, if fraction A is thrice fraction B.
A. $\frac{7}{9}$ B. $\frac{5}{7}$ C. $\frac{3}{7}$ D. $\frac{5}{9}$
23. The sum of numerator and denominator of a fraction is 30. If 2 is added to numerator and 2 is subtracted from denominator, then it becomes $\frac{2}{3}$. Find the fraction.
A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{2}{3}$ D. $\frac{1}{4}$
24. The denominator of a fraction is 2 more than numerator. If the numerator as well as denominator is increased by 4, the fraction becomes $\frac{8}{10}$. Find the original fraction.
A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{4}{7}$ D. $\frac{2}{5}$
25. The sum of the two numbers is 50 and their product is 624. Find out the numbers.
A. 23, 27 B. 24, 26 C. 22, 28 D. 18, 32
26. The difference between a positive proper fraction and its reciprocal is $\frac{17}{72}$. The fraction is:
A. $\frac{9}{8}$ B. $\frac{7}{8}$ C. $\frac{6}{9}$ D. $\frac{7}{9}$
27. The difference between the place value and face value of 9 in the 63894531 is:
A. 89981 B. 89991 C. 88999 D. 89999
28. $\frac{3}{4}$ th of a number exceeds its $\frac{2}{3}$ rd by 8. What is the number?
A. 96 B. 36 C. 72 D. 144

29. $\frac{6}{7}$ of a certain number is 96. Find quarter of that number.

A. 112

B. 32

C. 56

D. 28

30. The difference between the number and its three fifth is 40, what is the number? A. 80 B. 100 C. 90 D.

120

LEVEL - II

1. A shopkeeper contains apples, oranges and bananas in the ratio 5:7:8. There is a demand to increase their quantity by 50%, 60% and 70% respectively. What will be ratio of the increased quantity?

A] 25:75:100

B] 26:72:112

C] 75:112:136

D] 76:100:201

2. A packet of sweets is distributed among A, B, C, D in the proportion of 6:8:5:4. If B gets 10 sweets more than D then what is A's share?

A] 16

B] 17

C] 15

D] 18

3. If A's 60% of salary is equal to two-third of B's salary. Now find the ratio of A's salary to B's salary.

A] 9:10

B] 10:9

C] 11:12

D] 13:11

4. In a mixture of 45 litres, the ratio of milk and water is 3 : 2. How much water must be added to make the ratio 9 : 11?

A] 10 liters

B] 15 liters

C] 17 liters

D] 20 liters

5. Seats of physics, Chemistry and Mathematics in a school are in the ratio 4: 5: 6. There is a proposal to increase these seats by 75 in each department. What were the total number of seats in the school finally?

A] 600

B] 750

C] 900

D] Data Inadequate

6. 60 kg of an alloy A is mixed with 100 kg of alloy B. If alloy A has lead and tin in the ratio 3 : 2 and Alloy B has tin and copper in the ration 1 : 4, then the amount of tin in the new alloy is

A] 36 kg

B] 44 kg

C] 53 kg

D] 80 kg

7. A diamond falls and breaks into three pieces whose weights are in the ration 1 : 3 : 6. The value of the diamond is proportional to the square of its weight. If the original value is Rs. 30,000. What is the loss in the value due to the breakage?

A] Rs. 13,000

B] Rs. 16,200

C] Rs. 18,600

D] Rs.19,400

8. W varies inversely as the square of t. If W = 12 when t = 2. Find t when W = 27.

A] $27t = 24$

B] $27t^2 = 48$

C] $108 = 12t^2$

D] $12t = 54$

9. Find the mean proportion to 36 and 16?

A] 24

B] 36

C] 18

D] 26

10. Find a if, $4 : a :: a : 9$
 A] ± 7 B] ± 8 C] ± 6 D] ± 9
11. If the number is decreased by 5 and divided by 7 the result is 7. What would be the result if 4 is subtracted and divided by 10?
 A. 4 B. 7 C. 8 D. 5
12. The sum of squares of three numbers is 138 and the sum of their products taken two at a time is 131. Find their sum.
 A. 35 B. 42 C. 20 D. 18
13. Find a positive number which when increased by 11 is equal to 60 times the reciprocal of the number
 A. 3 B. 4 C. 6 D. 9
14. 1ab8 is 4-digit number divisible by 24. If the number formed from the two digits ab is a multiple of 9. Then $a + b = ?$
 A. 9 B. 8 C. 12 D. 14
15. A positive integer which when added to 100, gives a sum which is greater than when it is multiplied by 100. What is the number?
 A. 5 B. 2 C. 3 D. 1
16. 36 is divided in 2 parts such that 8 times the first part added to 3 times the second part makes 203. What is the first part?
 • 15 B. 19 C. 23 D. None of these
17. A number whose one-fourth part is increased by 5 is equal to the third part diminished by 5.
 A. 100 B. 80 C. 120 D. 60
18. When 20 is subtracted from a number, it reduces to seven-twelfth of the number. What is the sum of the digits of the number?
 A. 4 B. 8 C. 10 D. 12
19. When I add 3 times my age 3 years from now to 4 times my age 4 years from now, I get 8 times my current age. How old will I be 3 years from now?
 • 28 years B. 24 years C. 20 years D. 32 years
20. Three years ago, the average age of A, B and C was 27 years and that of B and C, 5 years ago was 20 years. A's present age is?
 • 30 yrs B. 35 yrs C. 40 yrs D. 48 yrs

LEVEL – III

1. A, B and C enter into partnership with capitals in the ratio of 2 : 5 : 3. B joins a few months later than A and C joins 2 months further than B and withdraws from the business some time before the year ended. If the ratio in which profits are distributed is 8 : 10 : 3, how many months later does B join?
A] 5 months B] 4 months C] 6 months D] 7 months
2. The ratio of the amount of money with A & B is 3 : 1. If we add amount of money with C that is Rs.20 to A's amount and subtract amount of money held by D that is Rs.10 from B's amount, we get another ratio which is $\frac{1}{15}$ more than the original ratio. What is the sum of money held by A & B?
A] Rs.2441 B] Rs.3045 C] Rs.3040 D] Rs.3042
3. The ratio of marks obtained by Tom and Julia in English are 3 : 4. From the data on marks obtained, it can be inferred that Tom and Julia have got 3 and 6 marks respectively in English per 5 marks obtained in Mathematics. If the marks obtained by Julia in English are 60, then what is the sum total of marks obtained by Tom in both the subjects?
A] 75 B] 120 C] 60 D] 105
4. The monthly incomes of A and B are in the ratio 4 : 3 and the expenses are in the ratio 6 : 5. If their savings are Rs.5000 and Rs.3000 respectively, find their respective monthly incomes.
A] Rs.14000, Rs.10500 B] Rs.24000, Rs.18000
C] Rs.16000, Rs.12000 D] None of these
5. Rohit and Mohit started a business in partnership. They earned a profit of Rs.40000 as a whole at the end of first year. If Rohit took two-fifths of the share of the profit and his investment was Rs.50000, find the investment done by Mohit.
A] Rs.40000 B] Rs.60000 C] Rs.75000 D] Rs.100000
6. Ram, Shyam and Mohan start a business by investing Rs.12000, Rs.16000 and Rs.10000 respectively. Ram is a working partner and gets one-fifth of the total profit for his services while the remaining profit is divided amongst the three in proportion to their investments. If Ram gets Rs.2000 for his services, then what are the shares of profit of Ram, Shyam and Mohan respectively?
A] Rs.4526, Rs.3369, Rs.2105 B] Rs.2526, Rs.3369, Rs.2105
C] Rs.3158, Rs.4210, Rs.2632 D] None of these
7. Rs.9700 has been divided among X, Y and Z such that if their shares are reduced by Rs.30, Rs.20 and Rs.50 the balance is in the ratio of 3 : 4 : 5. What is Y's share?
A] Rs.3180 B] Rs.3220 C] Rs.3253.33 D] Rs.3200
8. On a certain day, the ratio of the passenger in the 1st class and the second class travelling by train is 1:3. The ratio of the fares collected from each first class and second class passengers is 30:1. If the total amount collected from all the passengers is Rs 1,320. Find the amount in Rs, collected from the second class passengers.
A] 240 B] 360 C] 480 D] 120

9. By mistake, instead of dividing Rs 117 among three persons P, Q and R in the ratio $(\frac{1}{2}, \frac{1}{3}, \frac{1}{4})$, it was divided in the ratio 2:3:4. Who gains the most and how much?
 A] P, Rs 28 B] Q, Rs 35 C] R, Rs 25 D] R, Rs 27
10. Three dogs are roaming in a zoo in such a way that when dog A takes 5 steps, B takes 6 steps and C takes 7 steps. But the 6 steps of A are equal to the 7 steps of B and 8 steps of C. what is the ratio of their speeds?
 A] 140 : 144 : 147 B] 40 : 44 : 47 C] 15 : 21 : 28 D] 252: 245: 240
11. A number when divided by 9 leaves a remainder 5. When the square of the number is divided by 9, the remainder is:
 A. 4 B. 5 C. 6 D. 7
12. In a two digit prime number, if 18 is added, we get another prime number with reversed digits. How many such numbers are possible?
 A. 4 B. 3 C. 2 D. 5
13. A 3-digit number $4a5$ is added to another 3-digit number 675 to give a 4-digit number $11b0$, which is divisible by 3. How many possible combinations for a, b can be possible?
 A. 3 B. 4 C. 2 D. 1
14. A number consists of 3 digits whose sum is 10. The middle digit is equal to the sum of the other two and the number will be increased by 99 if its digits are reversed. The number is:
 A. 145 B. 253 C. 370 D. 352
15. The product of two numbers is 9375 and the quotient, when the larger one is divided by the smaller, is 15. The sum of the numbers is:
 A. 380 B. 395 C. 400 D. 425
16. The difference between a two-digit number and the number obtained by interchanging the digits is 36. What is the difference between the sum and the difference of the digits of the number if the ratio between the digits of the number is 1 : 2 ?
 A. 4 B. 8 C. 16 D. None of these
17. The difference between a two-digit number and the number obtained by interchanging the two digits is 63. Which is the smaller of the two numbers?
 • 12 B. 15 C. 17 D. None of these
18. Lucia is a wonderful grandmother. Her age is b/w 50 to 70. Each of her sons having as many sons as they have brothers. Their combined number gives Lucia's present age. What is her age?
 A. 85 B. 55 C. 84 D. 64

19. A person is 80 years old in 490 BC and only 70 years old in 500 BC. In which year is he born?

- A. 400 BC B. 550 BC C. 570 BC D. 440 BC

20. A man spent $\frac{1}{6}$ th of his life in child hood, $\frac{1}{12}$ th of his life as youngster and $\frac{1}{7}$ th of his life as abachelor. After five years of his marriage a son was born to him. The son died four years before the father died and at the time of his death his age was half the total age of his father. What is the age of the father?

- A. 84 B. 48 C. 72 D. 64

MIXTURE AND ALLIGATION

Mixtures

Mixing of two or more qualities of things produces a mixture. When two items of different qualities are thus mixed, the quality of the resultant mixture lies in between the qualities of the original constituent items, i.e. it will be higher than the lowest quality and lower than the highest quality of the items being mixed.

Here, the average quality is essentially the weighted average of the two constituent items. If q_1 is the quality (or number of items) of one particular item of quality p_1 and q_2 be the quantity (or number of items) of the second item of quality p_2 are mixed together to give a new mixture, then the weighted average value (p) of the quality of the mixture is given by

$$p = \frac{p_1q_1 + p_2q_2}{q_1 + q_2}$$

A mixture can also be a solution – that is, a liquid mixed with another liquid which is normally water. The concentration of the solution is expressed as the proportion (or percentage) of the liquid in the total solution.

For example, if 10 litres of pure alcohol is mixed with 40 litres of water, then in a total solution of 50 litres. There is 10 litres of alcohol. Hence the concentration of this solution is $0.2 (= 10/50)$ or 20%.

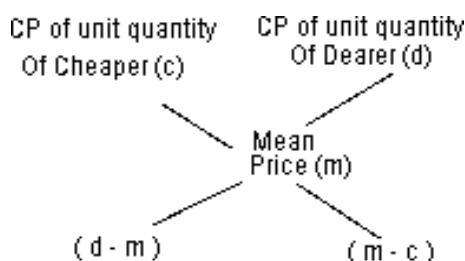
Alligation

It is the rule that enables us to find the ratio in which two or more ingredients at the given price must be mixed to produce a mixture of a desired price.

Mean price: The cost price of a unit quantity of the mixture is called the mean price.

Rule of Alligation: If gradients are mixed in a ratio then we can write

$$\frac{\text{Quantity of cheaper}}{\text{Quantity of dearer}} = \frac{\text{CP of dearer} - \text{Mean Price}}{\text{Mean Price} - \text{CP of cheaper}}$$



$$\therefore (\text{Cheaper Quantity}) : (\text{Dearer Quantity}) = (d - m) : (m - c).$$

Concept of Replacement:

Suppose a container contains a solution from which some quantity of solution is taken out and replaced with one of the ingredients. This process is repeated n times then,

Final Amount of ingredient that is not replaced =

$$\text{Initial Amount} \times (\text{Vol. after removal} / \text{Vol. after replacing})^n$$

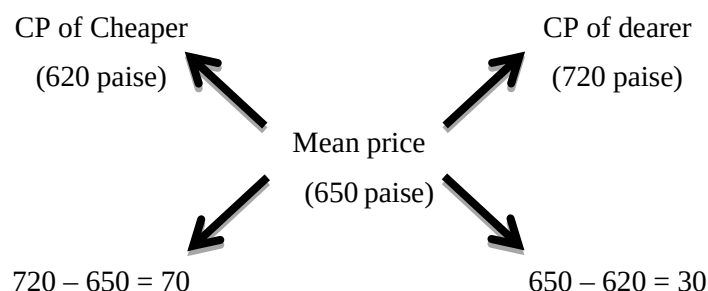
We can also write it as;

Suppose a container contains x units liquid from which y units are taken out and replaced by water.

After n operations, the quantity of pure liquid = $\left[x \left(1 - \frac{y}{x} \right)^n \right]$ units.

Example 1: In what proportion must wheat at R 6.20 per kg must be mixed with wheat at R 7.20 per kg so that the mixture is worth R 6.50?

Solution: According to the rule of mixture/alligation,



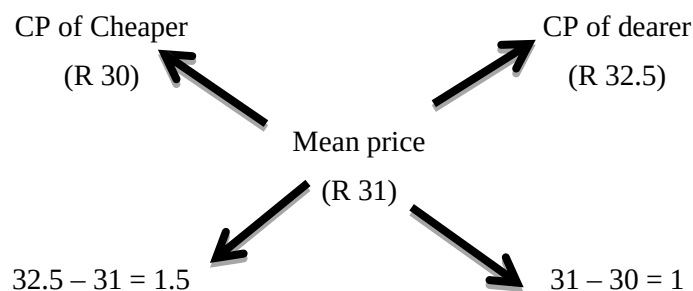
∴ Required ratio = 70 : 30 = 7 : 3

Example 2: In what ratio must a grocer mix rice worth Rs. 30 per kg and Rs. 32.5 per kg so that by selling the mixture at Rs. 34.10 per kg, he may gain 10%?

Solution: SP of 1 kg mixture = R 34.10, gain = 10%

$$\text{CP of 1 kg of mixture} = \frac{100}{110} \times 34.10 = \text{R } 31$$

According to the rule of mixture/alligation,



∴ Required ratio = 1.5 : 1 = 3 : 2

Example 3: Two vessels contain mixtures of milk and water in the ratio of 4:9 in the first vessel and in the ratio of 2:7 in the second. In what ration should the contents of these two vessels be mixed such that the resultant mixture has milk and water in the ratio of 2:5?

Solution. Here, we can apply allegation rule taking the concentration of the mixtures, the concentration of the milk in the first vessel as $\frac{4}{13}$ and that in second is $\frac{2}{9}$.

$$\begin{array}{ccc}
 4/13 & & 2/9 \\
 & \searrow \quad \swarrow & \\
 & 2/7 & \\
 & \swarrow \quad \searrow & \\
 2/7 - 2/9 = 4/63 & & 4/13 - 2/7 = 2/91
 \end{array}$$

The ratio in which the 2 mixtures should be mixed is $4/63 : 2/91$
 $= 4/9 : 2/13 = 52 : 18 = 26 : 9$

Example 4: A vessel has 300 ml of pure milk. Thirty millilitres of milk is removed and 30 ml of water is poured into the vessel (bringing the volume of mixture in the vessel back to 300 ml). If this operation is repeated another 2 times. What is the percentage of milk in the vessel at the end?

Solution: To solve this problem, we can look at a formula (to be remembered by the students) which will greatly simplify the solution.

In this problem,
 $P = 300$ ml, $Q = 30$ ml and $n = 3$
 Hence concentration of milk
 $= \left(\frac{300 - 30}{300} \right)^3 = \left(\frac{270}{300} \right)^3 = \left(\frac{9}{10} \right)^3 = 0.729$

$\therefore$ Percentage of milk is 72.9% volume.

Example 5: In what proportion must water be mixed with milk to gain 20 % by selling it at cost price?

Solution: Let cost price of milk be Rs 1 per liter, then S.P of mixture is also Rs 1 per liter

Now CP of mixture be $= 1 - (20 \% \text{ of Rs } 1) = 1 - (20 * 1 / 100) = 80 / 100 = \text{Rs } 4 / 5$

$$\begin{array}{ccc}
 1 & & 0 \\
 & \searrow \quad \swarrow & \\
 & 4/5 & \\
 & \swarrow \quad \searrow & \\
 (4/5) - 0 & & 1 - (4/5)
 \end{array}$$

Required ratio = 1 : 4

LEVEL – I

1. In what ratio must rice at Rs. 9.30 per kg be mixed with rice at Rs. 10.80 per kg so that the mixture be worth Rs10 per kg?
A] 8:7 B] 7:9 C] 5:9 D] 7:5
2. How much water must be added to 60 litres of milk at $\frac{3}{2}$ litres for Rs. 20 so as to have a mixture worth Rs. $\frac{32}{3}$ a litres?
A] 12 L B] 15 L C] 18 L D] 20 L
3. How many kg of wheat costing R8 must be mixed with 36 kg of rice costing Rs. 5.40 per kg so that 20% gain may be obtained by selling the mixture at Rs. 7.20 per kg?
A] 10kg B] 12kg C] 9.8kg D] 10.8kg
4. The milk & water in two vessels A & B are in the ratio 4:3 & 2:3 resp. In what ratio, the liquids in both the vessels are mixed to obtain a new mixture in vessel C containing half milk & half water?
A] 7:5 B] 7:9 C] 5:9 D] 3:7
5. In what ratio must a grocer mix two varieties of pulses costing Rs.15 & Rs. 20 per kg respectively so as to get a mixture worth Rs. 16.50 per kg?
A] 3:7 B] 5:7 C] 7:3 D]7:5
6. Find the ratio in which rice at Rs. 7.20 a kg be mixed with the rice at Rs. 5.70 a kg to produce a mixture worth Rs. 6.30 a kg
A] 1:3 B] 3:2 C] 3:4 D] 4:5
7. The cost of A chocolate is Rs. 14 per kg & B chocolate is Rs. 20 per kg. If both A & B are mixed in the ratio 2:3, then the price per kg of the mixed variety of chocolate is:
A] Rs. 17.6 B] Rs. 18.50 C] Rs. 19 D]Rs. 19.50
8. In what ratio must a grocer mix two varieties of tea worth Rs. 60 a kg & Rs. 65 a kg so that by selling a mixture at Rs. 68.20 a kg, he may gain 10%?
A] 3:2 B] 3:4 C] 3:5 D] 4:5
9. Two vessels A & B contain spirit & water mixed in the ratio 5:2 & 7:6 respectively. Find the ratio in which these mixture be mixed to obtain a new mixture in vessel C containing spirit & water in the ratio 8:5?
A] 4:3 B] 3:4 C] 5:6 D] 7:9
10. A container contains 40L of milk. From this container 4L of milk was taken out & replaced by water. This process was repeated further 2 times. How much milk is now contained in the container?
A] 26.34L B] 27.36L C] 28L D] 29.16L
11. How many kg of brand A coffee must be mixed with 126 kg of brand B coffee such that the cost of the mixture is Rs. 24 per kg.? The per kg prices of A and B are Rs. 36.60 and Rs. 17.10 respectively.
A] 42 kg B] 23 kg C] 69 kg D] 63.5 kg
12. 16 litres of draught beer contains 216 calories and 16 litres of Kingfisher contains 174 calories. If an 8 litre mixture of both the beers contains 101 calories, the fraction of Kingfisher beer in the mixture is
A] $\frac{2}{3}$ B] $\frac{1}{2}$ C] $\frac{1}{3}$ D] $\frac{1}{4}$

13. A merchant purchased two qualities of pulses at the rate of Rs. 200 per quintal and Rs. 260 per quintal. In 52 quintals of the second quality, how much pulse of the first quality should be mixed so that by selling the resulting mixture at Rs. 300 per quintal, he makes a profit of 25%?
A] 100 quintals B] 104 quintals C] 26 quintals D] None of these
14. Ravi covered a distance of 300 km in 20 hours in 2 parts. He covered the first part by a car which was driven at a speed of 20 km/hr and the second part by an auto rickshaw that was driven at a speed of 12 km/hr. The ratio of the distances covered by Ravi in the two parts is:
A] 1 : 1 B] 2 : 3 C] 5 : 3 D] 3 : 5
15. Two vessels contain a mixture of diesel and kerosene. In the first vessel, the ratio of diesel to kerosene is 8 : 3 and in the second vessel the ratio is 5 : 1. A 35 litre drum is filled from these vessels so as to contain a mixture of diesel and kerosene in the ratio 4 : 1. How many liters are taken from the second vessel?
A] 11 liters B] 35 liters C] 24 liters D] 17.5 liters
16. Alloy 'CuZ' has Cu : Zn :: 4 : 1 while alloy 'ZiC' has Cu : Zn :: 1 : 3. One melting 10 kg of CuZ with 16 kg of ZiC and some amount of pure Cu, a new alloy 'Cuzec' with Cu : Zn :: 3 : 2 is obtained. The weight of Cuzec is
A] 9 kg B] 35 kg C] 30 kg D] None of these
17. How many kgs of Basmati rice costing Rs. 42/kg should a shopkeeper mix with 25 kgs of ordinary rice costing Rs. 24 per kg so that he makes a profit of 25% on selling the mixture at Rs. 40/kg?
A] 20 kgs B] 12.5 kgs C] 16 kgs D] 200 kgs
18. From 40 liters of 2: 3 milk-water solutions, 5 liters are withdrawn and replaced by 5 liters of water. Now, 8 liters are withdrawn and replaced by 8 liters of water. Finally, 10 liters are withdrawn and replaced by 10 liters of water. What is the ratio of water to milk in the solution now?
A] 21: 79 B] 79 : 21 C] 4 : 9 D] 1 : 4
19. 18 liters are drawn from a bucket full of pure milk, and it is then completely filled with water. 18 litres of the mixture are again drawn and the bucket is again filled completely with water. The ratio of the volume of water now left in the bucket to that of the milk in it is 9 : 16. What is the total capacity (in litres of the bucket)?
A] 80 B] 45 C] 90 D] 18
20. From 90 liters of 4: 5 milk-water mixtures, 15 liters are taken out and replaced by same amount of water. This process is repeated once more. Now, 9 liters of the mixture is taken out and replaced by same amount of milk. The amount of water at the end of third cycle is how much percent of the amount of water at the end of second cycle?
A] 10% B] 90% C] 201.6% D] 49.6%

LEVEL – II

1. From a cask of milk containing 30 litres, 6 litres are drawn out and the cask is filled up with water. If the same process is repeated a second, then a third time, what will be the number of litres of milk left in the cask?
A] 0.512 liter B] 12 liters C] 14.38 liters D] 15.36 liters

2. There are some shepherds and their sheep in a grazing field. The no. of total heads are 60 and total legs are 168 including both men and sheep. The no. of sheep is
A] 18 B] 26 C] 24 D] 36
3. Gold is 19 times as heavy as water and copper 9 times. In what ratio should these metals be mixed so that the mixture may be 15 times as heavy as water?
A] 1 : 2 B] 3 : 2 C] 2 : 3 D] 4 : 5
4. In a class of 30 students, the average weight of boys is 20 kg and the average weight of the girls is 25 kg. The fraction of boys out of the total students of the class is
A] $\frac{4}{5}$ B] $\frac{5}{6}$ C] $\frac{3}{4}$ D] Data insufficient
5. A jar full of whisky contains 40% alcohol. A part of this whisky is replaced by another containing 19% alcohol and now the percentage of alcohol was found to be 26%. The quantity of whisky replaced is:
A] $\frac{1}{3}$ B] $\frac{2}{3}$ C] $\frac{2}{5}$ D] $\frac{3}{5}$
6. A dishonest grocer professes to sell pure butter at cost price, but he mixes it with adulterated fat and thereby gains 25%. Find the percentage of adulterated fat in the mixture assuming that adulterated fat is freely available.
A] 20% B] 25% C] 33.33% D] 40%
7. A housewife has 1 litre of solution that contains milk and water in the ratio 3:1. She adds 250 ml of 3:2 solutions of milk and water to it and then uses 250 ml of the combined mixture to make curd. How much of pure milk is she left with?
A] 1,000 ml B] 912.5 ml C] 750 ml D] 720 ml
8. A jar full of milk contains 30% water. A part of this milk is replaced by containing 20% water and now the percentage of water is found to be 26%. The quantity of milk replaced is
A] $\frac{2}{5}$ B] $\frac{1}{5}$ C] $\frac{3}{7}$ D] $\frac{4}{7}$
9. The amount of water (in ml) that should be added to reduce 9 ml lotion, containing 50% alcohol, to a lotion containing 30% alcohol, is
A] 5 ml B] 4 ml C] 3 ml D] 6 ml
10. Three vessels contain equal mixtures of milk and water in the ratio 6:1, 5:2 and 3:1 respectively. If all the solutions are mixed together, the ratio of milk to water in the final mixture will be
A] 64:65 B] 65:64 C] 19:65 D] 65:19
11. Gold is 20 times as heavy as water and copper is 10 times as heavy as water. In what ratio these two metals be mixed so that the alloy is 13 times as heavy as water is
A] 2:7 B] 3:7 C] 1:2 D] 2:3
12. The average age of boys in class is 16.66, while the average age of girls is 18.75. Thus the average age of all the 40 students of the class is 17.5. If the difference between the no. of boys and girls is 8, then the no. of girls in the class is:
A] 12 B] 16 C] 18 D] Data insufficient

13. A vessel contains 125 litres of wine. 25 litres of wine was taken out of the vessel and replaced by water. Then, 25 litres of mixture was withdrawn and again replaced by water. The operation was repeated for third time. How much wine is now left in the vessel?
A] 64 Litres. B] 41 litres C] 52 litres D] 45 litres
14. Three equal glasses are filled with mixture of milk and water. The proportion of milk and water in each glass is as follows: In the first glass as 3 : 1, in the second glass as 5 : 3 and in the third as 9 : 7. The contents of the three glasses are emptied into a single vessel. What is the proportion of milk and water in it?
A] 31 : 17 B] 32 : 21 C] 45 : 11 D] 11 : 25
15. Three glasses of sizes 3 litres, 4 litres and 5 litres contain mixture of milk and water in the ratio 2: 3, 3: 7 and 4: 11, respectively. The contents of all the three glasses are poured into a single vessel. Find the ratio of milk.
A] 14: 31 B] 15: 21 C] 16: 17 D] 18: 19
16. In a mixture of 45 litres, the ratio of milk and water is 3 : 2, How much water must be added to make the ratio 9 : 11?
A] 10 litres B] 15 litres C] 17 litres D] 20 litres
17. A bartender stole champagne from a bottle that contained 50% of spirit and he replaced what he had stolen with champagne having 20% spirit. The bottle then contained only 25% spirit. How much of the bottle did he steal?
A] 80% B] 83.33% C] 85.71% D] 88.88%
18. In three vessels each of 10 litres capacity, mixture of milk and water is filled. The ratios of milk and water are 2 : 1, 3 : 1 and 3 : 2 in the three respective vessels. If all the three vessels are emptied into a single large vessel, find the proportion of milk and water in the mixture.
A] 181 : 49 B] 101 : 49 C] 121 : 59 D] 131 : 69
19. A vendor mixes 12 kg of type I, 10 kg of type II and 14 kg of type III coffee whose costs are Rs.75 per kg, Rs.80 per kg and Rs.90 per kg respectively. He, then, sells the resultant mixture at Rs.85 per kg. Find his profit or loss per kg of coffee.
A] Rs.2.78 profit B] Rs.3.5 profit C] Rs.2.8 loss D] Rs.4.2 profit
20. Two tanks of equal volume contain chemical A & B. The first tank contains half as much A as B. The second tank contains one-fourth as much chemical B as A. The mixtures of the two tanks are mixed in the third tank in the ratio 3 : 5. What is the ratio of A : B finally?
A] 4 : 1 B] 5 : 3 C] 4 : 3 D] 1 : 5

LEVEL – III

1. Ian Holm mixes two varieties of rice of price Rs. 36 per kg and Rs. 24 per kg in the ratio of 1: 2, respectively. He sold the mixture for Rs. 2100 to earn a profit of 25%, and then find the quantity of rice (in Kg) of cost Rs. 24 per kg used in the mixture. (First Naukri)
(a) 20 (b) 40 (c) 45 (d) None of these
2. Cary Elwes mixes 45 liters of 80% pure milk with 60% pure milk and sells the mixture at the cost price of pure milk making a profit of 33.33%. What is the quantity of 60% pure milk (in litres) he mixed? (First Naukri)
(a) 14 (b) 15 (c) 16 (d) 17

3. Jeffrey Wright has 2 types of mixture of tea with him. In the 56 kg of the first mixture the ratio of tea to impurity is 5 : 2 and in 44 kg of the second mixture the ratio of tea to impurity is 3 : 1. If he mixes these two mixtures with 17 kg of pure tea in a large container, then find the ratio of tea to impurity in the large container. (First Naukri)
(a) 2:3 (b) 1:9 (c) 16:9 (d) 10:3
4. Steve Coogan was having drinks where he was mixing Rum and water, the concentration of Rum is 56%. 200 ml of this mixture was mixed with another mixture P of Rum and water, to get 300 ml mixture of Rum and water. The concentration of rum in the new mixture becomes 40%. What should be the concentration of Water in the mixture P?(First Naukri)
(a) 72% (b) 75% (c) 76% (d) 92%
5. Cillian Murphy has a mixture containing milk and water in the ratio 7 : 4. If 22 litres of mixture is replaced with water the ratio of milk and water becomes 5 : 6. Find the quantity of milk (litres) in the original mixture. (First Naukri)
(a) 15 (b) 16 (c) 45 (d) 49
6. From a 20 lit Jug, if 5 lit of water is taken out and the same amount of Lime soda is poured into it, Find the ratio of Water and Soda. (HCL)
(a) 5:2 (b) 3:1 (c) 3:2 (d) 4:1
7. A fixed amount of water is mixed with Sugar costing Rs 14 per liter. The rate of mixture is Rs 9 per liter. Find out the proportion of water and Sugar in the new combination. (HCL)
(a) 7:8 (b) 5:9 (c) 5:11 (d) 5:12
8. A pot contains 10 liters of Mint juice. 3 liters of Mint juice is replaced with Honey. This is done two times more. Find out the quantity of Mint juice which is in the pot now? (HCL)
(a) 2.6 (b) 2.7 (c) 3.43 (d) 4.24
9. A vessel contains 50 liters of lemonade. Firstly, 10 liters of lemonade was replaced by water. Further this is done 2 more times. How much lemonade is the vessel left with? (HCL)
(a) 24.6 (b) 25.6 (c) 26.6 (d) 27.6
10. Ajar has 210 lit of limewater solution. Present ratio of lime and water is 4:3. Find the amount of lime needed to make the ratio. (HCL)
(a) 50 ltr (b) 51 ltr (c) 52 ltr (d) 60 ltr
11. How many litres of a 90% of concentrated acid needs to be mixed with a 75% solution of concentrated acid to get a 30 litre solution of 78% concentrated acid? (TCS)
(a) 6 (b) 7 (c) 8 (d) 9
12. Two alloys A and B are composed of two basic elements. The ratios of the compositions of the two basic elements in the two alloys are 5:3 and 1:2, respectively. A new alloy X is formed by mixing the two alloys A and B in the ratio 4:3. What is the ratio of the composition of the two basic elements in alloy X? (TCS)
(a) 1:1 (b) 2:3 (c) 5:2 (d) 4:3
13. Mixture of milk and water has been kept in two separate containers. ratio of milk to water in one of the containers is 5:1 and that in the other container is 7:2. in what ratio the mixtures of these two containers should be added together so that the quantity of the milk in the new mixture may become 80%? (TCS)
(a) 7:5 (b) 2:3 (c) 3:2 (d) 2:1
14. A mixture of a certain quantity of milk with 16 litres of water is worth 90p per litre. if pure milk be worth Rs.1.08 per litre, how much milk is there in the mixture? (TCS)
(a) 75 (b) 54 (c) 60 (d) 80
15. A beaker contains 180 litres of alcohol. On day 1, 60 litres of alcohol is replaced with water. On 2nd and 3rd days 60 litres of the mixture in the beaker is replaced with water. What will be the quantity of alcohol in the beaker after 3rd day? (TCS)
(a) 41.33 (b) 53.33 (c) 61.33 (d) 80.33

PERMUTATION AND COMBINATION

These two words permutation and combination, at the initial level are very confusing and are generally used interchangeably. So let's take them one by one and understand them.

Combination

Combination means from the given certain objects (may be alike or different) selecting one or more objects. Combination can also be replaced by the words – selection, collection or committee.

For Example: Combination of top 5 cricket players from the team of 11 players is the selection of 5 players (in any order).

The sequence in which they have to be selected is not important here. Also we can say that the order of selection is not the concern in the case of combination

Permutation

Permutation means arrangement of the alike or different objects taken some or all at a time. So we can observe the word 'arrangement' used in the definition of permutation. Here the arrangement means selection as well as ordering. That means the order in which the objects are selected have also been taken care of in this case.

For Example: The number of 5 digit numbers which can be formed using the digits 0, 1, 2, 3, 4 and 5. In this example, we just not have to select the 5 digits out of given 6 digits but also have to see the number of possible cases for the different arrangement. So the numbers 34251, 21034, 42351 are all different cases.

Factorial

In Mathematics, the factorial is represented by the symbol '!' i.e. if we have to write 5 factorial, so it will be written as 5! So in general factorial of any positive number n will be represented by n!

Mathematically,

$$n! = \begin{cases} 1 & \text{if } n = 0 \\ (n-1)! \times n & \text{if } n > 0 \end{cases} \quad \text{where n is any positive integer.}$$

Similarly we can say for any positive integer 'n'

$$n! = n \times (n-1) \times (n-2) \times \dots \times 3 \times 2 \times 1.$$

i.e. the product of all the positive integers less than or equal to n.

Just see below for the factorial of few frequently used numbers.

$$0! = 1$$

$$1! = 1$$

$$2! = 2 \times 1 = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120 \text{ and so on.}$$

Difference between Permutation and Combination

1. The very basic difference in permutation and combination is the order of the objects considered. In combination, the order is not considered at all while for permutation it is must. So the permutation is the ordered arrangement while the combination is the unordered selection.

From the three alphabets A, B and C, the permutation of these 3 letters will be ABC, ACB, BAC, BCA, CBA and CAB. While the combination of 3 letters will be just (A, B, C).

2. Permutation gives the answer to the number of arrangements while the combination explains the possible number of selections.

3. Permutation of a single combination can be multiple but the combination of a single permutation is unique (considering all at a time).

Fundamental Principles of Counting

Multiplication Theorem

If an operation can be performed in m different ways and following which a second operation can be performed in n different ways, then the two operations in succession can be performed in $m \times n$ different ways.

Example 1: In a class of 5 girls and 4 boys, the teacher has to select 1 girl AND 1 boy. In how many ways can she make her selection?

Solution: Here the teacher has to choose the pair of a girl AND a boy
For selecting a boy she has 4 options/ways AND that for a girl 5 options/ways
For 1st boy ----- any one of the 5 girls ----- 5 ways
For 2nd boy ----- any one of the 5 girls ----- 5 ways....
For 4th boy ----- any one of the 5 girls ----- 5 ways
Total number of ways $5 + 5 + 5 + 5 = 20$ ways OR $5 \times 4 = 20$ ways.

Addition Theorem

If an operation can be performed in m different ways and a second independent operation can be performed in n different ways, either of the two operations can be performed in $(m + n)$ ways.

Example 2: In a class of 5 girls and 4 boys, the teacher has to select either a girl OR a boy. In how many ways can he make his selection?

Solution: Here the teacher has to choose either a girl OR a boy (Only 1 student)
For selecting a boy he has 4 options/ways OR that for a girl 5 options/ways.
The first of these can be performed in 4 ways and the second in 5 ways.
Therefore, by fundamental principle of addition either of the two jobs can be performed in $(4 + 5)$ ways.
Hence, the teacher can make the selection of a student in 9 ways.

Permutations (Arrangement)

The different arrangements which can be made by taking some or all of the given things or objects at a time is called Permutation. In permutations the order of arrangement is taken into account; When the order is changed, a different permutation is obtained.

Eg. A number of permutations of three elements a, b, c by taking three at a time will be $abc, acb, bac, bca, cab, cba$.

The formula of calculating number of permutations is ${}^n P_r$, i.e. number of all permutations all n distinct things taken r at a time ($0 \leq r \leq n$) = ${}^n P_r = \frac{n!}{(n-r)!}$

A permutation of n taken r at a time is defined as an ordered selection of r out of the n items. The total number of all the possible permutations is denoted as:

$n P_r = n! (n-1)(n-2) \dots (n-r+1)!$ Where $n \geq r$

Note: This is valid only when repetition is not allowed.

Example 3: In how many different ways would you arrange 5 persons on 3 chairs?

Solution: Here $n = 5$ and $r = 3$

Arranging 5 persons on 3 chairs is same as filling 3 places when we have 5 different things at our disposal. The first place can be filled in 5 ways (by Fundamental Principle of Multiplication).

After filling it, there are 4 things left and any one of these 4 things can be used to fill second place. So the second place can be filled in 4 ways.

Hence by fundamental principle of multiplication, the first two places can be filled in 5×4 ways.

Now, there are 3 things left, so that the third place be filled in 3 ways
So the total number of arrangements will be $5 * 4 * 3$.

As per the formula of permutation it will be ${}^5P_3 = 5! / (5-3)! = 5! / 2! = 5 * 4 * 3$

Rule 1: Continuing in this manner we can say that number of permutations (or arrangements) of n things taken all at a time will be n!

e.g. In how many ways can 6 persons stand in a queue? Here $n = r = 6$ so total number of permutations will be $6! = 720$

Example 4: How many four digit numbers are there with distinct digits $n = 10$ i.e. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 and $r = 4$

Solution: Total Number of arrangements = $10! / (10-4)!$

But these arrangements also include those numbers which have zero (0) at thousand's place.

Such numbers are not four digit numbers and hence need to be excluded.

When 0 is fixed at thousand's place, we have to arrange remaining 9 digits by taking 3 at a time in a way $9! / (9-3)!$

Hence total number of four digit numbers = $10! / (10-4)! - 9! / (9-3)! = 5040 - 504 = 4536$

Example 5: There are 6 periods in each working day of a school. In how many ways can one arrange 5 subjects such that each subject is allowed at least one period?

Solution: 5 periods can be arranged in 6 periods in 6P_5 ways.

Now one period is left and it can be allotted to any one of the 5 subjects. So number of ways in which remaining one period can be arranged is 5.

Total Number of arrangements = ${}^6P_5 * 5 = 3600$

Permutations under Certain Conditions

1. Permutations of n different objects taken r at a time, when a particular object is to be always included in each arrangement is $r \cdot {}^{(n-1)}P_{(r-1)}$

2. Permutations of n different objects taken r at a time, when a particular object is never taken in each arrangement is ${}^{(n-1)}P_r$

Example 6: Make all arrangement of letters of the word PENCIL so that

i) N is always next to E

ii) N and E are always together.

Solution: i) Let's keep EN together and consider it one letter.

Now we have 5 letters which can be arranged in a row in ${}^5P_5 = 5! = 120$ ways.

ii) Solve as above. Just keep in mind that now E and N can interchange their places in $2!$ ways. So total arrangements = $5! * 2! = 240$

Rule 2: Permutation of n different things taken r at a time when repetition is allowed
 $n \times n \times n \times \dots r \text{ times} = n^r \text{ ways}$

Rule 3: The number of mutually distinguishable permutations of n things, taken all at a time, of which p are alike of one kind, q are alike of second such that $p + q = n$, is $n! / (p! \times q!)$

Rule 4: The total number of permutations of n things, of which p are alike of one kind, q are alike of second kind and remaining all are distinct, is $n! / (p! \times q!)$

Example 7: How many words would you form with the letters of the word MISSISSIPPI ?

Solution: Total letters = 11, S = 4 times, I = 4 times, P = 2 times
 So, total number of permutations = $11! / (4! \times 4! \times 2!) = 34650$

Rule 5: Sum of all the numbers which can be formed by using the n digits without repetition is: $(n-1)! \times (\text{sum of the digits}) \times (111\dots n \text{ times})$.

Rule 6: Sum of all the numbers which can be formed by using the n digits (repetition being allowed) is: $n^{n-1} \times (\text{sum of the digits}) \times (111\dots n \text{ times})$.

Example 8: Find the sum of all the numbers that can be formed with the digits 2, 3, 4, and 5 taken all at a time.

(i) If repetition is not allowed

(ii) If repetition is allowed

Solution: i) (sum of digits) $(n-1)!$ (1111... n times)
 $= (2 + 3 + 4 + 5) (4-1)! (1111) = 93324$
 ii) $n^{n-1} \times (\text{sum of the digits}) \times (111\dots n \text{ times})$
 $= 4^4 \times (2 + 3 + 4 + 5) \times 1111 = 896 \times 1111 = 995456$

Circular permutations

If the objects are arranged in a circular manner this distinguished ordering no longer exists, that is, there is no "first element" in the arrangement, any element can be considered as the start of the arrangement. The arrangements of objects in a circular manner are called circular permutations.

The number of circular permutations of a set S with n elements is $(n - 1)!$

The basic difference between linear permutations and circulations is that every linear arrangement has a beginning and an end, but there is nothing like beginning or end in a circular permutation. Thus, in circular permutations, we consider one object as fixed and the remaining objects are arranged as in case of linear arrangement.

Rule 7: There are certain arrangements in which clockwise and anticlockwise arrangements are not distinct. e.g. arrangements of beads in a necklace, arrangement of flowers in a garland etc.

In such cases number of circular permutations of n distinct objects is $1/2 \times [(n-1)!]$

Example 9: i) Arrange 6 persons around a circular table.

ii) In how many of these arrangements will two particular persons be next to each other?

Solution: i. $(6-1)! = 5! = 120$
 ii. Consider two particular persons as one person. We have 5 persons in all. These 5 persons can be seated around a circular table in $(5-1)! = 4!$ Ways. But two Particular persons can be arranged between themselves in $2!$ Ways
 So Total number or arrangements = $4! \times 2! = 48$

Combinations (Selection)

Each of the different selections or groups which are made by taking some or all of a number of things or objects at a time is called combination. ${}^nC_r = n! / \{r! (n-r)!\}$

Suppose there are three objects namely x, y, and z. Now we are asked to calculate the combinations (selections) of these objects taking 2 at a time.

x y, y z, x z = Total 3 Combinations.

Note the important difference here. In later case we did not differ between x y and y x, as we did in the first case. This is the only difference between Permutations and Combinations.

In Combinations we find different ways of choosing r objects from n given objects while in Permutations we find different ways of choosing r objects from n given objects and ways of arranging these r objects.

The formula of permutations (nP_r) itself says first selection (nC_r) and then arrangement (r!) i.e.

$${}^nP_r = {}^nC_r * r!$$

Other Important Rules:

i. If ${}^nC_x = {}^nC_y$, then either $x = y$ or $x + y = n$

ii. ${}^nC_r = {}^nC_{(n-r)}$

iii. ${}^nC_n = 1$

iv. ${}^nC_0 = 1$

v. Selection of any number of things out of n distinct things. $= 2^n$

vi. Selection of any number of things out of n identical things $= n+1$

Zero ball selected = 1 way

One ball selected = 1 way

Two balls selected = 1 way

Total ways $= (2 + 1) = 3$ ways.

vii. Number of diagonals of n sided polygon $= {}^nC_2 - n = n(n-3) / 2$

viii. Number of straight lines formed by n points of which r are collinear $= {}^nC_2 - {}^rC_2 + 1$

ix. Number of triangles formed by n points of which r are collinear $= {}^nC_3 - {}^rC_3$

x. With m parallel lines intersected by n parallel lines number of parallelograms can be formed =

$${}^mC_2 * {}^nC_2 = m n(m-1)(n-1) / 4$$

Example 10: Find the number of ways in which Asha can eat sweets out of 5 distinct sweets at a party?

Solution: Number of ways Asha can or cannot eat sweets at a party out of 5 distinct sweets available at party $= 2^5 = 32$

Corollary: When zero selections are not allowed i.e. If Asha has asked to select at least one sweet $= 2^n - 1 = 2^5 - 1 = 31$

Example 11: 3 men and 3 women are candidates for 2 vacancies. A voter has to vote for 2 candidates. In how many ways can one cast her/his vote?

Solution: In all there are 6 candidates and a voter has to vote for any 2 of them.

So he can select 2 candidates from 6 candidates in 6C_2 ways

$$= 6! / 4! * 2! = (6 * 5) / 2 = 5 * 3 = 15$$

Example 12: In how many ways can a cricket eleven be chosen out of a batch of 15 players if

Solution: i) There is no restriction on the selection $= {}^{15}C_{11}$

ii) A Particular Player is always chosen $= {}^{14}C_{10}$

iii) A Particular Player is never chosen $= {}^{14}C_{11}$

Division and Distribution

A. Distinct Objects

Case 1: Number of ways in which n distinct things can be divided into r unequal groups containing $a_1, a_2, a_3, \dots, a_r$ things (different number of things in each group and the groups are unmarked, i.e., not distinct)

$$= {}^nC_{a_1} \times {}^{(n-a_1)}C_{a_2} \times \dots \times {}^{(n-a_1-a_2-\dots-a_{r-1})}C_{a_r} \\ = n!/(a_1! a_2! a_3! \dots a_r!)$$

(Here $a_1 + a_2 + \dots + a_r = n$)

Case 2: Number of ways in which n distinct things can be distributed among r persons such that some person get a_1 things, another person get a_2 things.....and similarly someone gets a_r things (each person gets different number of things)

= Number of ways in which n distinct things can be divided into r unequal groups containing $a_1, a_2, a_3, \dots, a_r$ things (different number of objects in each group and the groups are numbered, i.e., distinct)

$$= n!/(a_1! a_2! a_3! \dots a_r!)$$

(Here $a_1 + a_2 + a_3 + \dots + a_r = n$)

Case 3: Number of ways of grouping dissimilar things

In how many ways can you divide 4 different things (say a, b, c and d) into two groups having two things each? Your answer would be to select two things out of the four and two would be left behind for selecting next 2, i.e. ${}^4C_2 \times {}^2C_2 = 6$. But are there really 6 ways?

Actually, there are only 3 ways of dividing the four things into two groups of two. When you selected two things out of the four, the things selected were ab, ac, ad, bc, bd , and cd . But the last three groups are already formed when you select the first three groups, i.e. when you select ab , you automatically get cd . When you select ac , you automatically get bd , and so on.

If ' $n \times a$ ' different things are divided into n groups of ' a ' things each, the number of ways of grouping = ${}^nC_a \times {}^{(n-a)}C_a \times {}^{(n-2a)}C_a \times \dots \times {}^2C_a = n!/(a!)^n$

Case 4: Number of ways in which $m \times n$ distinct things can be distributed equally among n persons (each person gets a number of things)

= Number of ways in which $m \times n$ distinct things can be divided equally into n groups (each group will have m things and the groups are numbered, i.e., distinct) = $(n \times a)! / (a!)^n$

B. Identical Objects

Case 1: Number of ways of Non negative integral distribution of n identical items among r persons i.e. each one of whom can receive 0,1,2 or more items ($\leq n$) = ${}^{(n+r-1)}C_{(r-1)}$

Case 2: Number of ways in which n identical things can be distributed among r persons, each one of whom can receive at least 1 item ($\leq n$) i.e. positive integral distribution = ${}^{(n-1)}C_{(r-1)}$

LEVEL – I

1. In how many ways can the letters of the word SPECIAL be arranged using all the letters?
A. 5010 B. 5020 C. 5040 D. 5080
2. In how many ways can the letters of the word SPECIAL be arranged using only 4 letters at a time?
A. 810 B. 850 C. 830 D. 840
3. How many distinguishable permutations of the letters in the word BANANA are there?
A. 720 B. 120 C. 60 D. 360
4. How many ways a 6 member team can be formed having 3 men and 3 ladies from a group of 6 men and 7 ladies?
A. 700 B. 720 C. 120 D. 500
5. The value of ${}^{75}C_2$ is:
A. 2775 B. 2315 C. 1215 D. 1675
6. What is the number of possible words that can be made using the word “EASYQUIZ” such that the vowels always come together?
A. 120 B. 720 C. 2880 D. 4320
7. What is the number of possible words that can be made using the word “QUIZ” such that the vowels never come together?
A. 8 B. 12 C. 16 D. 24
8. How many words can be made from the word “APPLE” using all the alphabets with repetition and without repetition respectively?
A. 1024, 60 B. 60, 1024 C. 1024, 1024 D. 240, 1024
9. In how many different ways can the alphabets of the word ‘SCORING’ be arranged so that the vowels always come together?
A. 120 B. 720 C. 240 D. 1440
10. In how many ways can the alphabets of the word ‘DERAIL’ be arranged so that the vowels come at the odd positions only?
A. 12 B. 18 C. 24 D. 36
11. In how many ways can an interview panel of 3 members be formed from 3 engineers, 2 psychologists and 3 managers if at least 1 engineer must be included?
A. 30 B. 15 C. 46 D. 45
12. Out of 7 boys and 4 girls, how many queues of 3 boys and 2 girls can be formed?
A. 120 B. 25200 C. 24800 D. 1440

13. A box contains 2 red coins, 3 green coins and 4 blue coins. In how many ways can 3 coins be chosen such that at least one coin is green?
A. 16 B. 32 C. 64 D. 128
14. Out of 6 engineers and 4 doctors, how many groups of 4 professionals can be formed such that at least 1 engineer is always there?
A. 129 B. 109 C. 229 D. 209
15. From a group of 9 different books, 4 books are to be selected and arranged on a shelf. How many arrangements are possible?
A. 3023 B. 3024 C. 3025 D. 3026
16. How many different combinations are possible if 10 numbers are grouped five at a time?
A. 252 B. 242 C. 232 D. 282
17. There are 3 questions in a question paper. If the questions have 4, 3 and 2 solutions respectively, find the total number of solutions.
A. 22 B. 23 C. 24 D. 28
18. In how many ways can a party of 4 men and 4 women be seated at a circular table so that no two women are adjacent?
A. 576 B. 144 C. 36 D. 16
19. There are 5 boys and 5 girls. In how many ways they can be seated in a row so that all the girls do not sit together?
A. $5! \times 5!$ B. $10! - 5! \times 5!$ C. $6! \times 5!$ D. $10! - 6! \times 5!$
20. In a party, every guest shakes hand with every other guest. If there were total of 66 handshakes in the party, find the number of persons present in the party?
A. 12 B. 33 C. 4376 D. 66

LEVEL – II

1. How many 4-digit numbers can be formed from the digits 1, 2, 3, 4, 5, 6 and 7 which are divisible by 5 when none of the digits are repeated?
A. 120 B. 35 C. 24 D. 720
2. In how many ways can 20 boys and 18 girls make a queue such that no two girls are together?
A. $20! \times {}^{20}C_{18}$ B. $20! \times {}^{20}P_{18}$ C. $20! \times {}^{21}C_{18}$ D. $20! \times {}^{21}P_{18}$
3. In how many ways can 3 prizes be distributed among 4 boys,
i) When no boy gets more than one prize?
A. 12 B. 64 C. 24 D. 56
ii) When a boy may get any number of prizes?
A. 12 B. 64 C. 24 D. 56

iii) When no boy gets all the prizes?

- A. 64 B. 63 C. 12 D. 60

4. i. How many different arrangements can be made by using all the letters in the word MATHEMATICS?

- A. $11! / (2!)^3$ B. $11! / (2!)^2$ C. $11!$ D. $11! / (2!)^3 \times 5!$

ii. How many of the words begin with I from the word MATHEMATICS?

- A. $11! / (2!)^2$ B. $10! / (2!)^3$ C. $10!$ D. $10! / (2!)^3 \times 4!$

5. In how many ways can 6 similar toffees be distributed to 4 boys, such that every boy receives at least one toffee?

- A. 10 B. 18 C. 5 D. 13

6. How many 3 digit numbers can be formed whose unit digit is a prime number? (Repetition is allowed).

- A. 500 B. 450 C. 400 D. 360

7. If there are 4 roads from A to B, 5 roads from B to C and 3 roads from C to D, then how many combinations of roads are there from A to D via B and C?

- A. 48 B. 60 C. 72 D. 24

8. There are 10 steamers plying between Kerala and Mumbai. In how many ways can a person go from Kerala to Mumbai and return by a different steamer?

- A. 100 B. 90 C. 19 D. 99

9. In how many different ways a person can choose his dress among 3 shirts, 4 jeans and 5 trousers? A dress is formed by either a shirt and a jean or a shirt and a trouser.

- A. 17 B. 60 C. 27 D. 12

10. Find the number of ways in which a six-lettered code can be formed using the English alphabets and the digits from 0 to 9 such that the first three places and the last three places are to be filled with the numerals and alphabets respectively.

- A. $9(10)^2(26)^3$ B. $(234)^3$ C. $(260)^3$ D. $(36)^6$

11. The letter of the word LABOUR are permuted in all possible ways and the words thus formed are arranged as in a dictionary. What is the rank of the word LABOUR?

- A. 275 B. 251 C. 240 D. 242

12. A standard deck of cards consists of 4 suits (spades, hearts, diamonds, and clubs) of 13 cards each. In how many ways can 5-card hands be dealt that include 3 diamonds and 2 cards from other suits?

- A. 211906 B. 211916 C. 211926 D. 211936

13. The Indian Cricket team consists of 16 players. It includes 2 wicket keepers and 5 bowlers. In how many ways can a cricket eleven be selected if we have to select 1 wicket keeper and at least 4 bowlers?

- A. 1011 B. 1092 C. 2092 D. 3092

14. A class photograph has to be taken. The front row consists of 6 girls who are sitting. 20 boys are standing behind. The two corner positions are reserved for the 2 tallest boys. In how many ways can the students be arranged?
 A. $9! \times 1440$ B. $10! \times 1440$ C. $11! \times 1440$ D. $18! \times 1440$
15. How many words can be formed from the letters of the word "ENGINEERING", so that vowels always come together?
 A. 4200 B. 420 C. $7! \times 5!$ D. $7! \times 5! / (2! \times 3!)$
16. Ram goes to a fruit-seller who is left with 4 apples, 5 bananas and 6 guavas only. In how many ways can Ram make a purchase from the shop?
 A. 120 B. 209 C. 119 D. 210
17. How many three-digit numbers divisible by 3 may be formed out of the digits 2, 3, 4 and 6 if the digits are not to be repeated?
 A. 24 B. 6 C. 12 D. 36
18. In how many ways 6 girls out of 12 girls in a class may be selected for a team so that 2 particular girls (captain and vice-captain) are always there?
 A. 360 B. 210 C. 24 D. 120
19. How many 5 digit numbers can be formed by using the first 9 natural numbers, such that all the 5 digits of the number formed are in ascending order and at the hundredth place the digit is '5'?
 A. 34 B. 31 C. 37 D. 36
20. In how many different ways can five players A, B, C, D and E be arranged in a line such that A is always to the left of B?
 A. 60 B. 120 C. 48 D. 24
21. Find the number of squares on a chessboard?
 A. 204 B. 100 C. 1296 D. 64
22. A telegraph has 5 arms and each arm has four distinct positions including the position of rest. What is the total number of signals that can be made?
 A. 511 B. 625 C. 19 D. 1023
23. Out of the twenty six letters of the alphabet, in how many ways can a five letter word (with all letters distinct) be made if three letters of the word must be L, A and W?
 A. 50,400 B. 30,360 C. 2,42,880 D. None of these
24. How many five-digit numbers divisible by 11 can be formed out of 3, 4, 5, 6 and 7 if the digits are not to be repeated?
 A. 10 B. 12 C. 13 D. None of these
25. If ${}^nC_4 = 70$, find n.
 A. 5 B. 8 C. 4 D. 7
26. If $5 \times {}^nP_3 = 4 \times {}^{n+1}P_3$ find n?
 A. 10 B. 12 C. 11 D. 14

27. There are 12 yes or no questions. How many ways can these be answered?
 A. 4096 B. 2048 C. 1024 D. 144
28. If ${}^6P_r = 360$ and ${}^6C_r = 15$ find r ?
 A. 3 B. 5 C. 4 D. 6
29. How many diagonals can be drawn in a pentagon?
 A. 5 B. 7 C. 8 D. 10
30. The number of the triangles that can be formed joining the angular points of decagon, is
 A. 30 B. 45 C. 90 D. 120

LEVEL – III

1. There are 5 floating stones on a river. A man wants to cross the river. He can move either 1 or 2 steps at a time. Find the number of ways in which he can cross the river?
 A. 11 B. 12 C. 13 D. 14
2. How many integers, greater than 999 but not greater than 4000, can be formed with the digits 0, 1, 2, 3 and 4, if repetition of digits is allowed?
 A. 366 B. 356 C. 376 D. 396
3. Each of the 11 letters A, H, I, M, O, T, U, V, W, X and Z appears same when looked at in a mirror. They are called symmetric letters. Other letters in the alphabet are asymmetric letters. How many three letter computer passwords can be formed (no repetition allowed) with at least one symmetric letter?
 A. 12170 B. 12870 C. 12970 D. 12470
4. In how many ways can the letters of the word 'EQUIDISTANT' be arranged such that the vowels and consonants do not change their relative positions in the given word?
 A. $11!$ B. $11! / 2! * 2!$ C. $6! * 5! / 2! * 2!$ D. $11! / 4!$
5. In how many ways can a student attempt at least one question in a multiple choice question paper having 10 questions such that half of the questions have three answer options and the remaining questions have two answer options for question?(each question has only one correct answer)
 A. 12^5 B. $12^5 - 1$ C. $6^5 - 1$ D. 6^5
6. How many 4-digit numbers, with each digit being distinct, can be formed using natural numbers less than 5 such that the digit at unit's place is less than the sum of the digits at ten's and hundred's place?
 A. 12 B. 24 C. 16 D. 18
7. In how many ways can we select a pair of prime numbers from the first 50 natural numbers such that the sum of the two numbers is a composite number?
 A. 99 B. 210 C. 8 D. 14

8. Tanvi tore out several successive pages from a book. The number of the first page he tore out was 183, and it is known that the number of the last page is written with the same digits in some order. How many pages did Tanvi tear out of the book? (Both sides of the pages are numbered)
- A. 138 B. 231 C. 68 D. 190
9. A frog wants to reach to the top of a staircase that has 7 steps. In how many ways it can reach to the top of the staircase, if at least 3 jumps are required to reach the top and the frog is initially on the ground? (the frog doesn't jump back).
- A. 102 B. 57 C. 63 D. 28
10. If the letters of the word CHASM are rearranged to form 5 letter words such that none of the word repeat and the results arranged in ascending order as in a dictionary what is the rank of the word CHASM?
- A. 32 B. 42 C. 52 D. 62
11. There are 6 boxes numbered 1,2,... 6. Each box is to be filled up either with a red or a green ball in such a way that at least 1 box contains a green ball and the boxes containing green balls are consecutively numbered. The total number of ways in which this can be done is?
- A. 11 B. 31 C. 21 D. 51
12. If the total numbers of terms in the expansion of $(x - y - 2z)^n$ is 45. Then value of n is
- A. 8 B. 9 C. 7 D. 22
13. The maximum number of different permutations of 5 letters of the word 'MORADABAD' is
- A. 2150 B. 3640 C. 2930 D. 2250
14. The number of selections of four letters from the letters of the word ASSASSINATION is:
- A. 72 B. 71 C. 66 D. 52
15. The letters of the word SURITI are written in all possible orders and these words are written out as in a dictionary. Then the rank of the word SURITI is
- A. 236 B. 245 C. 307 D. 315
16. The sum of all numbers greater than 1000 formed by using digits 1, 3, 5, 7 no digit being repeated in any number is
- A. 72, 215 B. 83, 911 C. 106, 656 D. 114, 712
17. a,b,c,d and e are five natural numbers. Find the number of ordered sets (a,b,c,d,e) possible such that $a+b+c+d+e=64$.
- A. ${}^{64}C_5$ B. ${}^{63}C_4$ C. ${}^{65}C_4$ D. ${}^{63}C_5$
18. In how many ways can 15 people be seated around two round tables with seating capacities of 7 and 8 people?
- A. $15!8!$ B. $7! \times 8!$ C. ${}^{15}C_8 \times 6! \times 7!$ D. $2 \times {}^{15}C_7 \times 6! \times 7!$
19. A five-digit number is formed using digits 1, 3, 5, 7 and 9 without repeating any one of them. What is the sum of all such possible numbers?
- A. 6666600 B. 6666660 C. 6666666 D. None of these
20. A man has 9 friends, 4 boys and 5 girls. In how many ways can he invite them, if there have to be exactly 3 girls in the invitees?
- A. 320 B. 160 C. 80 D. 200

PROBABILITY

Probability or chance is a common term used in day-to-day life. For example, we generally say, 'it may rain today'. This statement has a certain uncertainty.

Probability is quantitative measure of the chance of occurrence of a particular event. Probability deals with the analysis of random phenomena. It is a way of assigning every event a value between zero and one, with the requirement that the event made up of all possible results is assigned a value of one.

Experiment: An operation which can produce some well-defined outcome, is an experiment.

Random Experiment: If each trial of an experiment is conducted under identical conditions, the outcome is not unique, but may be of any possible outcome then such experiment is known as random experiment.

Example: Tossing of a fair coin, throwing of an unbiased die.

Sample Space: The set of all possible outcomes in Random Experiment is known as Sample space, provided no two or more of these outcomes can occur simultaneously and exactly one of these outcomes must occur whenever the experiment is conducted.

Example: When a die is thrown, any one of the numbers 1, 2, 3, 4, 5, 6 can come up.

Therefore, sample space: $S = \{1, 2, 3, 4, 5, 6\}$

Sample Point or Event Point: Each element of the sample spaces is called a sample point or an event point.

Example: When a die is thrown, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$ where 1, 2, 3, 4, 5 and 6 are the sample points.

Event: Any subset of a sample space is called an event.

Types of Events

i] Simple Event or Elementary Event: An event is called a Simple Event if it is a singleton subset of the sample space S .

Example: When a coin is tossed, then the sample space is $S = \{H, T\}$

Then $A = \{H\}$ occurrence of head and

$B = \{T\}$ occurrence of tail are called Simple events.

ii] Mixed Event or Compound Event or Composite Event: A subset of the sample space S which contains more than one element is called a mixed event or when two or more events occur together, their joint occurrence is called a Compound Event.

Example: When a dice is thrown, then the sample space is $S = \{1, 2, 3, 4, 5, 6\}$

Then let $A = \{2, 4, 6\}$ is the event of occurrence of even and $B = \{1, 2, 4\}$ is the event of occurrence of exponent of 2 are Mixed events.

Compound events are of two types:

(a) Independent Events or Mutually Independent Events: Two or more events are said to be independent if occurrence or non-occurrence of any of them does not influence the occurrence or non-occurrence of the other events.

Example: Let bag contains 3 Red and 2 Black balls. Two balls are drawn one by one with replacement.

Let A is the event of occurrence of a red ball in first draw.

B is the event of occurrence of a black ball in second draw.

Then probability of occurrence of B has not been affected if A occurs before B. As the ball has been replaced in the bag and once again we have to select one ball out of $5(3R + 2B)$ given balls for event B.

(b) Dependent Events: Two or more events are said to be dependent if occurrence or non-occurrence of any of them influence the occurrence or non-occurrence of the other events.

Example: Two cards have been drawn from the deck of 52 cards without replacing the first one back.

Let A is the event of occurrence of getting first card as king

B is the event of occurrence of getting second card as queen.

Then probability of occurrence of B has been affected if A occurs before B.

iii] Certain and impossible events: If S is a sample space, the both S and null set ϕ both are events. S is called certain event and ϕ is called an impossible event.

iv] Equally Likely Events: The given events are said to be equally likely, if none of them is expected to occur in preference of the other.

v] Exhaustive events: In probability theory, system of events is called exhaustive, if at least one of the events of the system occurs. Eg: If a coin is tossed then Head and Tails forms exhaustive set of events.

vi] Mutually Exclusive events: Two or more events are said to be mutually exclusive if one of them occurs, others cannot occur. Thus if two or more events are said to be mutually exclusive, if not two of them can occur together.

Hence, $A_1, A_2, A_3, \dots, A_n$ are mutually exclusive if and only if $A_i \cap A_j = \phi$, for $i \neq j$

Example: When a coin is tossed the event of occurrence of a head and the event of occurrence of a tail are mutually exclusive events because we cannot have both head and tail at the same time.

Probability of Occurrence of an Event

Let S be the same space, then the probability of occurrence of an event E is denoted by $P(E)$ and is defined as

$P(E) = n(E) / n(S)$ = number of elements in E / number of elements in S

$P(E)$ = number of favourable (particular cases) / total number of cases

- Probability of the null event is 0.
- Probability of a sure event is 1.
- $0 \leq P(E) \leq 1$
- $\sum P(E) = 1$

Example 1: When a die is tossed, sample space $S = \{1, 2, 3, 4, 5, 6\}$

Let A is an event of occurrence of an odd number

And B is an event of occurrence of a number greater than 4

$A = \{1, 3, 5\}$ and $B = \{5, 6\}$

$P(A)$ = Probability of occurrence of an odd number = $n(A) / n(S)$
 $= 3/6 = 1/2$

$P(B)$ = Probability of occurrence of a number greater than 4 = $n(B) / n(S)$
 $= 2/6 = 1/3$

Given below are some important theorems that might help us in formulating solutions to the problems.

- If A is a subset of B then, $P(A) \leq P(B)$.
- $P(\phi) = 0$.
- $P(S) = 1$.
- $P(A^c) = 1 - P(A)$.
- $P(B-A) = P[B - (A \cap B)] = P(B) - P(A \cap B)$.

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. [Addition Theorem]
- $P(A \cup B) = P(A) + P(B)$ when $P(A \cap B) = \varnothing$.
- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$.

Conditional Probability: Let S be the sample space. Let A and B be any two events. $A \neq \varnothing$. Then, probability of the event B , if A have already been occurred, is called conditional probability of B restricted to the occurrence of A . It is represented as $P(A/B)$. Thus, the probability of the event B restricted to the occurrence of the event A is the same as the probability of event $A \cap B$ while A is considered as sample space.

$$P(B/A) = n(A \cap B) / n(A) = P(A \cap B) / P(A)$$

$$P(A \cap B) = P(A) * P(B/A)$$

If $A \neq \varnothing$ & $B \neq \varnothing$ then,

$$P(A \cap B) = P(A) * P(B/A) = P(A) * P(A/B).$$

Example 2: A card is drawn from the well-shuffled pack of card. What is the probability that the card drawn is of heart given that it is a king?

Solution: Let A = card is of heart and B = card is king
 Therefore, $P(A) = 13/52$; $P(B) = 4/52$
 Now $P(A \cap B) = 1/52$
 Thus, $P(A/B) = P(\text{King of heart}) = P(A \cap B) / P(A)$
 $= (1/52) / (4/52) = 1/4$

Independent Events: Two events are said to be independent if the probable occurrence or non-occurrence of any one is not affected by occurrence or non-occurrence of the other i.e. two events A and B are independent if

$$P(A/B) = P(A/B^c) = P(A)$$

$$\text{Or, } P(B/A) = P(B/A^c) = P(B)$$

$$\text{Or, } P(A \cap B) = P(A) * P(B)$$

Note: Three events A , B , and C are independent if

$$P(A \cap B \cap C) = P(A). P(B). P(C)$$

Three events A , B , and C are pairwise independent if

$$P(A \cap B) = P(A). P(B),$$

$$P(B \cap C) = P(B). P(C) \text{ \& } P(A \cap C) = P(A). P(C)$$

Relation between Independent and Mutually Exclusiveness of two events

- If two events $A \neq \varnothing$ and $B \neq \varnothing$ are independent, then they are not mutually exclusive.
- If two events $A \neq \varnothing$ and $B \neq \varnothing$ are mutually exclusive, then they are not independent.

Example 3: A problem is given to three students whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$, and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved?

Solution: Let there be three events A , B , and C where
 A is the event when student with probability $\frac{1}{2}$ solves the question,
 where B is the event when student with probability $\frac{1}{3}$ solves the question,
 where C is the event when student with probability $\frac{1}{4}$ solves the question,
 Therefore, $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, and $P(C) = \frac{1}{4}$
 Now let A^c , B^c and C^c is the event of not solving question by the students respectively.
 $P(A^c) = 1 - \frac{1}{2} = \frac{1}{2}$, $P(B^c) = 1 - \frac{1}{3} = \frac{2}{3}$, $P(C^c) = 1 - \frac{1}{4} = \frac{3}{4}$
 Therefore, $P(A^c \cap B^c \cap C^c) = P(A^c). P(B^c). P(C^c) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4}$

$$\begin{aligned}\text{Hence, the } P[\text{problem will be solved}] &= 1 - P[\text{none solves the problem}] \\ &= 1 - \frac{1}{4} = \frac{3}{4}\end{aligned}$$

Bayes' Theorem: Suppose $A_1, A_2, \dots, A_n$, are mutually exclusive and exhaustive set of events. Thus, they divide the sample into n parts and the event B occurs. Then the conditional probability that A_i happen given that B has happened is given by

$$\frac{P(A_i/B) \cdot P(B/A_i)}{\sum_{i=1}^n P(A_i) \cdot P(B/A_i)}$$

Example 4: An urn B1 contains 2 white and 3 black balls and another urn B2 contains 3 white and 4 black balls. One urn is selected at random and a ball is drawn from it. If the ball drawn is found black, find the probability that the urn chosen was B1.

Solution: Let E_1, E_2 denote the vents of selecting urns B1 and B2 respectively.

Then $P(E_1) = P(E_2) = 1/2$.

Let B denote the event that the ball chosen from the selected urn is black .

Then we have to find $P(E_1/B)$.

By hypothesis $P(B/E_1) = 3/5$ and $P(B/E_2) = 4/7$

By Bayes theorem $P(E_1/B) = P(E_1)P(B/E_1) / P(E_1)P(B/E_1) + P(E_2)P(B/E_2)$

$$= (1/2 \cdot 3/5) / (1/2 \cdot 3/5 + 1/2 \cdot 4/7)$$

$$= 21/41$$

Binomial Probability distribution

A binomial experiment is a probability experiment which satisfies the following requirements.

1. Each trial can have only two outcomes. These outcomes can be considered as either success or failure.
2. There must be a fixed number of trials.
3. The outcomes of each trial must be independent of each other.
4. The probability of a success must remain the same for each trial.

In a binomial experiment, The probability of achieving exactly r successes in n trials can be given by

$$P(r \text{ successes in } n \text{ trials}) = {}^nC_r \cdot p^r \cdot q^{n-r}$$

where p = probability of success in one trial

$q = 1 - p$ = probability of failure in one trial

$${}^nC_r = n! / r!(n-r)!$$

Odds of an event

Let E be an event associated with a random experiment. Let x outcomes are favourable to E and y outcomes are not favourable to E , then

Odds in favour of E are $x:y$, i.e., x/y

Odds against E are $y:x$, i.e., y/x

$$P(E \text{ (in favor)}) = x / x+y$$

$$P(E^c \text{ (against)}) = y / x+y$$

Example 5: What are the odds in favour of and against getting a 1 when a die is rolled?

Let E be an event of getting 1 when a die is rolled

Outcomes which are favourable to E , $x=1$

Outcomes which are not favourable to E , $y=5$

Odds in favour of getting 1 = $x/y = 1/5$

Odds against getting 1 = $y/x = 5/1$

LEVEL – I

1. Determine the probability that a digit chosen at random from the digits 1, 2, 3, ... 12 will be odd.
A. $\frac{1}{2}$ B. $\frac{1}{9}$ C. $\frac{5}{9}$ D. $\frac{4}{9}$
2. A dice is thrown, what is the probability that the number obtained is a prime number.
A. $\frac{1}{6}$ B. $\frac{1}{8}$ C. $\frac{1}{2}$ D. $\frac{1}{3}$
3. A card is drawn from a pack of 52 cards. What is the probability that the card is a Queen?
A. $\frac{1}{52}$ B. $\frac{1}{4}$ C. $\frac{1}{16}$ D. None of these
4. Determine the probability that a number chosen at random from the digits 1, 2, 3, ..., 10 will be a multiple of 4.
A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{1}{5}$ D. $\frac{1}{2}$
5. Two brother X and Y appeared for an exam. Let A be the event that X is selected and B is the event that Y is selected. The probability of A is $\frac{1}{7}$ and that of B is $\frac{2}{9}$. Find the probability that both of them are selected.
A. $\frac{1}{63}$ B. $\frac{2}{35}$ C. $\frac{2}{63}$ D. $\frac{9}{14}$
6. Determine the probability that a digit chosen at random from digits 1, 2, 3, ..., 13 will be even.
A. $\frac{1}{2}$ B. $\frac{1}{9}$ C. $\frac{5}{9}$ D. $\frac{6}{13}$
7. A coin is tossed four times, if H = head and T = tail, what is the probability of the tosses coming up in the order HTHH?
A. $\frac{3}{16}$ B. $\frac{1}{16}$ C. $\frac{5}{16}$ D. $\frac{7}{16}$
8. Find the probability of throwing a total of 8 in a single throw with two dice.
A. $\frac{1}{36}$ B. $\frac{5}{36}$ C. $\frac{25}{36}$ D. $\frac{12}{36}$
9. An urn contains 6 red, 5 blue and 2 green marbles. If 2 marbles are picked at random, what is the probability that both are red?
A. $\frac{6}{13}$ B. $\frac{5}{26}$ C. $\frac{5}{13}$ D. $\frac{7}{26}$
10. Four dice are thrown simultaneously. Find the probability that all of them show the same face.
A. $\frac{1}{216}$ B. $\frac{1}{36}$ C. $\frac{4}{216}$ D. $\frac{3}{216}$
11. Two dice are thrown, what is the probability that both the dices are not having the same number.
A. $\frac{1}{4}$ B. $\frac{5}{6}$ C. $\frac{1}{9}$ D. $\frac{1}{12}$
12. A die is thrown. Let A be the event that the number obtained is greater than 3. Let B be the event that the number obtained is less than 5. Then $P(A \cup B)$ is
A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. 0 D. 1
13. A five-digit number is formed by using digits 1, 2, 3, 4 and 5 without repetition. What is the probability that the number is divisible by 4?
A. $\frac{1}{5}$ B. $\frac{5}{6}$ C. $\frac{4}{5}$ D. None of these

14. A bag contains 5 red and 3 green balls. Another bag contains 4 red and 6 green balls. If one ball drawn from each bag. Find the probability that one ball is red and one is green.
A. $19/20$ B. $17/20$ C. $8/10$ D. $21/40$
15. A speaks truth in 75% of cases and B in 80% of cases. In what percent of cases are they likely to contradict each other in narrating the same event?
A. 35% B. 5% C. 45% D. 22.5%
16. A bag contains 6 white and 4 black balls .2 balls are drawn at random. Find the probability that they are of same colour.
A. $1/2$ B. $7/15$ C. $8/15$ D. $1/9$
17. In a charity show tickets numbered consecutively from 101 through 350 are placed in a box. What is the probability that a ticket selected at random (blindly) will have a number with a hundredth digit of 2?
A. 0.285 B. 0.40 C. $100/249$ D. $99/250$
18. A man and his wife appear in an interview for two vacancies in the same post. The probability of husband's selection is $(1/7)$ and the probability of wife's selection is $(1/5)$. What is the probability that only one of them is selected?
A. $2/7$ B. $1/7$ C. $3/4$ D. $4/5$
19. What is the probability of getting 53 Mondays in a leap year?
A. $1/7$ B. $3/7$ C. $2/7$ D. 1
20. In a class, there are 15 boys and 10 girls. Three students are selected at random. The probability that 1 girl and 2 boys are selected, is:
A. $21/46$ B. $1/5$ C. $3/25$ D. $1/50$

LEVEL – II

1. Two cards are drawn in succession from a pack of 52 cards, without replacement. What is the probability, that the first is a Queen and the second is a Jack of a different suit?
A. $1/52$ B. $1/13$ C. $4/13$ D. $1/221$
2. One ticket is selected at random from 50 tickets numbered 0, 01, 02,, 49. Then, the probability that the sum of the digits on the selected ticket is 8, given that the product of these digits is zero equals.
A. $1/14$ B. $1/7$ C. $5/14$ D. $1/50$
3. It is given that the events A and B are such that $P(A) = 1/4$, $P(A|B) = 1/2$ and $P(B|A) = 2/3$. Then $P(B|A) = 2/3$. Then $P(B)$ is
A. $1/2$ B. $1/6$ C. $1/3$ D. $2/3$
4. A pair of fair dice is thrown independently three times. The probability of getting a total of exactly 9 twice is
A. $1/729$ B. $8/9$ C. $8/729$ D. $8/243$

5. A bag contains 12 white and 18 black balls. Two balls are drawn in succession without replacement. What is the probability that first is white and second is black?
 A. 18/145 B. 18/29 C. 36/135 D. 36/145
6. The odds against an event are 5:3 and the odds in favour of another independent event are 7:5. Find the probability that at least one of the two events will occur.
 A. 52/96 B. 69/96 C. 71/96 D. 13/96
7. If the chance that a vessel arrives safely at a port is $\frac{9}{10}$ then what is the chance that out of 5 vessels expected at least 4 will arrive safely?
 A. $14 \times 9^4 / 10^5$ B. $15 \times 9^5 / 10^4$ C. $14 \times 9^3 / 10^4$ D. $14 \times 9^6 / 10^5$
8. Derek throws three dice in a special game. If he knows that he needs 15 or higher in this throw to win, then find the chance of his winning the game.
 A. 5/54 B. 17/216 C. 13/216 D. 15/216
9. An urn contains 6 red, 5 blue and 2 green marbles. If three marbles are picked at random, what is the probability that at least one is blue?
 A. 28/143 B. 115/197 C. 28/197 D. 115/143
10. Four cards are drawn at random from a pack of 52 playing cards. Find the probability of getting all the four cards of the same suit.
 A. 13/270725 B. 91/190 C. 178/20825 D. 44/4165
11. There are four hotels in a town. If 3 men check into the hotels in a day then what is the probability that each checks into a different hotel?
 A. 6/7 B. 1/8 C. 3/8 D. 5/9
12. Two teams Arrogant and Overconfident are participating in a cricket tournament. The odds that team Arrogant will be champion is 5 to 3, and the odds that team Overconfident will be the champion is 1 to 4. What are the odds that either Arrogant or team Overconfident will become the champion?
 A. 3 to 2 B. 5 to 2 C. 6 to 1 D. 33 to 7
13. A box contains 100 balls, numbered from 1 to 100. If three balls are selected at random and with replacement from the box, what is the probability that the sum of the three numbers on the balls selected from the box will be odd?
 A. 1/2 B. 3/4 C. 3/8 D. 1/8
14. A bag contains 3 white balls and 2 black balls. Another bag contains 2 white and 4 black balls. A bag and a ball are picked random. The probability that the ball will be white is:
 A. 7/11 B. 7/30 C. 5/11 D. 7/15
15. I forgot the last digit of a 7-digit telephone number. If I randomly dials the final 3 digits after correctly dialling the first four, then what is the chance of dialling the correct number?
 A. 1/1001 B. 1/1000 C. 1/999 D. 1/990

16. In his wardrobe, Dexter has three trousers. One of them is black the second is blue, and the third brown. In his wardrobe, he also has four shirts. One of them is black and the other 3 are white. He opens his wardrobe in the dark and picks out one shirt and one trouser pair without examining the colour. What is the likelihood that neither the shirt nor the trouser is black?
 A. $1/12$ B. $1/2$ C. $1/4$ D. $1/6$
17. A man can hit a target once in 4 shots. If he fires 4 shots in succession, what is the probability that he will hit his target?
 A. $175/256$ B. $1/256$ C. $81/256$ D. 1
18. The letters B,G,I,N and R are rearranged to form the word 'Bring'. Find its probability:
 A. $1/120$ B. $1/54$ C. $1/24$ D. $1/76$
19. Abhishek has 9 pairs of dark blue socks and 9 pairs of black socks. He keeps them all in the same bag. If he picks out three socks at random, then what is the probability that he will get a matching pair?
 A. 1 B. $2 \times {}^9C_2 \times {}^9C_1 / {}^{18}C_3$ C. ${}^9C_3 \times {}^9C_1 / {}^{18}C_3$ D. None of these
20. Four boys and three girls stand in queue for an interview. The probability that they stand in alternate positions is:
 A. $1/17$ B. $1/34$ C. $1/35$ D. $1/68$

LEVEL – III

1. Two aeroplanes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2, respectively. The second plane will bomb only if the first misses the target. The probability that the target is hit by the second plane, is
 A. 0.06 B. 0.14 C. 0.32 D. 0.7
2. Two urns contain 5 white and 7 black balls and 3 white and 9 black balls respectively. One ball is transferred to the second urn and then one ball is drawn from the second urn. Find the probability that the first ball transferred is black, given that the ball drawn is black?
 A. $13/23$ B. $11/23$ C. $14/23$ D. $7/23$
3. A bag contains 10 balls numbered from 0 to 9. The balls are such that the person picking a ball out of the bag is equally likely to pick anyone of them. A person picked a ball and replaced it in the bag after noting its number. He repeated this process 2 more times. What is the probability that the ball picked first is numbered higher than the ball picked second and the ball picked second is numbered higher than the ball picked third?
 A. $72/100$ B. $3/25$ C. $4/5$ D. $1/6$
4. There are three similar boxes, containing
 (i). 6 black and 4 white balls
 (ii). 3 black and 7 white balls
 (iii). 5 black and 5 white balls, respectively.

If you choose one of the three boxes at random and from that particular box picks up a ball at random, and find that to be black, what is the probability that the ball picked up from the second box?

- A. $14/30$ B. $3/14$ C. $7/30$ D. $7/14$

5. If an integer n is to be chosen at random from the integers 1 to 96, inclusive, what is the probability that $n(n+1)(n+2)$ will be divisible by 8?

- A. $1/4$ B. $3/8$ C. $1/2$ D. $5/8$

6. A bag contains blue and red balls. Two balls are drawn randomly without replacement. The probability of selecting a blue and then a red ball is 0.2. The probability of selecting a blue ball in the first draw is 0.5. What is the probability of drawing a red ball, given that the first ball drawn was blue?

- A. 0.4 B. 0.2 C. 0.1 D. 0.5

7. A die is rolled thrice. What is the probability that the sum of the rolls is atleast 5?

- A. $1/216$ B. $1/6$ C. $3/216$ D. $212/216$

8. Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

- A. $13/90$ B. $12/90$ C. $78/90$ D. $77/90$

9. There are 10 pair of socks in a cupboard from which 4 individual socks are picked at random. The probability that there is at least one pair is:

- A. $195/323$ B. $99/323$ C. $198/323$ D. $185/323$

10. In a purse there are 30 coins, twenty one-rupee and remaining 50-paise coins. Eleven coins are picked simultaneously at random and are placed in a box. If a coin is now picked from the box, find the probability of it being a rupee coin?

- A. $4/7$ B. $1/2$ C. $2/3$ D. $5/6$

11. In 2 bags, there are to be put together 5 red and 12 white balls, neither bag being empty. How must the balls be divided so as to give a person who draws 1 ball from either bag-

- (i) the least chance of drawing a red ball ?
(ii) the greatest chance of drawing a red ball ?

- A. $5/32$, $5/8$ B. $5/32$, $1/2$ C. $2/3$, $5/8$ D. $5/6$, $1/2$

12. In a zoo, there are 1 billion monkeys. Probability that a monkey has seen a banyan tree is 0.6. Prob. that monkey has seen a mango tree is 0.65. What is the minimum percentage of monkeys in the zoo who have seen both the trees?

- A. 25% B. 39% C. 40% D. 60%

13. There are 5 different boxes named a,b,c,d,e. there are two balls red and green.a box is to be filled with one ball either red or blue. what is the probability that no two blue balls are adjacent to each other.

- A. $11/32$ B. $1/2$ C. $13/32$ D. $15/16$

14. Raju and Reenu are standing in a row there are 9 peoples. what is the probability that at least 3 people will stand between Raju and Reenu?

- A. $11/12$ B. $5/12$ C. $5/32$ D. $11/16$

15. Here is 15 dots. If you select 3 dots randomly, what is the probability that 3 dots make a triangle?
A. $440/455$ B. $434/455$ C. $449/455$ D. $438/455$
16. A bag contains 9 red and 7 white balls. Four balls are drawn out one by one and not replaced. What is the probability that they are alternatively of different colours?
A. $9/65$ B. $6/65$ C. $9/130$ D. $8/130$
17. A basket contains 5 white and 9 black balls. There is another basket which contains 7 white and 7 black balls. One ball is to drawn from either of the two baskets. What is the probability of drawing a white ball?
A. $4/7$ B. $6/7$ C. $3/7$ D. $2/7$
18. In a box carrying one dozen of oranges, one third have become bad. If 3 oranges are taken out from the box at random, what is the probability that at least one oranges out of the three oranges picked up is good?
A. $1/55$ B. $54/55$ C. $45/55$ D. $3/55$
19. 8 persons are seated at a round table. What is the probability that 3 particular persons sit together?
A. $2/7$ B. $3/14$ C. $3/14$ D. $1/7$
20. One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is either a red or a king?
A. $6/13$ B. $1/2$ C. $7/13$ D. $27/52$

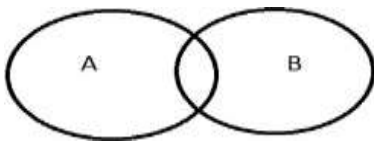
SYLLOGISM

The word 'Syllogism' is also referred to 'Logic'. Syllogism is an important section of logical reasoning and hence, a working knowledge of its rules is required to solve the problems. Two or more statements are given and one is supposed to find out all the possible conclusions from the given statements.

The first step is to make a Venn diagram. Second step is deriving the conclusion. Let's go to all possible concepts. (Concepts = Statements)

Rule 1:

Statement: Some A is B.



Definite Conclusions :

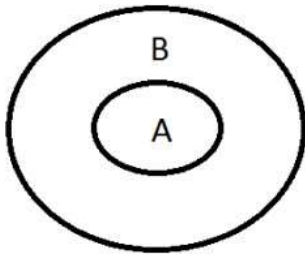
1. Some A is B
2. Some B is A

Possibility Conclusions :

1. All A's are B is a Possibility
2. All B's are A is a Possibility
3. Some A are not B is a possibility
4. Some B are not A is a possibility

Rule 2:

All A is B



Definite Conclusions :

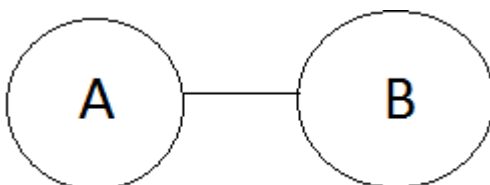
1. All A is B
2. Some A is B
3. Some B is A

Possibility Conclusions :

1. All B's are A is Possibility
2. Some B's are not A is a Possibility

Rule 3:

No A is B



Definite Conclusion :

1. No A is B

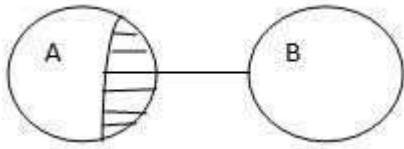
Possibility Conclusions:

No Possibility Conclusion

2. No B is A
3. Some A is not B
4. Some B is not A

Rule 4:

Some A's are Not B



Definite Conclusion :

1. Some A's are not B

Possibility Conclusions:

1. Except All A's are B is possibility.
All other possibilities conclusion follows

Shortcut:

All+All=All

All+No=No

All+Some=No Conclusion

Some+All=Some

Some+No= Some Not

Some+Some= No Conclusion

No +All = Some Not (Reversed)

No+Some=Some Not (Reversed)

No+No=No Conclusion

Some Not /Some Not Reversed +Anything = No Conclusion

Points to remember:

- First draw venn diagrams according to the statement
- If the definite conclusion doesn't satisfy the basic diagram then no need to check the possibility diagram
- If the definite conclusion satisfies the basic diagram then it must satisfy all possibility diagrams.
- Possibility conclusion can satisfy any one of the possibility Diagram

Conditions of Either Or (Complementary Pair):

- (1) Subject Predicate should be same in both statements
- (2) Complimentary pairs i.e. one should be positive and one should be negative
- (3) Maximum possibility i.e. maximum diagrams possibility should be covered
- (4) Individually both false
- (5) relation between subject and predicate should not be clear.
- (6) Either or condition not applicable between All and no type sentences.

Condition 1: Some + Some Not

Eg: Some A are B + Some A are not B

OR

Some B are A + Some A are not B

Condition 2: Some + No

Eg: Some A is B + No A is B

OR

Some B is A + No A is B

Condition 3: All + Some Not

Eg: All A is B + Some A are not B

Note: All A is B + Some B are not A (is not complementary pair)

Example1:

Statements

Some reds are crows.

All crows are yellows.

All yellows are rabbits.

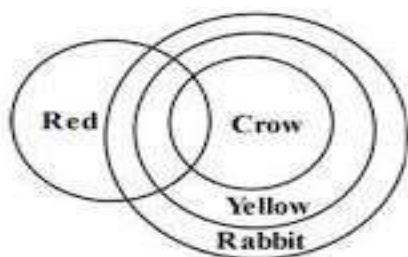
Conclusions

I. All crows are rabbits.

II. Some yellows are reds.

III. Some reds are rabbits.

This is the standard representation of all the statements taken together. Now we shall check the conclusions based on this representation.



From the diagram, we can see that the circle 'rabbit' engulfs the circle 'crow'. So conclusion I follow. Also, the circles 'yellow' and 'red' intersect each other. So conclusion II also follows. We can see that the circles 'rabbit' and 'red' intersect each other. So conclusion III also follows. Hence all of the given conclusions follow.

Example 2:

Statements

All cars are bikes

Some bikes are ships

All ships are planes

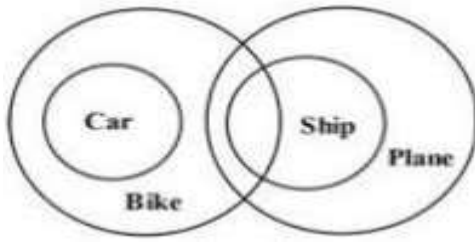
Conclusions

I. At least some planes are bikes.

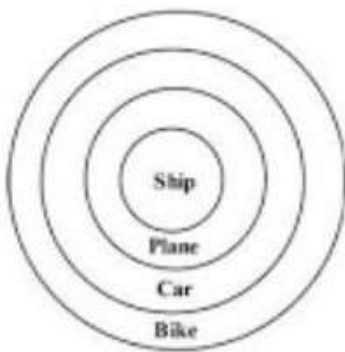
II. Some bikes are cars.

III. All planes are cars is a possibility.

From the given statements, the standard diagram can be drawn as:



From this, we can say that conclusions I and II follow, but III does not. Conclusion III says all planes are cars is a possibility, it does not mean that it has to be definitely true. So, we can have another representation of the given statements as below:



So, even if we have a single case where all planes are car, the conclusion follows. Hence all the conclusions follow.

Example 3:

Statements

Some cameras are laptops.

Some laptops are phones.

Some phones are tablets.

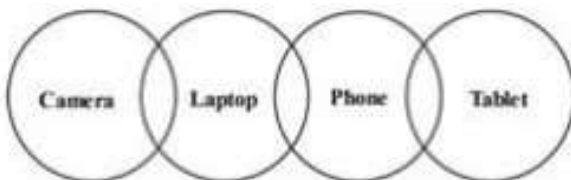
Conclusions

I. At least some tablets are cameras

II. There is a possibility that all tablets are laptops

III. No tablet is a laptop

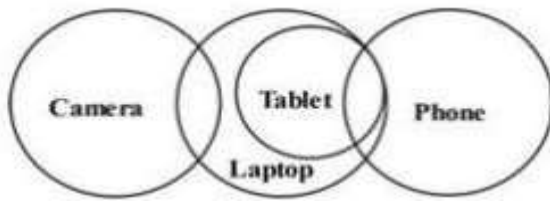
From the given statements, the standard diagram can be drawn as:



According to this diagram, conclusion III follows.

But there's a conclusion with the word 'possibility'. So we'll check if any such case is possible.

In that case, the diagram would look like:



But in this case, conclusion III does not follow.
Thus either conclusion II or conclusion III follows.

Example 4:

Statements

Some parrots are scissors.

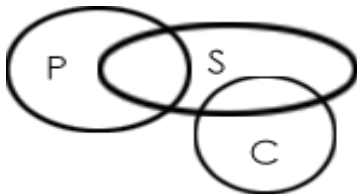
Some scissors are not combs.

Conclusions

Some scissors are parrots.

Some combs are parrots.

Now, in this case, the possible conclusion is: Some scissors are parrots (I to I), but with two particular statements only I is possible.



Therefore, only 1 conclusion is possible. Nothing else is possible.

Example 5:

Statements

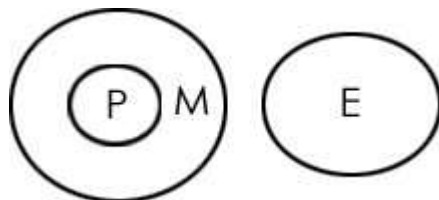
All prisoners are men.

No man is educated.

Conclusions

All prisoners are uneducated.

Some men are prisoners.



Conclusion I follows from the venn diagram, so conclusion I is true. Conclusion II also follows from the venn diagram, so conclusion II is also true. Therefore, both conclusions are true.

In each question below statements followed by two conclusions which is numbered as I, and II. You have to take the given statements to be true even if they seem to be at variance with commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follows from the given statements, disregarding commonly known facts.

Mark your answer as –

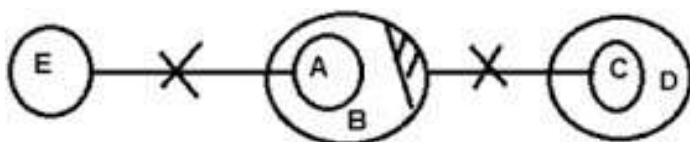
- (a) If only conclusion I follows.
- (b) If only conclusion II follows.
- (c) If either conclusion I or conclusion II follows.
- (d) If neither conclusion I nor conclusion II follows.
- (e) If both conclusion I and conclusion II follow.

Example 6:

Statements: All A is B. Some B is not C. All C is D. No A is E

Conclusions:

- I. All C can be B is a possibility.
- II. Some B can be D is a possibility.



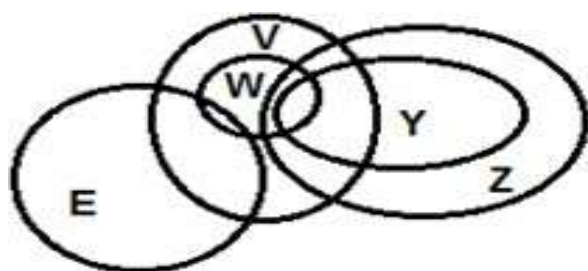
From the above venn diagram we can conclude that, in case of possibilities we can say both the conclusions are true.

Example 7:

Statements: All W is V. Some W is Y. All Y is Z. Some W is E

Conclusions:

- I. Some E is V
- II. Some E is Y



From the above venn diagram we can conclude that, it follows only conclusion I and does not follows second conclusion.

LEVEL - I

Directions for 1 to 20: Given two statements, verify the conclusions and mark the answer as given below.

Mark (a) - If only conclusion I follow

Mark (b) - If only conclusion II follows

Mark (c) - If either conclusion I or II follows

Mark (d) - If neither of the two conclusions follows

Mark (e) - If both conclusions follow

1. Statements:
I. Some cars are trucks
Conclusions:
I. Some cars are trucks
II. Some trucks are buses
II. Some cars are not trucks
2. Statements:
I. All locks are keys
Conclusions:
I. No lock is spoon.
II. No spoon is a lock
II. No key is a spoon
3. Statements:
I. All grasses are trees
Conclusions:
I. No grasses are shrubs
II. No tree is a shrub
II. Some shrubs are grasses
4. Statements:
I. Some boys are thieves
Conclusions:
I. Some boys are dacoits
II. All thieves are dacoits
II. All dacoits are thieves
5. Statements:
I. Some bottles are pencils
Conclusions:
I. No glass is bottle
II. Some pencils are glasses
II. Some bottles are glasses
6. Statements:
I. All fans are chairs
Conclusions:
I. No tables are chairs
II. No tables are fans
II. Some tables are chairs
7. Statements:
I. All women are mothers
Conclusions:
I. All womens are sisters
II. All mothers are sisters
II. Some women are not sisters

- | | | |
|-----|---|---|
| 8. | Statements:
I. All plays are stories
Conclusions:
I. Some poems are stories | II. Some poems are plays

II. All stories are poems |
| 9. | Statements:
I. All hunters are punters
Conclusions:
I. Some hunters are tigers | II. Some punters are tigers

II. Some punters are hunters |
| 10. | Statements:
I. All boxes are pens
Conclusions:
I. All boxes are dogs | II. All pens are dogs

II. Some dogs are pens |
| 11. | Statements:
I. Some boys are girls
Conclusions:
I. Some boys are cute | II. All girls are cute

II. All boys are cute |
| 12. | Statements:
I. All books are pens
Conclusions:
I. All books are pencils | II. All pens are pencils

II. Some pens are pencils |
| 13. | Statements:
I. Some singers are rockers
Conclusions:
I. Some rockers are dancers | II. All rockers are dancers

II. No singers are dancers |
| 14. | Statements:
I. No bars are coins
Conclusions:
I. All coins are books | II. All coins are books

II. Some books are not bars |
| 15. | Statements:
I. Some copies are desks
Conclusions:
I. Some copies are pen | II. No desks are pen

II. Some copies are not pen |
| 16. | Statements:
I. All boxes are pens
Conclusions:
I. No boxes are dogs | II. No dogs are pens

II. Some pens are boxes |

17. Statements:
 I. All apples are oranges II. Some oranges are papayas
 Conclusions:
 I. Some apples are papayas II. Some papayas are apples
18. Statements:
 I. Some dogs are bulls II. No tigers are dogs
 Conclusions:
 I. No dogs are tiger II. Some bulls are tiger
19. Statements:
 I. Some bottles are pencils II. Some pencils are glasses
 Conclusions:
 I. No glass is bottle II. Some bottles are glasses
20. Statements:
 I. All fans are chairs II. No tables are fans
 Conclusions:
 I. No tables are chairs II. Some tables are chairs

LEVEL - II

Directions (1-5): In each question below are given four statements followed by two conclusions which is numbered as I, and II. You have to take the given statements to be true even if they seem to be at variance with commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follows from the given statements, disregarding commonly known facts.

Mark your answer as –

- (A) If only conclusion I follows.
 (B) If only conclusion II follows.
 (C) If either conclusion I or conclusion II follows.
 (D) If neither conclusion I nor conclusion II follows.
 (E) If both conclusion I and conclusion II follow.

1. Statements: All A is B. Some B is not C. All C is D. No A is E

Conclusions:

- I. All C can be B II. Some B can be D

2. Statements: Some M is N. Some N is P. All P is Z. No Z is X

Conclusions:

- I. Some N is Z II. Some N is X

3. Statements: No J is K. Some K is D. Some D is E. No E is F

Conclusions:

- I. Some E is K II. Some D is F

4. Statements: All W is V. Some W is Y. All Y is Z. Some W is E

Conclusions:

I. Some E is V

II. Some E is Y

5. Statements: Only D is E. No E is F. All F is G. Some G is H

Conclusions:

I. Some G is F

II. Some G is not E

Directions (6-10): Question consists of five statements followed by five conclusions. Consider the given statements to be true even if they seem to be at variance with commonly known facts. Read all the conclusions and then decide which of the given conclusions does not logically follow from the given statements using all statements together.

6. Statements: All M are N. All N are C. No D is N. All K are F. Some K are D

Conclusions:

(a) Some K are not N.

(b) Some C can be K.

(c) Some F is C.

(d) No M are D

(e) All N can be K.

7. Statements: All D is E. Some D is P. No P is K. All K are L. Some L are T

Conclusions:

(a) All P can be T

(b) All L can be P

(c) All D can be P

(d) Some E is P

(e) Some T can be K.

8. Statements: Some A is B. Some B is K. All K are T. No M is K. Some M is N

Conclusions:

(a) Some T can be M

(b) All M being T is a possibility

(c) Some K can be M

(d) All M can be B

(e) Some B is A

9. Statements: Some W is X. All X is Y. No Y is T. No T is K. Some K is L.

Conclusions:

(a) All X can be T

(b) All L can be Y

(c) All X can be K

(d) Some Y is W

(e) No K is T.

10. Statements: No P is Q. Some Q is R. Some R is not S. Some S is T. All T is V

Conclusions:

(a) All S can be R

- (b) All S can be V
- (c) Some S is V
- (d) Some R is T
- (e) All Q can be V.

Directions (11-20): In each of the questions/set of questions below are given two statements followed by two conclusions numbered I and II. You have to assume everything in the statements to be true even if they seem to be at variance from commonly known facts and then decide which of the two given conclusions logically follows from the information given in the statement. Give answer

- A) if only conclusion I follows.
- B) if only conclusion II follows.
- C) if either conclusion I or conclusion II follows.
- D) if neither conclusion I nor conclusion II follows.
- E) if both conclusions I and II follow.

11. Statements:

All rings are circles.
All squares are rings.
No ellipse is a circle.

Conclusions:

- I. Some rings being ellipses is a possibility.
- II. At least some circles are squares.

12. Statements:

No house is an apartment.
Some bungalows are apartments.

Conclusions:

- I. No house is a bungalow.
- II. All bungalows are houses.

13. Statements:

Some gases are liquids.
All liquids are water.

Conclusions:

- I. All gases being water is a possibility.
- II. All such gases which are not water can never be liquids.

14. Statements:

All minutes are seconds.
All seconds are hours.
No second is a day.

Conclusions:

- I. No day is an hour.
- II. At least some hours are minutes.

15. Statements:

Some teachers are professors.

Some lecturers are teachers.

Conclusions:

I. All teachers as well as professors being lecturers is a possibility.

II. All those teachers who are lecturers are also professors.

16. Statements:

Some teachers are professors.

Some lecturers are teachers.

Conclusions:

I. No professor is a lecturer.

II. All lecturers being professors is a possibility.

17. Statements:

Some squares are circles.

Some circles are rectangles.

Conclusions:

I. At least some rectangles are squares.

II. No rectangle is a square.

18. Statements:

No office is a palace.

All colleges are palaces.

Conclusions:

I. All palaces are colleges.

II. No college is an office.

19. Statements:

All mountains are rivers.

All rivers are lakes.

Conclusions:

I. All mountains are lakes.

II. At least some lakes are rivers.

20. Statements:

Some wins are losses.

All trophies are losses.

Conclusions:

I. All trophies are wins.

II. All losses are trophies.

LEVEL - III

Directions (1-5): Question consists of Six statements followed by five conclusions. Consider the given statements to be true even if they seem to be at variance with commonly known facts. Read all the conclusions and then decide which of the given conclusions does not logically follow from the given statements using all statements together.

1. Statements: Some file is copy. All copy is kite. Some kite is joy. All joy is orange. All orange is toy. All file is black.

Conclusions:

- (a) Some kite is black
- (b) Some kite is orange
- (c) Some kite is toy
- (d) Some copy is black
- (e) Some toy is copy

2. Statements: Some mango is banana. No banana is apple. All grapes are apple. Some grapes are tree. No tree is green. All green is good.

Conclusions:

- (a) No grapes is banana.
- (b) Some mango is not apple.
- (c) All tree being good is a possibility
- (d) Some apple is tree
- (e) Some good are tree.

3. Statements: Some court is justice. No justice is faith. All faith is quick. No quick is decision. Some decision are jury. All jury is delay.

Conclusions:

- (a) Some court is not faith
- (b) All justice being quick is possibility.
- (c) No faith is delay.
- (d) No faith is decision
- (e) Some delay is decision

4. Statements: All heart is green. Some green is fire. All fire is tree. No fire is Lion. All mobile is lion. Some mobile is network.

Conclusions:

- (a) Some tree is green
- (b) Some network is lion
- (c) Some tree is not lion
- (d) Some green is not lion
- (e) Some Heart is fire

5. Statements: All time is unit. Some time is rock. All rock is stone. No stone is door. All door is car. No car is bus.

Conclusions:

- (a) Some time is stone.

- (b) Some unit is stone.
- (c) No rock is door.
- (d) No stone is car.
- (e) Some car is not bus.

Directions (6-10): In each of the questions below are given four statements followed by four conclusions numbered I, II, III & IV. You have to take the given statements to be true even if they seem to be at variance with commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follows from the given statements disregarding commonly known facts.

6. Statements:

Some trains are cars.
All cars are branches.
All branches are nets.
Some nets are dresses.

Conclusions:

- I. Some dresses are cars.
- II. Some nets are trains.
- III. Some branches are trains.
- IV. Some dresses are trains.

- (a) Only I and III follow
- (c) Only I and IV follow
- (e) None of these

- (b) Only II and III follow
- (d) Only II, III and IV follow

7. Statements:

Some pencils are kites.
Some kites are desks.
All desks are jungles.
All jungles are mountains.

Conclusions:

- I. Some mountains are pencils.
- II. Some jungles are pencils.
- III. Some mountains are desks.
- IV. Some jungles are kites.

- (a) Only I and III follow
- (c) Only III and IV follow
- (e) None of these

- (b) Only I, II and III follow
- (d) Only II, III and IV follow

8. Statements:

All papers are clips.
Some clips are boards.
Some boards are lanes.
All lanes are roads.

Conclusions:

- I. Some roads are boards.
- II. Some lanes are clips.
- III. Some boards are papers.

IV. Some roads are clips.

- (a) Only I and II follow
- (c) Only I, II and III follow
- (e) None of these

- (b) Only I and III follow
- (d) Only II, III and IV follow

9. Statements:

All pens are clocks.

Some clocks are tyres.

Some tyres are wheels.

Some wheels are buses.

Conclusions:

I. Some buses are tyres.

II. Some wheels are clocks.

III. Some wheels are pens.

IV. Some buses are clocks.

- (a) None follows
- (c) Only II follows
- (e) Only IV follows

- (b) Only I follows
- (d) Only III follows

10. Statements:

All stones are hammers.

No hammer is a ring.

Some rings are doors.

All doors are windows.

Conclusions:

I. Some windows are stones.

II. Some windows are rings.

III. No window is a stone.

IV. Some rings are stones.

- (a) Only I follows
- (c) Only III follows
- (e) Only either I or III and II follow

- (b) Only II follows
- (d) Only either I or III follows

Directions (11-15): Read all the conclusions and then decide which of the given conclusions logically follows from the given statements disregarding commonly known facts.

11. Statements: Some red are blue. Some blue are grey.

All grey are white. No white is black.

Conclusions: I. No black is grey.

II. Some blue are white.

III. Some black are red.

IV. No black is red.

- (a) Only I and II follow
- (c) Only I and either III or IV follow
- (e) None of these

- (b) Only either III or IV follows
- (d) Only I, II and either III or IV follow

12. Statements: All red are white. Some white are pink.

Some pink are yellow. No yellow is blue.

Conclusions: I. No blue is pink.

II. Some pink are red.

III. Some blue are red.

IV. Some blue are pink.

(a) None follows

(b) Only either I or IV follows

(c) Only I follows

(d) Only III & IV follow

(e) All follow

13. Statements: Some blue are black. Some black are grey.

All grey are red. All red are pink.

Conclusions: I. Some red are black.

II. Some pink are black.

III. Some pink are grey.

IV. Some red are blue.

(a) Only I & II follow

(b) Only II & III follow

(c) Only I, II and III follow

(d) All follow

(e) None of these

14. Statements: All green are pink. Some pink are black.

Some black are blue. All blue are white.

Conclusions: I. Some black are white.

II. Some blue are pink.

III. Some pink are green.

IV. No green is white.

(a) None follows

(b) Only I and III follows

(c) Only III follows

(d) Only either I or II follows

(e) None of these

15. Statements: Some blue are white. All white are red.

All red are pink. Some pink are yellow.

Conclusions: I. Some yellow are red.

II. Some yellow are white.

III. All red are white.

IV. Some yellow are blue.

(a) None follows

(b) Only I follows

(c) Only II follows

(d) Only II & III follow

(e) None of these

Directions (16-18): In each of the questions below are given three statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance with commonly known facts. Read all conclusions and then decide which of the given conclusions logically follows from the given statements, disregarding commonly known facts. Give answer-

(a) If only conclusion I follows

(b) If only conclusion II follows

(c) If only conclusion I or II follows

- (d) If neither conclusion I nor II follows
- (e) If both conclusions I and II follows

16. Statements: Some poor are rich. All rich are doctors. Some intelligent are doctors.

Conclusions: I. At least some poor are intelligent.

II. All intelligent being rich is a possibility.

17. Statements: Some poor are rich. All rich are doctors. Some intelligent are doctors.

Conclusions: I. All intelligent being doctors is a possibility.

II. Some poor are doctors.

18. Statements: All fans are bulbs. All wires are holders. Some wires are bulbs.

Conclusions: I. At least some fans are wires.

II. All holders being fans is a possibility.

Directions (19-20) : In each of the questions below are given three statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance with commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follows from the given statements, disregarding commonly known facts. Give answer-

- a) If only conclusion I follow.
- b) If only conclusion II follows.
- c) If either conclusion I or conclusion II follows.
- d) If neither conclusion I nor conclusion II follows.
- e) If both conclusions I and II follow.

19. Statements:

No fan is a light. All boards are fans. All fans are wires.

Conclusions:

I. All boards being wires is a possibility.

II. No boards is a light.

20. Statements:

No oil is a bread. No pot is a tap. All breads are pots.

Conclusions:

I. Some oils are not pots.

II. All taps being breads is a possibility.

DATA INTERPRETATION

Data interpretation is an act of analysing data with the objective to gain useful information from it. It is done to draw conclusions from the given data. Different statistical tools are used to represent the data in organized structures.

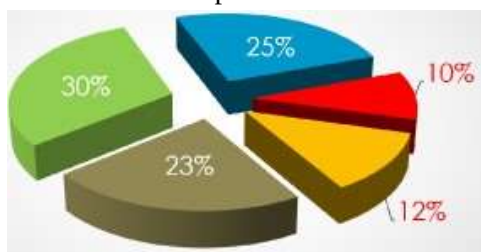
Different method in which data can be presented to solve Data Interpretation Questions:

1. Tables: Tables are the most convenient and versatile method to present data. Analyzing and drawing conclusions from tables is much easier than any other method. Tables are the fundamental method to represent data. In tables, the data is arranged in rows and columns which help us to scrutinize data efficiently.

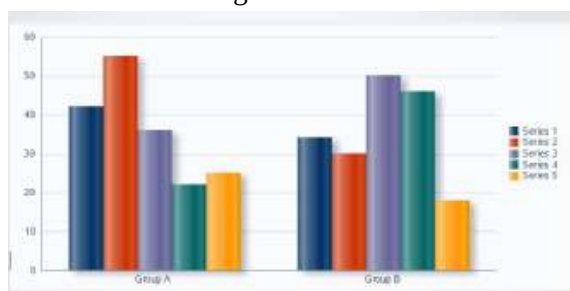
Standard	Boys	Girls
1	32	26
2	24	28
3	38	31
4	41	21

2. Pie chart: A Pie chart is a pictorial representation of data as part of a circle. The circle presents the total value and the different parts of the circle present certain portions of the data.

There are two types of pie chart: Normal and Exploded



3. Bar Graph: A bar graph is a way of representing data on the graph using X-axis and Y-axis. It is the most convenient way to present data. It is proven that using lengths in case of bar chart is a better indicator than pie charts where the data is categorized in terms of areas.



4. Line Graph: Line graph is the simplified version of the normal bar graph. It is simple to draw conclusions from Line Graph.



5. Mixed Graph: Pie Chart along with tabular data, bar graph along with tabular data.

Some important tricks to solve data interpretation questions:

1. Memorize some fractional and percentage values.
2. Don't waste time with lengthy calculations. Try to solve questions with approximation.
3. Use previous calculation if needed again.
4. Important topics to concentrate in data interpretation is Ratios, Percentages and Average.
5. Don't assume anything, There are few questions which ask to find out some data which is not available. Always be alert enough to see whether the data given is enough to answer the question or not and do not answer the question based on assumptions.

Example 1: Study the following table and answer the questions.

Classification of 100 Students based on the Marks Obtained by them in Physics and Chemistry in an Examination.

Subject	Marks out of 50				
	40 and above	30 and above	20 and above	10 and above	0 and above
Physics	9	32	80	92	100
Chemistry	4	21	66	81	100
Average (Aggregate)	7	27	73	87	100

I. What is the difference between the number of students passed with 30 as cut-off marks in Chemistry and those passed with 30 as cut-off marks in aggregate?

Solution: Required difference = (No. of students scoring 30 and above marks in Chemistry)
– (Number of students scoring 30 and above marks in aggregate)
= 27 – 21 = 6.

II. If at least 60% marks in Physics are required for pursuing higher studies in Physics, how many students will be eligible to pursue higher studies in Physics?

Solution:

$$\text{We have 60\% of 50} = \left(\frac{60}{100} \times 50 \right) = 30.$$

∴ Required number

= No. of students scoring 30 and above marks in Physics

= 32

III. The percentage of number of students getting at least 60% marks in Chemistry over those getting at least 40% marks in aggregate, is approximately?

Solution:

Number of students getting at least 60% marks in Chemistry

= Number of students getting 30 and above marks in Chemistry

= 21.

Number of students getting at least 40% marks in aggregate

= Number of students getting 20 and above marks in aggregate

= 73.

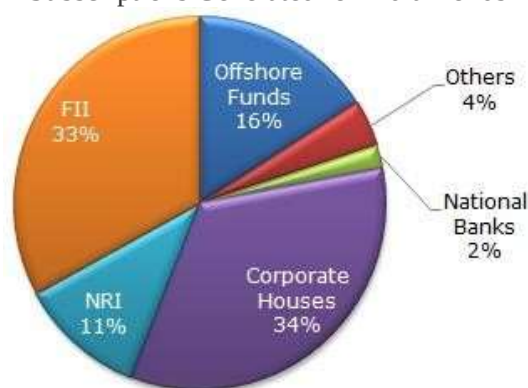
$$\text{Required percentage} = \left(\frac{21}{73} \times 100 \right) \%$$

$$= 28.77\%$$

$$\approx 29\%.$$

Example 2: The following pie chart shows the amount of subscriptions generated for India Bonds from different categories of investors.

Subscriptions Generated for India Bonds



I. In the corporate sector, approximately how many degrees should be there in the central angle?

Solution:

Since 1% = 3.6 degrees

Corporate Houses = 34%

$$34 \times 3.6 = 122.4^\circ$$

II. If the investment by NRI's are Rs 4,000 crore, then the investments by corporate houses and FII's together is:

Solution:

Corporate Houses and FII's together = 34 + 33 = 67%

NRI = 11%

Total investment by Corporate Houses and FII's together =

$$(67/11) \times 4000 = 24,363.6364$$

III. If the total investment other than by FII and corporate houses is Rs 335,000 crore, then the investment by NRI's and Offshore funds will be (approximately)?

Solution:

Investment other than NRI and corporate houses is 33% = 335000.

Also, investment by offshore funds and NRI's is equal to 27%.

$$\text{Hence, } 27 \times 335,000/33 = 274,090.909$$

Example 3: The following bar graph shows the Income and Expenditures (in million US \$) of five companies in the year 2001. The percent profit or loss of a company is given by



Income and Expenditure (in million US \$) of five companies in the year 2001

I. The companies M and N together had a percentage of profit/loss of?

Solution:

Total income of Companies M and N together

$$= (35 + 50) \text{ million US \$}$$

$$= 85 \text{ million US \$}$$

Total expenditure of Companies M and N together

$$= (45 + 40) \text{ million US \$}$$

$$= 85 \text{ million US \$}.$$

$\therefore$ Percent Profit/Loss of companies M and N together

$$\% \text{ Profit/Loss} = \left(\frac{85 - 85}{85} \times 100 \right) = 0\%.$$

Thus, there was neither loss nor profit for companies M and N together.

II. In 2001, what was the approximate percentage of profit/loss of all the five Companies taken together?

Solution:

Total income of all five companies

$$= (35 + 50 + 40 + 40 + 50) \text{ million US \$}$$

$$= 215 \text{ million US \$}.$$

Total expenditure of all five companies

$$= (45 + 40 + 45 + 30 + 45) \text{ million US \$}$$

$$= 205 \text{ million US \$}.$$

$$\therefore \% \text{ Profit} = \left[\frac{(215 - 205)}{205} \times 100 \right] \% = 4.88\% \approx 5\%.$$

III. Which company earned the maximum percentage profit in the year 2001?

Solution:

The percentage profit/loss in the year 2001 for various companies are:

$$\text{For M} = \left[\frac{(35 - 45)}{45} \times 100 \right] \% = -22.22\% \text{ i.e., Loss} = 22.22\%.$$

$$\text{For N} = \left[\frac{(50 - 40)}{40} \times 100 \right] \% = 25\% \text{ i.e., Profit} = 25\%.$$

$$\text{For P} = \left[\frac{(40 - 45)}{45} \times 100 \right] \% = -11.11\% \text{ i.e., Loss} = 11.11\%.$$

$$\text{For Q} = \left[\frac{(40 - 30)}{30} \times 100 \right] \% = 33.33\% \text{ i.e., Profit} = 33.33\%.$$

$$\text{For R} = \left[\frac{(50 - 45)}{45} \times 100 \right] \% = 11.11\% \text{ i.e., Profit} = 11.11\%.$$

Clearly, the Company Q earned the maximum profit in 2001.

IV. For Company R, if the expenditure had increased by 20% in year 2001 from year 2000 and the company had earned profit of 10% in 2000, what was the Company's income in 2000 (in million US \$)?

Solution:

Let the expenditure of Company R in 2000 be x million US \$.

Then, expenditure of Company R in 2001 = $\left(\frac{120}{100} \times x \right)$ million US \$.

$$\therefore \frac{120x}{100} = 45 \Rightarrow x = 37.5.$$

i.e., expenditure of Company R in 2000 = 37.5 million US \$.

Let the income of Company R in 2000 be I million US \$.

$$\text{Then, } 10 = \frac{(I - 37.5)}{37.5} \times 100 \quad [\because \% \text{ Profit in 2000} = 10\%]$$

$$\Rightarrow I - 37.5 = 3.75$$

$$\Rightarrow I = 41.25$$

i.e., Income of Company R in 2000 = 41.25 million US \$.

Example 4: The following line graph gives the percentage of the number of candidates who qualified an examination out of the total number of candidates who appeared for the examination over a period of seven years from 1994 to 2000.

Percentage of Candidates Qualified to Appeared in an Examination Over the Years



I. The difference between the percentages of candidates qualified to appear was maximum in which of the following pairs of years?

Solution: The differences between the percentages of candidates qualified to appear for the give pairs of years are:

$$\text{For 1994 and 1995} = 50 - 30 = 20$$

$$\text{For 1998 and 1999} = 80 - 80 = 0$$

$$\text{For 1994 and 1997} = 50 - 30 = 20$$

$$\text{For 1997 and 1998} = 80 - 50 = 30$$

$$\text{For 1999 and 2000} = 80 - 60 = 20$$

Thus, the maximum difference is between the years 1997 and 1998.

II. If the number of candidates qualified in 1998 was 21200, what was the number of candidates appeared in 1998?

Solution:

The number of candidates appeared in 1998 be x .

$$\text{Then, } 80\% \text{ of } x = 21200 \Rightarrow x = \frac{21200 \times 100}{80} = 26500 \text{ (required number).}$$

III. If the total number of candidates appeared in 1996 and 1997 together was 47400, then the total number of candidates qualified in these two years together was?

Solution: The total number of candidates qualified in 1996 and 1997 together, cannot be determined until we know at least, the number of candidates appeared in any one of the two years 1996 or 1997 or the percentage of candidates qualified to appeared in 1996 and 1997 together.

Hence, the data is inadequate

IV. The total number of candidates qualified in 1999 and 2000 together was 33500 and the number of candidates appeared in 1999 was 26500. What was the number of candidates in 2000?

Solution:

$$\text{The number of candidates qualified in 1999} = (80\% \text{ of } 26500) = 21200.$$

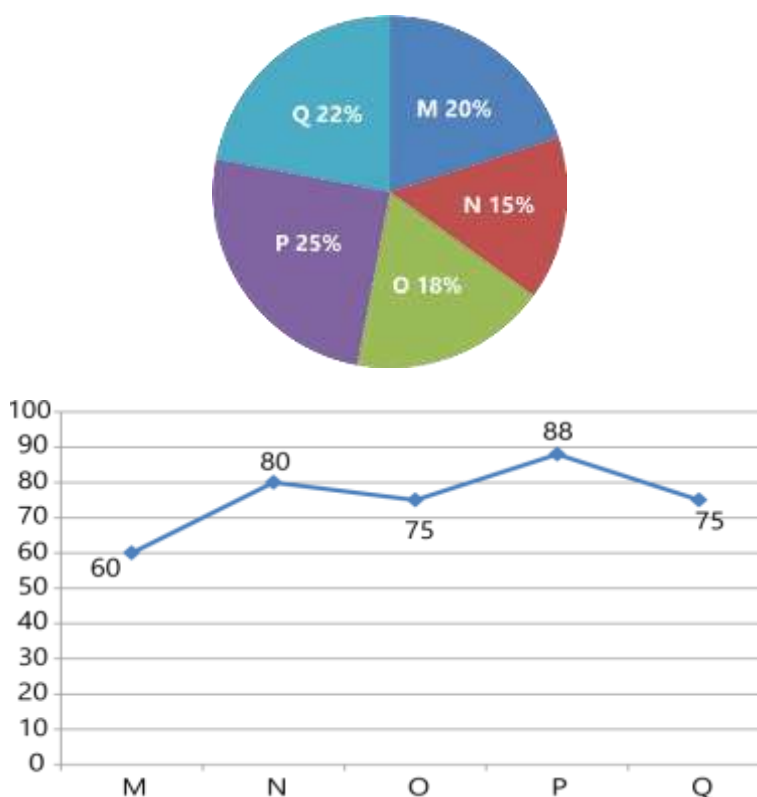
$$\therefore \text{Number of candidates qualified in 2000} = (33500 - 21200) = 12300.$$

Let the number of candidates appeared in 2000 be x .

$$\text{Then, } 60\% \text{ of } x = 12300 \Rightarrow x = \left(\frac{12300 \times 100}{60} \right) = 20500.$$

Example 5: Study the following information carefully and answer the question given below:
Pie chart shows the production of rice by 5 different factories and line chart shows the percentage of sale.

Total production = 3600 kg



I. In which of the following factory the sale of rice is the highest?

Solution: Sale of rice by factory M = $3600 \times 20\% \times 60\% = 432$ kg
 Sale of rice by factory N = $3600 \times 15\% \times 80\% = 432$ kg
 Sale of rice by factory O = $3600 \times 18\% \times 75\% = 486$ kg
 Sale of rice by factory P = $3600 \times 25\% \times 88\% = 792$ kg
 Sale of rice by factory Q = $3600 \times 22\% \times 75\% = 594$ kg
 Sale of rice of factory P is the highest.

II. Rice sold by factory M and N together is how much percent more than the production of rice by factory M?

Solution: Rice sold by factory M = $3600 \times 20\% \times 60\% = 432$ kg
 Rice sold by factory N = $3600 \times 15\% \times 80\% = 432$ kg
 Production of rice by factory M = $3600 \times 20\% = 720$ kg
 $\% \text{ more} = \{[432 + 432 - 720] \times 100\} / 720$
 $= (144 \times 100) / 720 = 20\%$

III. If the cost price of rice is Rs 45 per kg and profit earned by factory Q on 1 kg rice is 20% of the cost price, find the total profit earned by factory Q.

Solution: Rice sold by factory Q = $3600 \times 22\% \times 75\% = 594$ kg
 Profit on 1 kg rice = $45 \times 20\% = \text{Rs } 9$
 Total profit = $9 \times 594 = \text{Rs } 5346$

IV. What percent of total rice has been sold by all the factory?

Solution: Sale of rice by factory M = $3600 \times 20\% \times 60\% = 432$ kg
Sale of rice by factory N = $3600 \times 15\% \times 80\% = 432$ kg
Sale of rice by factory O = $3600 \times 18\% \times 75\% = 486$ kg
Sale of rice by factory P = $3600 \times 25\% \times 88\% = 792$ kg
Sale of rice by factory Q = $3600 \times 22\% \times 75\% = 594$ kg
Total sale = $432 + 432 + 486 + 792 + 594 = 2736$ kg
Required % = $(2736 \times 100) / 3600 = 76\%$

V. What is the ratio of rice sold by factory P and Q together to the rice sold by factory N and O together?

Solution: Rice sold by factory P = $3600 \times 25\% \times 88\% = 792$ kg
Rice sold by factory Q = $3600 \times 22\% \times 75\% = 594$ kg
Rice sold by factory N = $3600 \times 15\% \times 80\% = 432$ kg
Rice sold by factory O = $3600 \times 18\% \times 75\% = 486$ kg
Required Ratio = $792 + 594 : 432 + 486$
 $= 1386 : 918 = 77 : 51$

LEVEL – I

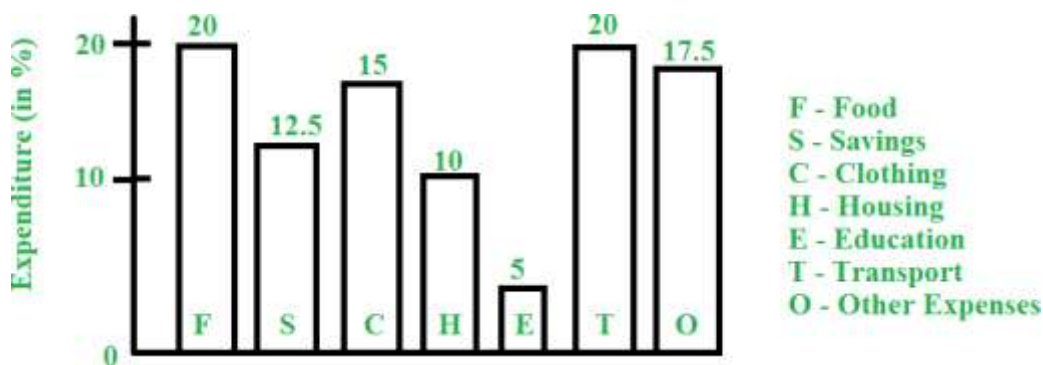
Directions (1-3): Classification of 100 students based on the marks obtained by them in Physics and Chemistry in an examination.

Marks out of 50 → Subject ↓	40 and above	30 and above	20 and above	10 and above	0 and above
Physics	9	32	80	92	100
Chemistry	4	21	66	81	100
(Aggregate) Average	7	27	73	87	100

- If at least 60% marks in Physics are required for pursuing higher studies in Physics, how many students will be eligible to pursue higher studies in Physics?
A. 27 B. 32 C. 34 D. 41
- The percentage of the number of students getting at least 60% marks in Chemistry over those getting at least 40% marks in aggregate, is approximately:
A. 21% B. 27% C. 29% D. 31%
- If it is known the at least 23 students were eligible for a Symposium on Chemistry, the minimum qualifying marks in Chemistry for eligibility to Symposium would lie in the range:
A. 40-50 B. 30-40 C. 20-30 D. Cannot be determined

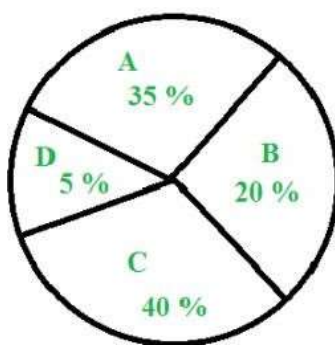
Directions (4-6): Study the following bar graph and answer the questions that follow:

Total monthly income = Rs. 50,000



- What amount is spent on food?
A. Rs. 10,000 B. Rs. 15,000 C. Rs. 20,000 D. Rs. 30,000
- How much more money is spent on clothing and housing together than on transportation?
A. Rs. 1,000 B. Rs. 5,000 C. Rs. 2,000 D. Rs. 2,500
- What percent of amount on food is spent as amount on education?
A. 10% B. 40% C. 25% D. 20%

Directions (7-9): Study the pie chart below and answer the questions that follow:



The above pie chart shows the sales of four different types of articles in a shop.

7. What is the central angle of type A?

- A. 128 degrees B. 126 degrees C. 136 degrees D. 140 degrees

8. If the total sale is 1200, what is the sale of B?

- A. 120 B. 360 C. 260 D. 240

9. What is the difference between the central angle of A and D?

- A. 108 degrees B. 120 degrees C. 96 degrees D. 130 degrees

Directions (10-14): The following table shows the number of boys and girls of different schools that have participated in a scholarship test over five years.

School	A		B		C		D	
	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls
2001	300	80	280	60	320	84	450	70
2002	320	70	300	80	424	100	320	60
2003	340	90	420	120	230	70	360	90
2004	370	100	480	140	360	120	500	120

10. How many girls more participated in year 2004 as compared to 2003 for all schools taken together?

- A. 110 B. 370 C. 480 D. 210

11. The ratio of boys: girls for school B over the years 2001-2004 is?

- A. 17:34 B. 14:37 C. 37:10 D. 34:17

12. The percentage increase in participation of boys from school B in 2004 over those in 2001 is nearly?

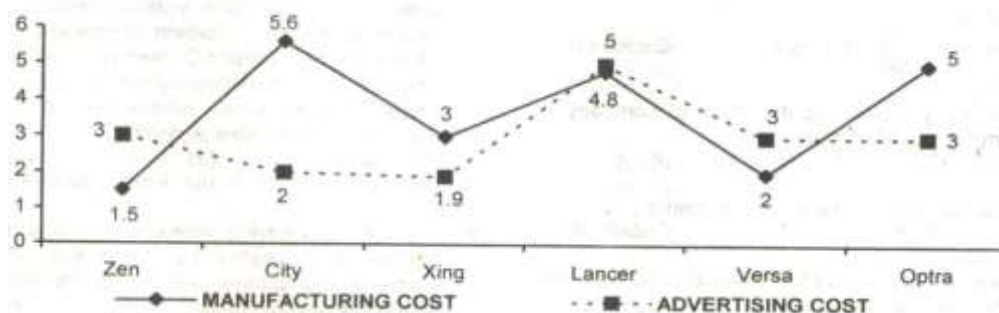
- A. 28% B. 71% C. 72% D. 72%

13. The total number of participants in year 2003 is

- A. 2000 B. 1830 C. 1720 D. 1640

14. In year 2004 the number of participating girls is what percent of number of participating boys?
 A. 17:56 B. 56:17 C. 57:16 D. 16:57

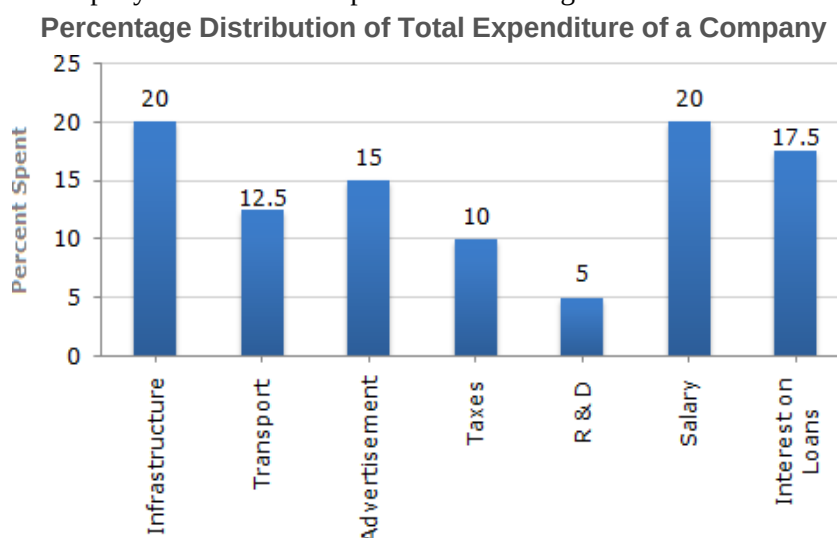
Directions (15-17): These questions are based on the line graph below.
 (in Rs. lakhs)



Total Cost = Manufacturing Cost + Advertising Cost

15. For which of the following cars is the manufacturing cost as a percentage of advertising cost the least?
 A. Xing B. City C. Lancer D. Zen
16. In a certain year, 30,000 "City" cars, are produced, and are sold at Rs. 9.3 lacs/car. If 2% of the total profit is given as a bonus to the 2,040 engineers, the amount received by each engineer as bonus is (in Rs.)
 A. 5,000 B. 50,000 C. 5 lacs D. None of these
17. The company that manufactures Zen produces 500 Zen cars per day while the company that manufactures City produces 600 City per day. They sell them at Rs. 6 lacs/car and Rs. 8.4 lacs/car respectively. The profit made by the former is approximately what percentage of that of the latter?
 A. 100% B. 156% C. 250% D. None of these

Directions (18-20): The bar graph given below shows the percentage distribution of the total expenditures of a company under various expense heads during 2003.



18. The expenditure on the interest on loans is by what percent more than the expenditure on transport?

- A. 5% B. 40% C. 20% D. 10%

19. If the interest on loans amounted to Rs. 2.45 crores then the total amount of expenditure on advertisement, taxes and research and development is?

- A. Rs. 7 crores B. Rs. 5.4 crores C. Rs. 4.2 crores D. Rs. 3 crores

20. The total amount of expenditures of the company is how many times of expenditure on research and development?

- A. 20 B. 27 C. 18 D. 8

LEVEL – II

Directions (1-4): Study the following table carefully to answer these questions:

Number of workers employed in six units of a factory over the years.

UnitYear	A	B	C	D	E	F
1998	145	88	115	120	140	136
1999	128	76	122	112	152	132
2000	136	96	132	124	158	140
2001	183	92	125	135	166	126
2002	160	107	140	118	170	146
2003	152	110	148	128	175	150

1. In the year 2000 the number of workers employed by unit 'C' is what percent of the total number of workers employed by all the units in the same year (rounded off to two places of decimal)?

- A. 16.39 B. 17.21 C. 16.88 D. None of these

2. For all the given years what is the difference between the average number of workers in units D and E?

- A. 37 B. 33 C. $32\frac{2}{3}$ D. $37\frac{1}{3}$

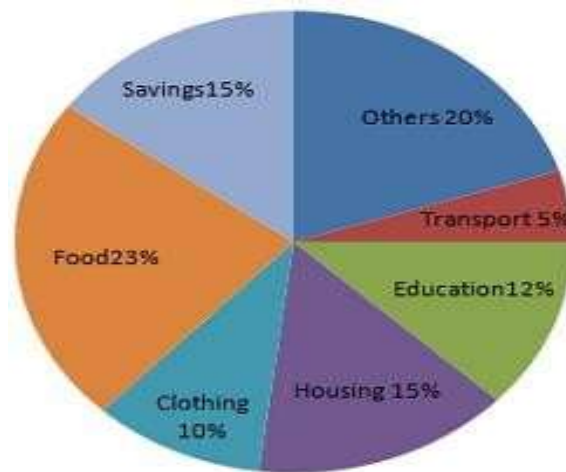
3. In which year is the percentage increase/decrease in the number of workers employed the minimum for unit 'F'?

- A. 2003 B. 1999 C. 2002 D. 2000

4. What is the approximate ratio of the number of workers employed in all the units in 1998 to that in 1999?

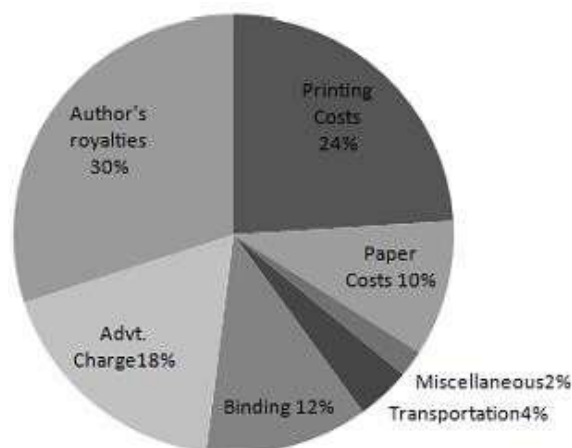
- A. 13:14 B. 37:36 C. 10:9 D. 13:11

Directions (5-7): Pie chart shows the percent of money spent by family on various item during 1999. Study the graph and answer these questions.



5. If the total amount spent was Rs. 46,000, how much money was spent on clothing and housing together?
 A. Rs. 11500 B. Rs. 1150 C. Rs. 10000 D. Rs. 15000
6. The ratio of the total amount of money spent on housing to that spent on education was
 A. 5:2 B. 2:5 C. 4:5 D. 5:4
7. If the total expenditure of the family for the year 1999 was Rs. 46,000, the family saved during the year
 A. Rs. 1500 B. Rs. 15000 C. Rs. 6900 D. Rs. 3067 approx.

Directions (8-10): These questions are based on the circle graph which shows the expenditure incurred in bringing out a book by a publisher.

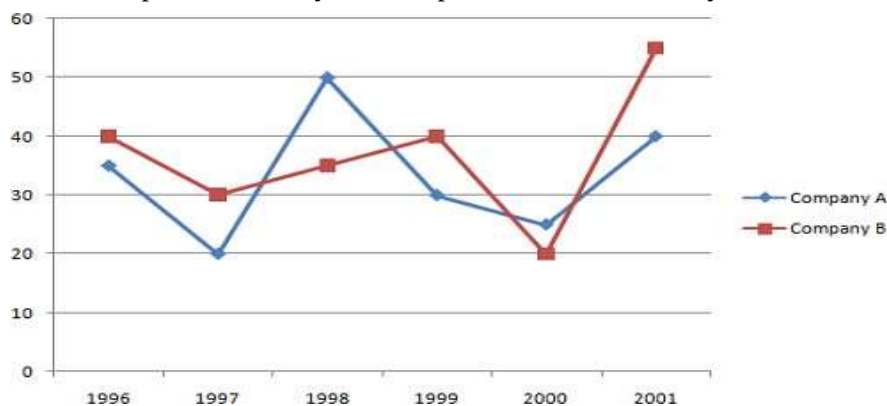


8. What should be the central angle of the sector for transportation charges?
 A. 4° B. 14.4° C. 12.4° D. 8.4°
9. If the advertisement charges amount to Rs. 18,000, the total expenditure incurred in bringing out the book is Rs.
 A. 60,300 B. 63,000 C. 9,000 D. 1,00,000

10. If the author's royalties amount to Rs. 30,000, the binder's charges amount to Rs.
A. 6,000 B. 10,500 C. 12,000 D. 15,000

Directions (11-15): Study the following graph to answer these questions:

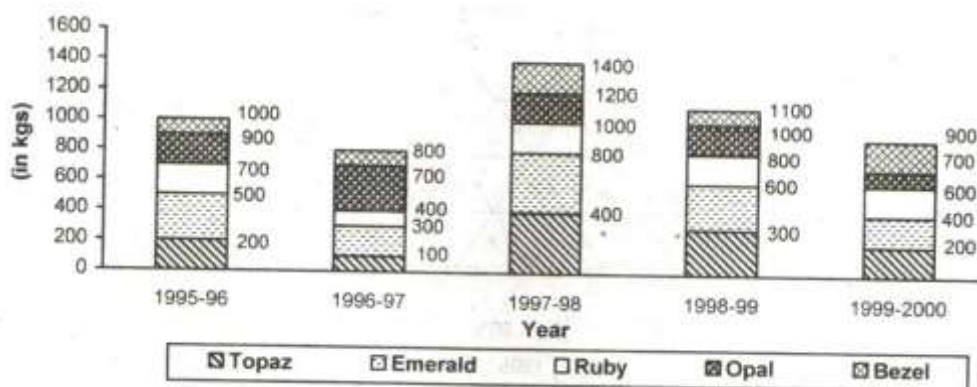
Percent profit earned by two companies A & B over the years.



$$\text{Profit} = \text{Income} - \text{Expenditure}$$
$$\% \text{ Profit} = \frac{\text{Profit}}{\text{Expenditure}} \times 100$$

11. If the income of company A in 1998 was Rs. 1,42,500 what was its expenditure in that year?
A. 1,05,000 B. 95,000 C. 99,500 D. 1,05,555
12. Expenditure of company B in 1999 was 90% of its expenditure in 1998. Income of company B in 1999 was what percent of its income in 1998?
A. 130.5 B. 96 (2/3) C. 121.5 D. 99 (1/3)
13. If the expenditure of company A in 1997 was Rs. 70 lakhs and income of company A in 1997 was equal to its expenditure in 1998, what was the total income (in Rs. lakh) of the company A in 1997 & 1998 together?
A. 175 B. 131.25 C. 218.75 D. Cannot be determined
14. Expenditure of company B in years 1996 and 1997 were in the ratio of 5:7 respectively. What was the respective ratio of their incomes?
A. 10:13 B. 8:13 C. 13:14 D. 11:14
15. Total expenditure of companies A & B together in 2001 was Rs. 13.5 lakhs. What was the total income of the two companies (in Rs. lakh) in that year?
A. 19.575 B. 20.25 C. 19.75 D. Cannot be determined

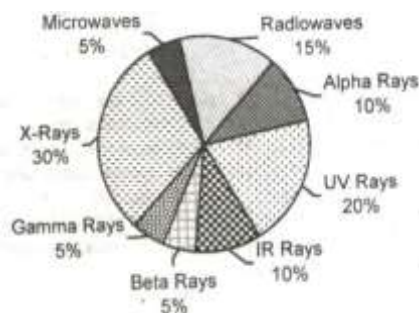
Directions (16-19): The following questions are based on the stacked bar graph given below.
Sales of various precious stones in India for the period of 1995-1996 to 1999-2000



16. What is the total sale of ruby as a percent of the total sales of precious stones for the given period?
A. 17.3% B. 19.23% C. 23.1% D. None of these
17. By what percent is the average annual sale of Emerald for the given period more than the sales of Opal in 1998-1999?
A. 120% B. 50% C. 25% D. 40%
18. For how many years is the sale of Bezel as a percentage of the total sales of precious stones less than that of Topaz?
A. One B. Two C. Three D. Four
19. If the sales of Topaz increased from 1994-1995 to 1995-1996 by 25% and increased from 1999-2000 to 2000-01 by 50%, then what is the difference between the sales of Topaz in 1994-95 and that in 2000-01?
A. 50, 000 tonnes B. 100, 000 tonnes C. 140, 000 tonnes D. 160, 000 tonnes

20. The question is based on the pie chart given below.

Constituents of Sunrays received in 1 minute



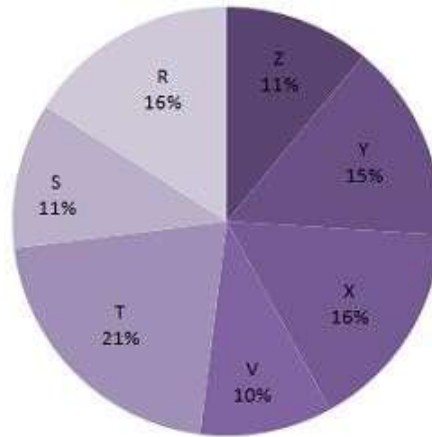
Total sunrays received in 1 minute = 3600 units

If the human body can withstand a maximum of 9720 units of IR rays, when exposed to the sun continuously, then what is the maximum time (in minutes) that any person could stand in the sun without crossing the threshold limit of IR rays?

- A. 19 B. 23 C. 27 D. 29

LEVEL – III

Directions (1-5): Study the following pie-chart and the table and answer the question based on them.
Proportion of population of seven villages in 1997



Village %	Population Below Poverty Line
X	38
Y	52
Z	42
R	51
S	49
T	46
V	58

- Find the population of village S if the population of village X below poverty line in 1997 is 12160.
A. 18500 B. 20500 C. 22000 D. 26000
- The ratio of population of village T below poverty line to that of village Z below poverty line in 1997 is?
A. 11:23 B. 13:11 C. 23:11 D. 11:13
- If the population of village R in 1997 is 32000, then what will be the population of village Y below poverty line in that year?
A. 14100 B. 15600 C. 16500 D. 17000
- If in 1998, the population of village Y and V increase by 10% each and the percentage of population below poverty line remains unchanged for all the villages, then find the population of village V below poverty line in 1998, given that the population of village Y in 1997 was 30000.
A. 11250 B. 12760 C. 13140 D. 13780
- If in 1999, the population of village R increased by 10% while that of village Z reduces by 5% compared to that in 1997 and the percentage of population below poverty line remains unchanged for all the villages, then find the approximate ratio of population of village R below poverty line to the ratio of population of village Z below poverty line for the year 1999.
A. 2:1 B. 3:2 C. 4:3 D. 5:4

Directions (6-10): These questions are based on the following graph.

The Country wise break up Exports/Import of Country 'XYZ' in 1996

(in Rs. thousand crores)



Trade Surplus = Exports – Imports; Trade Deficit = Imports – Exports

6. The cumulative trade deficit of country XYZ is approximately what percent of its average imports from each of the above mentioned countries?

- A. 65% B. 9% C. 6.5% D. 0.6%

7. If the average cost of exports is Rs. 2000 per ton and that of imports of Rs. 3000 per ton, then by what percent is the total tonnage of exports more/less than the total tonnage of imports

- A. 33.3% more B. 49% less C. 32.8 % more/less D. 49% more

8. By what percentage are the imports from the country to which the exports are the highest more than the exports to the country from which the imports are the least?

- A. 175% B. 80% C. 55.55% D. 125%

9. Which of the following statements is definitely true?

- A. Country XYZ has a cumulative trade surplus of Rs. 1 crore
 B. The cumulative trade deficit of country XYZ is approximately one-fifteenth of its total imports.
 C. The trade deficit of country XYZ considering its trade with China alone is 300% more than its cumulative trade deficit/surplus.
 D. The difference between the highest exports to any country and the lowest imports from any country is equal to the average of the exports to Brazil and Germany.

10. What is the ratio of the total imports from Brazil, Japan, South Africa, Russia and China, to the total exports to the other five countries?

- A. 0.975 B. 1.026 C. 0.96 D. None of these

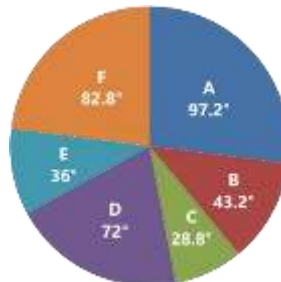
Directions (11-15): In the following pie-charts, pie chart I shows the angular distribution of the total number of employees among six companies and pie-chart II shows the angular distribution of the total number female employees among these companies.

Chart I



Total Female Employees = 14000

Chart II

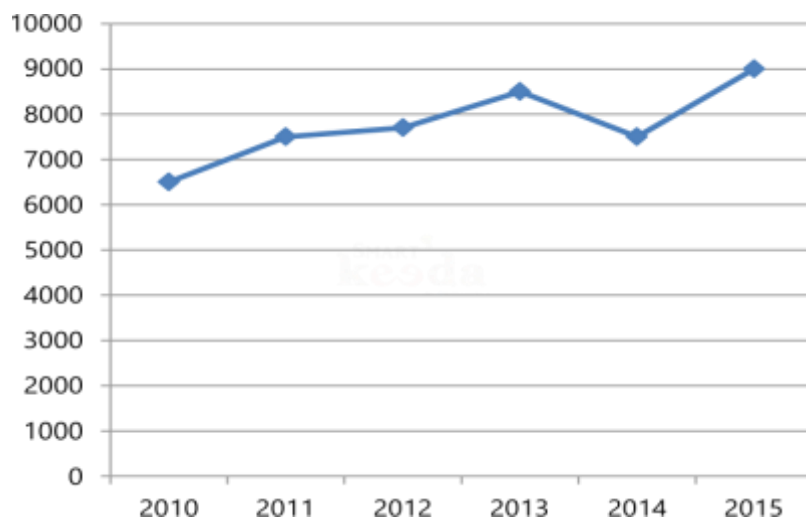


Total Female Employees = 6000

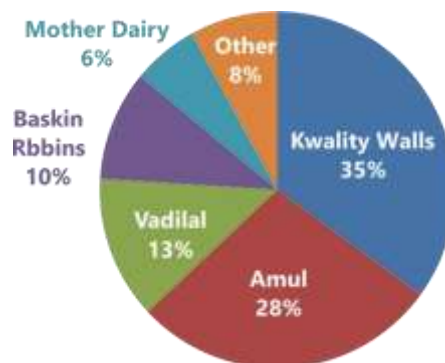
11. What is the total number of male employees in Company B?
 A. 720 B. 1040 C. 1260 D. 1380
12. What is the difference between the total number of male employees and the total number of female employees in Company A?
 A. 220 B. 240 C. 260 D. 280
13. The total number of female employees of Company C is approximately what per cent of the total number of employees of Company B?
 A. 17% B. 23% C. 27% D. 30%
14. What is the difference between the total number of female employees of Company F and the total number of female employees of Company E?
 A. 740 B. 760 C. 780 D. 820
15. The total number of female employees of Company A is approximately what per cent of the total number female employees of Company D?
 A. 74% B. 120% C. 135% D. 150%

Directions (16-20): Refer to the line graph and pie-chart below and answer the question that follows. The line graph shows Ice-cream market size (in crores) in India from the year 2010 to 2015 and the pie chart depicts percentage market shares of different Ice-cream brands in 2013.

Ice-cream market in India



Market Share of various companies in 2013



16. If the market share of Vadilal increases by 38% from 2013 to 2015, what would be the approximate market share of Vadilal in 2015?
A. 12% B. 17% C. 14% D. 21%
17. What is the difference between the market share of Amul and that of Mother Dairy in 2013?
A. ₹ 1650 crore B. ₹ 1950 crore C. ₹ 1870 crore D. ₹ 1750 crore
18. If the market share of all the companies remains the same in 2014, what was the increase/decrease in the turnover of Amul from 2013 to 2014?
A. decreased by 280 crore B. increased by 520 crore
C. decreased by 420 crore D. increased by 620 crore
19. What is the market share of Mother Dairy in 2016, if the market share of each company remains the same as in 2013 and the ice-cream market increased by 20% from 2013 to 2016?
A. ₹ 640 B. ₹ 720 C. ₹ 612 D. ₹ 900
20. If the market share of the companies in 2015 is the same as in 2013 and in 2016 the ice-cream market increases by 10% from the previous year but the share of Kwality Walls falls by 20%. What is the percentage change in the market share of Kwality Walls from 2015 to 2016?
A. 18.24% B. 13.14% C. 12.54% D. 9.54%

ANSWER KEYS

Number System									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	A	3	D	4	B	5	A
6	C	7	B	8	C	9	A	10	A
11	C	12	C	13	B	14	A	15	D
16	D	17	B	18	A	19	C	20	A
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	B	2	C	3	B	4	C	5	B
6	C	7	B	8	A	9	B	10	B
11	D	12	C	13	C	14	C	15	B
16	C	17	B	18	B	19	C	20	A
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	D	2	C	3	C	4	C	5	D
6	D	7	D	8	C	9	C	10	C
11	B	12	A	13	C	14	C	15	C
16	D	17	D	18	B	19	D	20	D

Average									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	B	3	B	4	A	5	D
6	C	7	B	8	A	9	D	10	A
11	A	12	C	13	D	14	C	15	C
16	B	17	A	18	A	19	D	20	C
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	A	2	D	3	B	4	D	5	B
6	D	7	C	8	B	9	C	10	A
11	B	12	C	13	A	14	C	15	D
16	C	17	A	18	D	19	D	20	B
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	B	2	A	3	D	4	D	5	D
6	B	7	D	8	A	9	D	10	B
11	D	12	C	13	D	14	B	15	D
16	A	17	A	18	A	19	B	20	C

Chapter 4 – Percentage									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	B	2	B	3	A	4	C	5	C
6	D	7	A	8	A	9	D	10	A
11	C	12	B	13	C	14	A	15	D
16	B	17	B	18	A	19	B	20	A
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	B	2	D	3	C	4	A	5	D
6	D	7	B	8	D	9	C	10	C
11	A	12	C	13	D	14	B	15	D
16	D	17	C	18	B	19	B	20	A
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	A	2	A	3	D	4	A	5	B
6	C	7	A	8	D	9	C	10	B
11	B	12	A	13	B	14	A	15	C
16	C	17	D	18	B	19	D	20	C
Chapter 5 – Profit and Loss									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	B	2	B	3	C	4	B	5	C
6	A	7	C	8	C	9	C	10	D
11	A	12	D	13	C	14	A	15	D
16	B	17	B	18	C	19	B	20	C
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	A	2	D	3	B	4	B	5	A
6	B	7	A	8	B	9	A	10	B
11	C	12	A	13	D	14	B	15	A
16	C	17	A	18	B	19	B	20	B
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	A	2	D	3	C	4	B	5	B
6	A	7	A	8	D	9	C	10	D
11	D	12	C	13	C	14	C	15	C
16	B	17	A	18	C	19	A	20	C

Chapter 6- Direction Sense Test									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	D	2	B	3	D	4	B	5	C
6	C	7	A	8	C	9	D	10	A
11	B	12	B	13	C	14	A	15	B
16	B	17	A	18	B	19	A	20	B
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	D	3	C	4	D	5	D
6	D	7	A	8	A	9	B	10	B
11	D	12	A	13	D	14	C	15	A
16	A	17	D	18	B	19	D	20	C
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	D	2	B	3	C	4	C	5	C
6	A	7	A	8	C	9	C	10	B
11	B	12	A	13	B	14	D	15	D
16	A	17	D	18	D	19	B	20	C
Chapter 7- Blood Relation									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	A	2	B	3	B	4	A	5	A
6	A	7	D	8	C	9	C	10	A
11	B	12	D	13	C	14	A	15	D
16	A	17	A	18	A	19	C	20	C
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	A	2	B	3	B	4	C	5	C
6	B	7	D	8	B	9	D	10	D
11	C	12	B	13	A	14	B	15	A
16	D	17	D	18	C	19	A	20	A
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	D	3	D	4	A	5	D
6	B	7	B	8	D	9	D	10	C
11	B	12	D	13	B	14	B	15	B
16	D	17	B	18	D	19	B	20	C

Chapter 11- Number, Ranking and Time Sequence									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	B	2	A	3	D	4	A	5	C
6	C	7	C	8	B	9	C	10	C
11	C	12	A	13	C	14	B	15	B
16	B	17	C	18	C	19	D	20	D
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	B	3	D	4	B	5	D
6	C	7	B	8	D	9	B	10	A
11	C	12	B	13	C	14	D	15	A
16	D	17	D	18	C	19	A	20	C
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	D	2	D	3	C	4	B	5	A
6	B	7	D	8	B	9	A	10	D
11	A	12	B	13	A	14	A	15	D
16	C	17	C	18	D	19	B	20	A

Chapter 9 - Ratio and Proportion, Variation and Partnership									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	D	3	B	4	B	5	A
6	C	7	C	8	A	9	C	10	A
11	B	12	C	13	B	14	A	15	A
16	C	17	D	18	B	19	A	20	D
21	A	22	C	23	A	24	A	25	B
26	A	27	B	28	A	29	D	30	B
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	C	3	B	4	B	5	D
6	B	7	B	8	B	9	A	10	C
11	D	12	C	13	B	14	A	15	D
16	B	17	C	18	D	19	A	20	C
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	C	3	B	4	A	5	C
6	A	7	B	8	D	9	C	10	A
11	D	12	C	13	A	14	B	15	C
16	B	17	D	18	D	19	C	20	A

Chapter 10 - Mixture and Alligation									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	A	2	B	3	D	4	A	5	C
6	B	7	A	8	A	9	D	10	D
11	C	12	C	13	C	14	C	15	C
16	B	17	A	18	B	19	C	20	B
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	D	2	C	3	B	4	D	5	B
6	A	7	D	8	A	9	D	10	D
11	B	12	D	13	A	14	A	15	A
16	B	17	B	18	C	19	A	20	B
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	B	2	D	3	D	4	D	5	D
6	B	7	B	8	C	9	B	10	C
11	A	12	A	13	B	14	D	15	B

Chapter 15 – Permutation and Combination									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	D	3	C	4	A	5	A
6	C	7	B	8	A	9	D	10	D
11	C	12	B	13	C	14	D	15	B
16	A	17	C	18	B	19	D	20	A
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	A	2	D	3	C, B, D	4	A, B	5	A
6	D	7	B	8	B	9	C	10	C
11	D	12	C	13	B	14	D	15	A
16	B	17	C	18	B	19	D	20	A
21	A	22	D	23	B	24	B	25	B
26	D	27	A	28	C	29	A	30	D
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	C	3	B	4	C	5	B
6	D	7	A	8	C	9	B	10	A
11	C	12	A	13	D	14	A	15	A
16	C	17	B	18	C	19	A	20	B

Chapter 16 – Probability									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	A	2	C	3	D	4	C	5	C
6	D	7	B	8	B	9	B	10	A
11	B	12	D	13	A	14	D	15	A
16	B	17	B	18	A	19	C	20	A
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	D	2	A	3	C	4	D	5	D
6	C	7	A	8	A	9	D	10	D
11	C	12	D	13	A	14	D	15	B
16	B	17	A	18	A	19	A	20	C
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	C	3	B	4	B	5	D
6	A	7	D	8	D	9	B	10	C
11	A	12	B	13	C	14	B	15	C
16	A	17	C	18	B	19	D	20	C

Chapter 14 - Syllogism									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	A	2	E	3	A	4	A	5	C
6	D	7	A	8	A	9	B	10	E
11	A	12	E	13	A	14	E	15	B
16	E	17	D	18	D	19	D	20	D
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	E	2	A	3	D	4	A	5	E
6	C	7	B	8	C	9	A	10	D
11	B	12	D	13	E	14	B	15	A
16	B	17	C	18	B	19	E	20	D
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	E	2	E	3	C	4	E	5	D
6	B	7	C	8	E	9	A	10	E
11	D	12	B	13	C	14	B	15	A
16	B	17	E	18	B	19	B	20	D

Chapter 17 – Data Interpretation									
Level – I									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	B	2	C	3	C	4	A	5	D
6	C	7	B	8	D	9	A	10	A
11	C	12	B	13	C	14	D	15	D
16	B	17	D	18	B	19	C	20	A
Level – II									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	D	2	D	3	A	4	B	5	A
6	D	7	C	8	B	9	D	10	C
11	B	12	D	13	C	14	A	15	D
16	D	17	C	18	C	19	A	20	C
Level – III									
Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	C	2	C	3	B	4	B	5	A
6	C	7	D	8	B	9	D	10	A
11	D	12	C	13	B	14	C	15	C
16	B	17	C	18	A	19	C	20	D